MILITARY SPECIFICATION

* MODULES, STANDARD ELECTRONIC

LOGIC UNIT, ARITHMETIC, DIGITAL, KEY CODE RDH

This specification is approved for use by Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the procurement of an electronic arithmetic logic unit/look-ahead carry generator (ALU/LCG) module.

1.2 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table IV.

Name	Arithmetic logic unit
Family	Digital
Key code	RDH
Size	1A
Weight	35 grams maximum
Failure rate	1.186 failures/10^6 hours maximum
* Environment	Class I as specified in MIL-STD-1389
Maximum power dissipation	1,755 mW (outputs not loaded)
Supply voltage	5 Vdc ±10 percent
Input voltage	
Low level	-0.5 V minimum, 0.8 V maximum
High level	2.0 V minimum, V_{CC} maximum
Input current, M, C_N	
Low level (V_{IL} = 0.0 V)	-2.2 mA maximum
(V_{IL} = 0.5 V)	-2.0 mA maximum
High level (V_{IH} = 2.5 V)	50.0 µA maximum
(V_{IH} = 5.5 V)	1.0 mA maximum

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Electronic Systems Command, ATTN. ELEX 5043, Department of the Navy, Washington, D.C. 20360 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Input current, $\overline{P}3$		
Low level	(V_{IL} = 0.0 V)	-4.4 mA maximum
	(V_{IL} = 0.5 V)	-4.0 mA maximum
High level	(V_{IH} = 2.5 V)	100 µA maximum
	(V_{IH} = 5.5 V)	1.0 mA maximum
Input current, A, B, $\overline{P}2$		
Low level	(V_{IL} = 0.0 V)	-6.6 mA maximum
	(V_{IL} = 0.5 V)	-6.0 mA maximum
High level	(V_{IH} = 2.5 V)	150 µA maximum
	(V_{IH} = 5.5 V)	1.0 mA maximum
Input current, S0-S3, $\overline{G}3$, $\overline{P}0$, $\overline{P}1$		
Low level	(V_{IL} = 0.0 V)	-8.8 mA maximum
	(V_{IL} = 0.5 V)	-8.0 mA maximum
High level	(V_{IH} = 2.5 V)	200 µA maximum
	(V_{IH} = 5.5 V)	1.0 mA maximum
Input current, $\overline{C}_{IN}$		
Low level	(V_{IL} = 0.0 V)	-11.0 mA maximum
	(V_{IL} = 0.5 V)	-10.0 mA maximum
High level	(V_{IH} = 2.5 V)	250 µA maximum
	(V_{IH} = 5.5 V)	1.0 mA maximum
Input current, $\overline{G}0$, $\overline{G}2$		
Low level	(V_{IL} = 0.0 V)	-15.4 mA maximum
	(V_{IL} = 0.5 V)	-14.0 mA maximum
High level	(V_{IH} = 2.5 V)	350 µA maximum
	(V_{IH} = 5.5 V)	1.0 mA maximum
Input current, $\overline{G}1$		
Low level	(V_{IL} = 0.0 V)	-17.6 mA maximum
	(V_{IL} = 0.5 V)	-16.0 mA maximum
High level	(V_{IH} = 2.5 V)	400 µA maximum
	(V_{IH} = 5.5 V)	1.0 mA maximum
Output voltage (except EQ)		
Low level	(I_{OL} = 20 mA)	0.5 V maximum
High level	(I_{OH} = -1.0 mA)	2.5 V minimum
Output voltage, EQ (open collector output)		
Low level	(I_{OL} = 20 mA)	0.5 V maximum

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Output leakage current, EQ (open collector output)	
High level (V_{OH} = 5.5 V)	250 µA maximum
Output drive current (except EQ) maximum capability	
Low level (V_{OL} = 0.5 V)	20 mA maximum
High level (V_{OH} = 2.5 V)	-1.0 mA maximum
Propagation delay time (t_{PLH})	
A or B to F (SUM)	
(50 pF load)	3.0 ns minimum, 23.0 ns maximum
(150 pF load)	4.0 ns minimum, 26.0 ns maximum
B to F with carry (SUM)	
(50 pF load)	3.0 ns minimum, 34.0 ns maximum
(150 pF load)	4.0 ns minimum, 37.0 ns maximum
A or B to F (DIFF)	
(50 pF load)	3.0 ns minimum, 26.0 ns maximum
(150 pF load)	4.0 ns minimum, 30.0 ns maximum
B to F with carry (DIFF)	
(50 pF load)	3.0 ns minimum, 37.0 ns maximum
(150 pF load)	4.0 ns minimum, 40.0 ns maximum
B to F (LOGIC)	
(50 pF load)	3.0 ns minimum, 26.0 ns maximum
(150 pF load)	4.0 ns minimum, 30.0 ns maximum
A to $\overline{C}_O$ (SUM)	
(50 pF load)	3.0 ns minimum, 28.0 ns maximum
(150 pF load)	4.0 ns minimum, 35.0 ns maximum
B to $\overline{C}_O$ (DIFF)	
(50 pF load)	3.0 ns minimum, 32.0 ns maximum
(150 pF load)	4.0 ns minimum, 40.0 ns maximum
A to CG or CP (SUM)	
(50 pF load)	3.0 ns minimum, 17.0 ns maximum
(150 pF load)	4.0 ns minimum, 20.0 ns maximum
B to CG or CP (DIFF)	
(50 pF load)	3.0 ns minimum, 21.0 ns maximum
(150 pF load)	4.0 ns minimum, 24.0 ns maximum
B to EQ (DIFF)	
(50 pF load)	3.0 ns minimum, 31.0 ns maximum
(150 pF load)	4.0 ns minimum, 38.0 ns maximum
$\overline{C}_{IN}$ to $\overline{C}_O$	
(50 pF load)	3.0 ns minimum, 17.0 ns maximum
(150 pF load)	4.0 ns minimum, 24.0 ns maximum
$\overline{C}_{IN}$ to F	
(50 pF load)	3.0 ns minimum, 17.0 ns maximum
(150 pF load)	4.0 ns minimum, 21.0 ns maximum

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Propagation delay time (t_{PLH}) - Continued

S to F	
(50 pF load)	3.0 ns minimum, 45.0 ns maximum
(150 pF load)	4.0 ns minimum, 55.0 ns maximum
S to CG or CP or $\overline{C}_0$, M to F	
(50 pF load)	3.0 ns minimum, 45.0 ns maximum
(150 pF load)	4.0 ns minimum, 55.0 ns maximum
$\overline{P}0$-$\overline{P}2$ or $\overline{G}0$-$\overline{G}2$ to C_{N+X} or C_{N+Y} or C_{N+Z}	
(50 pF load)	1.0 ns minimum, 9.5 ns maximum
(150 pF load)	3.0 ns minimum, 14.0 ns maximum
P0-P3 to $\overline{P}$, $\overline{P}1$-$\overline{P}3$ to $\overline{G}$, $\overline{G}3$ to $\overline{G}$	
(50 pF load)	1.0 ns minimum, 11.0 ns maximum
(150 pF load)	3.0 ns minimum, 15.0 ns maximum
C_N to C_{N+X} or C_{N+Y} or C_{N+Z}	
(50 pF load)	1.0 ns minimum, 12.0 ns maximum
(150 pF load)	3.0 ns minimum, 18.0 ns maximum

Propagation delay time (t_{PHL})

A or B to F (SUM)	
(50 pF load)	3.0 ns minimum, 23.0 ns maximum
(150 pF load)	4.0 ns minimum, 26.0 ns maximum
B to F with carry (SUM)	
(50 pF load)	3.0 ns minimum, 34.0 ns maximum
(150 pF load)	4.0 ns minimum, 37.0 ns maximum
A or B to F (DIFF)	
(50 pF load)	3.0 ns minimum, 28.0 ns maximum
(150 pF load)	4.0 ns minimum, 32.0 ns maximum
B to F with carry (DIFF)	
(50 pF load)	3.0 ns minimum, 39.0 ns maximum
(150 pF load)	4.0 ns minimum, 42.0 ns maximum
B to F (LOGIC)	
(50 pF load)	3.0 ns minimum, 28.0 ns maximum
(150 pF load)	4.0 ns minimum, 32.0 ns maximum
A to $\overline{C}_0$ (SUM)	
(50 pF load)	3.0 ns minimum, 28.0 ns maximum
(150 pF load)	4.0 ns minimum, 35.0 ns maximum
B to $\overline{C}_0$ (DIFF)	
(50 pF load)	3.0 ns minimum, 32.0 ns maximum
(150 pF load)	4.0 ns minimum, 40.0 ns maximum
A to CG or CP (SUM)	
(50 pF load)	3.0 ns minimum, 17.0 ns maximum
(150 pF load)	4.0 ns minimum, 20.0 ns maximum
B to CG or CP (DIFF)	
(50 pF load)	3.0 ns minimum, 21.0 ns maximum
(150 pF load)	4.0 ns minimum, 24.0 ns maximum

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Propagation delay time (t_{PHL}) - Continued	
B to EQ (DIFF)	
(50 pF load)	3.0 ns minimum, 42.0 ns maximum
(150 pF load)	4.0 ns minimum, 47.0 ns maximum
$\overline{C}_{IN}$ to $\overline{C}_O$	
(50 pF load)	3.0 ns minimum, 17 0 ns maximum
(150 pF load)	4.0 ns minimum, 24 0 ns maximum
$\overline{C}_{IN}$ to F	
(50 pF load)	3.0 ns minimum, 17.0 ns maximum
(150 pF load)	4.0 ns minimum, 21.0 ns maximum
S to F	
(50 pF load)	3.0 ns minimum, 45.0 ns maximum
(150 pF load)	4.0 ns minimum, 55.0 ns maximum
S to CG or CP or $\overline{C}_O$, M to F	
(50 pF load)	3 0 ns minimum, 45.0 ns maximum
(150 pF load)	4.0 ns minimum, 55 0 ns maximum
$\overline{P}0$-$\overline{P}2$ or $\overline{G}0$-$\overline{G}2$ to C_{N+X} or C_{N+Y} or C_{N+Z}	
(50 pF load)	1 0 ns minimum, 11 5 ns maximum
(150 pF load)	3.0 ns minimum, 17 0 ns maximum
P0-P3 to $\overline{P}$, $\overline{P}1$-$\overline{P}3$ to $\overline{G}$, $\overline{G}3$ to $\overline{G}$	
(50 pF load)	1.0 ns minimum 13 0 ns maximum
(150 pF load)	3.0 ns minimum, 19 0 ns maximum
C_N to C_{N+X} or C_{N+Y} or C_{N+Z}	
(50 pF load)	1.0 ns minimum, 15.5 ns maximum
(150 pF load)	3.0 ns minimum, 20 0 ns maximum

* 1.3 Module function. This module contains one arithmetic logic unit (ALU) and one look-ahead carry generator (LCG) The ALU performs either logic functions (M = H) or arithmetic functions (M = L) The function performed on the A and B inputs is determined by the logic state of the select inputs (S0, S1, S2, and S3) Logic functions are performed on a bit basis. All arithmetic functions are performed on a word basis where each word consists of four bits The A word is composed of bits A0, A1, A2, and A3, where A0 is the least significant bit. The B word is composed of bits B0, B1, B2, and -B3, where B0 is considered the least significant bit

a. The carry input ($\overline{C}_{IN}$) and the ripple carry output ($\overline{C}_O$) are provided for ripple-carry arithmetic operations. In addition, the propagate (CP) and generate (CG) outputs are used with the LCG to provide look-ahead carry for fast arithmetic functions. Placing the ALU/LCG in the subtract mode (S0 = L, S1 = H, S2 = H, and S3 = L) produces a $\overline{C}_O$ output that represents A and B word magnitude comparisons. The truth table shown in table III exists for the magnitude comparison.

b When two words of equal magnitude are applied to the A and B inputs, the EQ output will go to a high level. For all other cases, the EQ output will be a low level

c. When the ALU is operated in the logic mode (M = H), the $\overline{C}_O$ output is considered a "don't care condition "

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d. The module also includes a look-ahead carry generator (LCG) capable of generating a carry across four binary adders or groups of adders. The LCG can be cascaded to perform full look-ahead across N-bit adders.

Positive logic equations for the LCG are:

(1) $C_{N+X} = \overline{G0} + \overline{P0}\ C_N$
(2) $C_{N+Y} = \overline{G1} + \overline{P1}\ \overline{G0} + \overline{P1}\ \overline{P0}\ C_N$
(3) $C'_{N+Z} = \overline{G2} + \overline{P2}\ \overline{G1} + \overline{P2}\ \overline{P1}\ \overline{G0} + \overline{P2}\ \overline{P1}\ \overline{P0}\ C_N$
(4) $\overline{G} = \overline{G3}\ (\overline{P3} + \overline{G2})\ (\overline{P3} + \overline{P2} + \overline{G1})\ (\overline{P3} + \overline{P2} + \overline{P1} + \overline{G0})$
(5) $\overline{P} = \overline{P3}\ \overline{P2}\ \overline{P1}\ \overline{P0}$.

1.3.1 Input to output relationship. Each ALU/LCG has the input to output relationship of table I for the ALU and table II for the LCG for positive logic.

1.3.2 Parallel outputs. Paralleling outputs is not permissible except for the EQ output which may be paralleled with the EQ outputs of other modules to allow comparison of words with more than four bits. The EQ output is an open collector output provided to accomplish this purpose. The R output is a pullup resistor of 750 ohms connected to V_{cc} that can be connected to the EQ to pullup the output to a high level.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARD

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test equipment for the Standard Electronic Modules Program.

* (Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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TABLE I. Input to output relationship (ALU).

Item	Selection				M = H Logic functions	M = L Arithmetic operations	
	S3	S2	S1	S0		$\overline{C}_{IN}$ = H	$\overline{C}_{IN}$ = L
1	L	L	L	L	F = $\overline{A}$	F = A	F = A plus 1
2	L	L	L	H	F = $\overline{A + B}$	F = A + B	F = (A + B) plus 1
3	L	L	H	L	F = $\overline{A} \cdot B$	F = A + $\overline{B}$	F = (A + $\overline{B}$) plus 1
4	L	L	H	H	F = L	F = Minus 1 (two's complement)	F = Zero
5	L	H	L	L	F = $\overline{A \cdot B}$	F = A plus A · $\overline{B}$	F = A plus A · $\overline{B}$ plus 1
6	L	H	L	H	F = $\overline{B}$	F = (A + B) plus A · $\overline{B}$	F = (A + B) plus A · $\overline{B}$ plus 1
7	L	H	H	L	F = A ⊕ B	F = A minus B minus 1	F = A minus B
8	L	H	H	H	F = A · $\overline{B}$	F = A · $\overline{B}$ minus 1	F = A $\overline{B}$
9	H	L	L	L	F = $\overline{A}$ + B	F = A plus A · B	F = A plus A · B plus 1
10	H	L	L	H	F = $\overline{A \oplus B}$	F = A plus B	F = A plus B plus 1
11	H	L	H	L	F = B	F = (A + $\overline{B}$) plus A · B	F = (A + $\overline{B}$) plus A · B plus 1
12	H	L	H	H	F = A B	F = A · B minus 1	F = A · B
13	H	H	L	L	F = H	F = A plus A (note 1)	F = A plus A plus 1
14	H	H	L	H	F = A + $\overline{B}$	F = (A + B) plus A	F = (A + B) plus A plus 1
15	H	H	H	L	F = A + B	F = (A + $\overline{B}$) plus A	F = (A + $\overline{B}$) plus A plus 1
16	H	H	H	H	F = A	F = A minus 1	F = A

NOTES:
1. Each bit is shifted to the next more significant position.
2. ⊕ EXCLUSIVE-OR operator.
3. + OR operator.
4. - Negation operator.
5. · AND operator.
6. H = High-voltage level
7. L = Low-voltage level

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TABLE II Input to output relationship (LCG)

Inputs									Outputs				
C_N	$\overline{G}0$	$\overline{P}0$	$\overline{G}1$	$\overline{P}1$	$\overline{G}2$	$\overline{P}2$	$\overline{G}3$	$\overline{P}3$	C_{N+X}	C_{N+Y}	C_{N+Z}	$\overline{G}$	$\overline{P}$
X L X H	H H L X	H X X L							L L H H				
X X L X X H	X H H X L X	X H X X X L	H H H L X X	H X X X L L						L L L H H H			
X X X L X X X H	X X H H X X L X	X X H X X X X L	X H H H X L X X	X H X X X X L L	H H H H L X X X	H X X X X L L L					L L L L H H H H		
	X X X H X X X L		X X H H X X L X	X X H X X X X L	X H H H X L X X	X H X X X X L L	H H H H L X X X	H X X X X L L L				H H H H L L L L	
		H X X X L		X H X X L		X X H X L		X X X H L					H H H H L

NOTES:
1. H = High-voltage level
2. L = Low-voltage level.
3. X = Don't care condition

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TABLE III. Truth table

Word comparisons	Input	Output
	$\overline{C}_{IN}$	$\overline{C}_{O}$
A ≤ B	H	H
A < B	L	H
A > B	H	L
A ≥ B	L	L

3. REQUIREMENTS

3.1 General requirements The module shall comply with the requirements of MIL-M-28787 except as modified herein

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

* 3 3 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 1.186 failures per million operating hours when predicted in accordance with MIL-STD-1389.

* 3.4 Performance requirements. The module shall conform to the electrical requirements specified in table IV and shall operate within the environment of design class I as specified in MIL-STD-1389.

3 5 Module configuration requirements The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code The module key code shall be RDH

3.5.2 Size. The module size shall be 1A.

3 5.3 Weight The module shall have a maximum weight of 35 grams.

3 5 4 Contact pins The module shall have 40 contact pins positioned A1 through A40

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A

3 5 6 Depth. The module shall have no additional depth requirements

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

* 3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/79-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3 7 Acceptance test All production modules shall be tested in accordance with notes 1 and 2 of table IV

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3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed two years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table IV.

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table IV for a power supply voltage (V_{CC}) of 5.0 ±0.5 Vdc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 Vdc ±1 percent.

3.10 Functional tests. All modules shall be subjected to functional tests which conform to the requirements stated in 3.10.1 through 3.10.3.

3.10.1 Test patterns. A set of unique input patterns shall be generated capable of detecting all nonredundant failures in the module logic network. The test patterns shall be generated by assuming "stuck at H state" and "stuck at L state" failures in each of the basic logic functions and applying input patterns to detect these failures. The unique test patterns necessary to satisfy these requirements shall be submitted with the qualification samples. If the tests of tables IV and X satisfy or exceed these unique test patterns, no additional tests shall be added; otherwise, the needed tests shall be added to table X. "Stuck at H state" indicates a failure resulting in a steady high-level output. "Stuck at L state" indicates a failure resulting in a steady low-level output. Prior to any change in module design which would necessitate a different set of test patterns to detect a "stuck at H state" or a "stuck at L state", the vendor shall be required to submit to the procuring activity and Standard Electronic Modules Program (SEMP) Quality Assurance Activity the proposed new set of test patterns for approval.

3.10.2 Input requirements. The following input voltage levels shall be used for test patterns:

Low level-input voltage shall be 0.8 V ±1 percent.
High level-input voltage shall be 2.0 V ±1 percent.

3.10.3 Output requirements. The output requirements for the test patterns shall be as follows

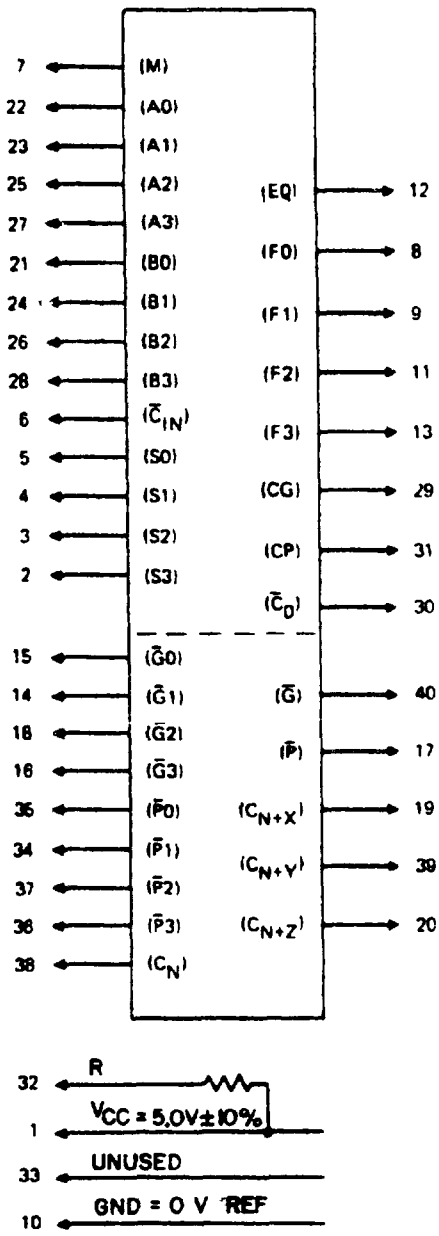
Low level-output voltage shall be 0 V minimum, 0.5 V maximum.
High level-output voltage shall be 2.5 V minimum, V_{CC} maximum.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

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NOTES:
1. S0-S3 = Select inputs.
2. $\overline{M}$ = Mode input
3. $\overline{C}_{IN}$ = Carry input
4. A0-A3, B0-B3 = Bit inputs
5. $\overline{G}$0-$\overline{G}$3 = Carry generate inputs.
6. $\overline{P}$0-$\overline{P}$3 = Carry propagate inputs.
7. C_N = Ripple-carry input.
8. EQ = Equality comparison output.
9. F0-F3 = Function outputs.
10. CG = Carry generate output (ALU).
11. $\overline{CP}$ = Carry propagate output (ALU).
12. $\overline{C}_0$ = Ripple-carry output.
13. $\overline{G}$ = Carry generate (LCG).
14. $\overline{P}$ = Carry propagate (LCG).
15. C_{N+X}, C_{N+Y}, C_{N+Z} = Carry outputs.
16. R = Pullup resistor

FIGURE 1. RDH module function diagram, ALU/LCG.

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TABLE IV. Module electrical requirements

Test condition	Parameter	Ref test table	Test type (see notes)	Acceptance requirements 25° ±5°C Ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1	Isolation	V	S	10	∞	1.0	∞	10	∞	1.0	∞	MΩ
A2	Isolation, unused pin	V	S	10	∞	1.0	∞	10	∞	1 0	∞	MΩ
B1-B13	V_{OUT} (low-loaded)	VI	S	0.01	1.0	0 01	1.0	0 01	1.0	0.01	1.0	V
C1-C23	Input breakdown current	VII	S	0.0	1.0	0 0	1 0	0 0	1.0	0.0	1.0	mA
D1	Output leakage current, high level, EQ	VIII	X	0 0	0.25	0.0	0.25	0.0	0.25	0.0	0.25	mA
D2-D13	Output leakage current, high level, all except EQ	VIII	X	0.0	1.0	0.0	1 0	0.0	1.0	0.0	1.0	mA
E1	Output short circuit current, R	IX	X	-7.0	-7.7	-6.4	-8.6	-6 4	-8.6	-6.1	-9.2	mA
E2-E20	Output short circuit current, all except R and EQ	IX	X	-40	-100	-40	-100	-40	-100	-40	-100	mA
F1-F25	Functional	X	X	Test vectors must produce output vectors as specified in table X.								
G1-G23	Input clamp diode voltage	XI	X	-0 4	-1.2	-0.4	-1.2	-0 4	-1.2	-0.4	-1.2	V
H1-H13	Static output voltage, low level	XII	X	0 0	0 5	0 0	0.5	0.0	0.5	0 0	0.5	V
J1-J13	Static output voltage, high level	XIII	X	2 5	4 5	2 5	4 5	2 5	4.5	2 5	4 5	V
K1	Input current, high level, M	XIV	X	0	50	0	50	0	50	0	50	µA
K2-K9	Input current, high level, A, B	XIV	X	0	150	0	150	0	150	0	150	µA
K10-K13	Input current, high level, S	XIV	X	0	200	0	200	0	200	0	200	µA
K14	Input current, high level, $\overline{C}_{IN}$	XIV	X	0	250	0	250	0	250	0	250	µA
K15	Input current, high level, C_N	XIV	X	0	50	0	50	0	50	0	50	µA

See notes at end of table.

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TABLE IV. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type (see notes)	Acceptance requirements 25° ±5°C Ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
K16	Input current, high level, $\overline{P3}$	XIV	X	0	100	0	100	0	100	0	100	µA
K17	Input current, high level, $\overline{P2}$	XIV	X	0	150	0	150	0	150	0	150	µA
K18-K20	Input current, high level, $\overline{P1}$, $\overline{P0}$, $\overline{G3}$	XIV	X	0	200	0	200	0	200	0	200	µA
K21	Input current, high level, $\overline{G2}$	XIV	X	0	350	0	350	0	350	0	350	µA
K22	Input current, high level, $\overline{G1}$	XIV	X	0	400	0	400	0	400	0	400	µA
K23	Input current, high level, $\overline{G0}$	XIV	X	0	350	0	350	0	350	0	350	µA
L1, L2	Input current, low level, M, C_N	XV	X	-0.1	-2 2	-0.1	-2.2	-0.1	-2.2	-0.1	-2.2	mA
L3	Input current, low level, $\overline{P3}$	XV	X	-0.1	-4.4	-0.1	-4.4	-0.1	-4.4	-0.1	-4.4	mA
L4-L12	Input current, low level, $\overline{P2}$, A, B	XV	X	-0.1	-6.6	-0.1	-6.6	-0.1	-6.6	-0.1	-6.6	mA
L13-L19	Input current, low level, S, $\overline{P1}$, $\overline{P0}$, $\overline{G3}$	XV	X	-0.1	-8.8	-0.1	-8.8	-0.1	-8.8	-0.1	-8.8	mA
L20	Input current, low level, $\overline{C}_{IN}$	XV	X	-0.1	-11.0	-0.1	-11.0	-0.1	-11.0	-0.1	-11.0	mA
L21, L22	Input current, low level, $\overline{G2}$, $\overline{G0}$	XV	X	-0.1	-15.4	-0.1	-15.4	-0.1	-15.4	-0.1	-15.4	mA
L23	Input current, low level, $\overline{G1}$	XV	X	-0.1	-17.6	-0.1	-17.6	-0.1	-17.6	-0.1	-17.6	mA
M1	Power supply current, low-level	XVI	X	100	319	100	319	100	319	100	319	mA
M2	Power supply current, high-level	XVI	X	90	270	90	270	90	270	90	270	mA
N1	Transition time (t_{TLH}), EQ, C_L = 50 pF	XVII	S	2.0	28.0	2.0	28.0	2.0	30.0	2.0	30.0	ns
N2-N13	Transition time (t_{TLH}), all except EQ, C_L = 50 pF	XVII	S	2.0	15.0	2.0	15.0	2.0	18.0	2.0	18.0	ns

See notes at end of table.

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TABLE IV. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type (see notes)	Acceptance requirements 25° ±5°C Ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
N14	Transition time (t_{TLH}), EQ, C_L = 150 pF	XVII	Q	2.0	80.0	2.0	80.0	2.0	85.0	2.0	85.0	ns
N15-N26	Transition time (t_{TLH}), all except EQ, C_L = 150 pF	XVII	Q	2.0	27.0	2.0	27.0	2.0	30.0	2.0	30.0	ns
P1	Transition time (t_{THL}), EQ, C_L = 50 pF	XVII	S	2.0	10.0	2.0	10.0	2.0	12.0	2.0	12.0	ns
* P2-P13	Transition time (t_{THL}), all except EQ, C_L = 50 pF	XVII	S	2.0	8.0	2.0	8.0	2.0	10.0	2.0	10.0	ns
P14	Transition time (t_{THL}), EQ, C_L = 150 pF	XVII	Q	2.0	20.0	2.0	20.0	2.0	25.0	2.0	25.0	ns
* P15-P26	Transition time (t_{THL}), all except EQ, C_L = 150 pF	XVII	Q	2.0	13.0	2.0	13.0	2.0	15.0	2.0	15.0	ns
Q1-Q5	Propagation delay time (t_{PLH}), SUM, A or B to F, C_L = 50 pF	XVIII	S	3.0	21.0	3.0	21.0	3.0	23.0	3.0	23.0	ns
Q6-Q8	Propagation delay time (t_{PLH}), SUM, B to F with carry, C_L = 50 pF	XVIII	S	3.0	31.0	3.0	31.0	3.0	34.0	3.0	34.0	ns
Q9-Q13	Propagation delay time (t_{PLH}), SUM, A or B to F, C_L = 150 pF	XVIII	Q	4 0	23.5	4.0	23.5	4.0	26.0	4.0	26.0	ns
Q14-Q16	Propagation delay time (t_{PLH}), SUM, B to F with carry, C_L = 150 pF	XVIII	Q	4.0	34.0	4.0	34.0	4.0	37.0	4.0	37.0	ns
Q17-Q21	Propagation delay time (t_{PLH}), DIFF, A or B to F, C_L = 50 pF	XVIII	S	3.0	24.0	3.0	24.0	3.0	26.0	3.0	26.0	ns
Q22-Q24	Propagation delay time (t_{PLH}), DIFF, B to F with carry, C_L = 50 pF	XVIII	S	3.0	34.0	3.0	34.0	3.0	37.0	3.0	37.0	ns
Q25-Q29	Propagation delay time (t_{PLH}), DIFF, A or B to F, C_L = 150 pF	XVIII	Q	4.0	27.5	4.0	27.5	4,0	30.0	4.0	30.0	ns

See notes at end of table.

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TABLE IV Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type (see notes)	Acceptance requirements 25° ±5°C Ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
Q30-Q32	Propagation delay time (t_{PLH}), DIFF, B to F with carry, C_L = 150 pF	XVIII	Q	4.0	37.0	4.0	37.0	4.0	40.0	4.0	40.0	ns
Q33-Q36	Propagation delay time (t_{PLH}), LOGIC, B to F, C_L = 50 pF	XVIII	S	3 0	24.0	3.0	24.0	3.0	26.0	3.0	26.0	ns
Q37-Q40	Propagation delay time (t_{PLH}), LOGIC, B to F, C_L = 150 pF	XVIII	Q	4.0	27.5	4.0	27.5	4.0	30.0	4.0	30.0	ns
Q41-Q44	Propagation delay time (t_{PLH}), DIFF, B to EQ, C_L = 50 pF	XVIII	S	3.0	29.0	3.0	29.0	3.0	31.0	3.0	31.0	ns
Q45-Q48	Propagation delay time (t_{PLH}), DIFF, B to EQ, C_L = 150 pF	XVIII	Q	4.0	35.5	4.0	35.5	4.0	38.0	4.0	38.0	ns
Q49-Q52	Propagation delay time (t_{PLH}), SUM, A to CG, C_L = 50 pF	XVIII	S	3.0	15.0	3.0	15.0	3.0	17.0	3.0	17.0	ns
Q53-Q56	Propagation delay time (t_{PLH}), SUM, A to CG, C_L = 150 pF	XVIII	Q	4.0	17.5	4.0	17.5	4.0	20.0	4.0	20.0	ns
Q57-Q60	Propagation delay time (t_{PLH}), DIFF, B to CG, C_L = 50 pF	XVIII	S	3.0	19.0	3.0	19.0	3.0	21.0	3.0	21 0	ns
Q61-Q64	Propagation delay time (t_{PLH}), DIFF, B to CG, C_L = 150 pF	XVIII	Q	4.0	21.5	4.0	21.5	4.0	24.0	4.0	24 0	ns
Q65-Q68	Propagation delay time (t_{PLH}), SUM, A to CP, C_L = 50 pF	XVIII	S	3.0	15.0	3 0	15.0	3.0	17.0	3.0	17.0	ns
Q69-Q72	Propagation delay time (t_{PLH}), SUM, A to CP, C_L = 150 pF	XVIII	Q	4.0	17 5	4 0	17.5	4.0	20.0	4.0	20.0	ns
Q73-Q76	Propagation delay time (t_{PLH}), DIFF, B to CP, C_L = 50 pF	XVIII	S	3.0	19 0	3.0	19.0	3 0	21.0	3.0	21 0	ns

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TABLE IV. Module electrical requirements - Continued

Test condition	Parameter	Ref test table	Test type (see notes)	Acceptance requirements 25° ±5°C Ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
Q77-Q80	Propagation delay time (t_{PLH}), DIFF, B to CP, C_L = 150 pF	XVIII	Q	4.0	21.5	4.0	21.5	4.0	24.0	4.0	24.0	ns
Q81-Q84	Propagation delay time (t_{PLH}), SUM, A to $\overline{C}_O$, C_L = 50 pF	XVIII	S	3.0	26.0	3.0	26.0	3.0	28.0	3.0	28.0	ns
Q85-Q88	Propagation delay time (t_{PLH}), SUM, A to $\overline{C}_O$, C_L = 150 pF	XVIII	Q	4 0	32.5	4.0	32.5	4.0	35.0	4.0	35.0	ns
Q89-Q92	Propagation delay time (t_{PLH}), DIFF, B to $\overline{C}_O$, C_L = 50 pF	XVIII	S	3.0	30.0	3.0	30 0	3.0	32.0	3.0	32.0	ns
Q93-Q96	Propagation delay time (t_{PLH}), DIFF, B to $\overline{C}_O$, C_L = 150 pF	XVIII	Q	4.0	37.5	4.0	37 5	4.0	40.0	4.0	40.0	ns
Q97-Q100	Propagation delay time (t_{PLH}), $\overline{C}_{IN}$ to F, C_L = 50 pF	XVIII	S	3.0	15.0	3.0	15.0	3.0	17.0	3.0	17.0	ns
Q101-Q104	Propagation delay time (t_{PLH}), $\overline{C}_{IN}$ to F, C_L = 150 pF	XVIII	Q	4.0	18.5	4.0	18.5	4.0	21.0	4.0	21 0	ns
Q105	Propagation delay time (t_{PLH}), $\overline{C}_{IN}$ to $\overline{C}_O$, C_L = 50 pF	XVIII	S	3.0	15.0	3.0	15.0	3.0	17.0	3.0	17.0	ns
Q106	Propagation delay time (t_{PLH}), $\overline{C}_{IN}$ to $\overline{C}_O$, C_L = 150 pF	XVIII	Q	4.0	21.5	4.0	21.5	4.0	24.0	4.0	24 0	ns
Q107-Q114	Propagation delay time (t_{PLH}), S to F, C_L = 50 pF	XVIII	S	3.0	40.0	3.0	40.0	3.0	45.0	3.0	45.0	ns
Q115-Q122	Propagation delay time (t_{PLH}), S to F, C_L = 150 pF	XVIII	Q	4.0	50.0	4.0	50.0	4.0	55.0	4.0	55.0	ns
Q123-Q130	Propagation delay time (t_{PLH}), S to CG or CP or $\overline{C}_O$, M to F, C_L = 50 pF	XVIII	S	3.0	40.0	3.0	40.0	3.0	45.0	3.0	45 0	ns

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TABLE IV Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type (see notes)	Acceptance requirements 25° ±5°C Ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
Q131-Q138	Propagation delay time (t_{PLH}), S to CG or CP or $\overline{C}_O$, M to F, C_L = 150 pF	XVIII	Q	4.0	50.0	4.0	50.0	4 0	55.0	4 0	55.0	ns
Q139-Q144	Propagation delay time (t_{PLH}), $\overline{P}0$-$\overline{P}2$ or $\overline{G}0$-$\overline{G}2$ to C_{N+X} or C_{N+Y} or C_{N+Z}, C_L = 50 pF	XVIII	S	1.0	8.5	1.0	8.5	1.0	9.5	1.0	9.5	ns
Q145-Q152	Propagation delay time (t_{PLH}), $\overline{P}0$-$\overline{P}3$ to $\overline{P}$, $\overline{P}1$-$\overline{P}3$ to $\overline{G}$, $\overline{G}3$ to $\overline{G}$, C_L = 50 pF	XVIII	S	1.0	10.5	1.0	10.5	1.0	11.0	1.0	11.0	ns
Q153-Q155	Propagation delay time (t_{PLH}), $\overline{G}1$ to C_{N+Y}, $\overline{G}2$ to C_{N+Z}, C_L = 50 pF	XVIII	S	1.0	8.5	1.0	8.5	1 0	9.5	1.0	9 5	ns
Q156-Q158	Propagation delay time (t_{PLH}), C_N to C_{N+X} or C_{N+Y} or C_{N+Z}, C_L = 50 pF	XVIII	S	1.0	11.0	1.0	11.0	1.0	12.0	1.0	12.0	ns
Q159-Q164	Propagation delay time (t_{PLH}), $\overline{P}0$-$\overline{P}2$ or $\overline{G}0$-$\overline{G}2$ to C_{N+X} or C_{N+Y} or C_{N+Z}, C_L = 150 pF	XVIII	Q	3.0	12.5	3.0	12.5	3.0	14.0	3 0	14.0	ns
Q165-Q172	Propagation delay time (t_{PLH}), $\overline{P}0$-$\overline{P}3$ to $\overline{P}$, $\overline{P}1$-$\overline{P}3$ to $\overline{G}$, $\overline{G}3$ to $\overline{G}$, C_L = 150 pF	XVIII	Q	3 0	13.0	3.0	13.0	3.0	15.0	3.0	15.0	ns
Q173-Q175	Propagation delay time (t_{PLH}), $\overline{G}1$ to C_{N+Y}, $\overline{G}2$ to C_{N+Z}, C_L = 150 pF	XVIII	Q	3.0	12.5	3.0	12.5	3.0	14.0	3.0	14.0	ns
Q176-Q178	Propagation delay time (t_{PLH}), C_N to C_{N+X} or C_{N+Y} or C_{N+Z}, C_L = 150 pF	XVIII	Q	3.0	16 0	3.0	16.0	3.0	18.0	3.0	18 0	ns

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TABLE IV Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type (see notes)	Acceptance requirements 25° ±5°C Ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
R1-R5	Propagation delay time (t_{PHL}), SUM, A or B to F, C_L = 50 pF	XVIII	S	3 0	21.0	3.0	21.0	3.0	23.0	3.0	23.0	ns
R6-R8	Propagation delay time (t_{PHL}), SUM, B to F with carry, C_L = 50 pF	XVIII	S	3 0	31.0	3.0	31.0	3.0	34.0	3.0	34.0	ns
R9-R13	Propagation delay time (t_{PHL}), SUM, A or B to F, C_L = 150 pF	XVIII	Q	4.0	23.5	4.0	23.5	4.0	26.0	4.0	26.0	ns
R14-R16	Propagation delay time (t_{PHL}), SUM, B to F with carry, C_L = 150 pF	XVIII	Q	4 0	34.0	4.0	34.0	4.0	37.0	4.0	37.0	ns
R17-R21	Propagation delay time (t_{PHL}), DIFF, A or B to F, C_L = 50 pF	XVIII	S	3 0	26 0	3.0	26.0	3.0	28.0	3.0	28.0	ns
R22-R24	Propagation delay time (t_{PHL}), DIFF, B to F with carry, C_L = 50 pF	XVIII	S	3.0	36.0	3.0	36.0	3.0	39 0	3.0	39.0	ns
R25-R29	Propagation delay time (t_{PHL}), DIFF, A or B to F, C_L = 150 pF	XVIII	Q	4 0	29.5	4 0	29.5	4.0	32 0	4.0	32.0	ns
R30-R32	Propagation delay time (t_{PHL}), DIFF, B to F with carry, C_L = 150 pF	XVIII	Q	4 0	39.0	4.0	39.0	4 0	42.0	4.0	42.0	ns
R33-R36	Propagation delay time (t_{PHL}), LOGIC, B to F, C_L = 50 pF	XVIII	S	3.0	26.0	3.0	26.0	3.0	28 0	3.0	28 0	ns
R37-R40	Propagation delay time (t_{PHL}), LOGIC, B to F, C_L = 150 pF	XVIII	Q	4.0	29.5	4.0	29.5	4.0	32.0	4.0	32.0	ns

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TABLE IV. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type (see notes)	Acceptance requirements 25° ±5°C Ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
R41-R44	Propagation delay time (t_{PHL}), DIFF, B to EQ, C_L = 50 pF	XVIII	S	3.0	40.0	3.0	40.0	3.0	42.0	3.0	42.0	ns
R45-R48	Propagation delay time (t_{PHL}), DIFF, B to EQ, C_L = 150 pF	XVIII	Q	4.0	44.5	4.0	44.5	4.0	47.0	4.0	47.0	ns
R49-R52	Propagation delay time (t_{PHL}), SUM, A to CG, C_L = 50 pF	XVIII	S	3.0	15.0	3.0	15 0	3.0	17.0	3.0	17.0	ns
R53-R56	Propagation delay time (t_{PHL}), SUM, A to CG, C_L = 150 pF	XVIII	Q	4.0	17.5	4.0	17.5	4.0	20.0	4 0	20.0	ns
R57-R60	Propagation delay time (t_{PHL}), DIFF, B to CG, C_L = 50 pF	XVIII	S	3.0	19.0	3.0	19.0	3 0	21.0	3 0	21 0	ns
R61-R64	Propagation delay time (t_{PHL}), DIFF, B to CG, C_L = 150 pF	XVIII	Q	4.0	21.5	4.0	21.5	4.0	24.0	4.0	24.0	ns
R65-R68	Propagation delay time (t_{PHL}), SUM, A to CP, C_L = 50 pF	XVIII	S	3.0	15.0	3.0	15.0	3 0	17.0	3.0	17.0	ns
R69-R72	Propagation delay time (t_{PHL}), SUM, A to CP, C_L = 150 pF	XVIII	Q	4.0	17.5	4.0	17.5	4.0	20.0	4.0	20.0	ns
R73-R76	Propagation delay time (t_{PHL}), DIFF, B to CP, C_L = 50 pF	XVIII	S	3.0	19.0	3.0	19.0	3.0	21 0	3.0	21.0	ns
R77-R80	Propagation delay time (t_{PHL}), DIFF, B to CP, C_L = 150 pF	XVIII	Q	4.0	21.5	4.0	21.5	4.0	24.0	4.0	24.0	ns

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TABLE IV Module electrical requirements - Continued

Test condition	Parameter	Ref test table	Test type (see notes)	Acceptance requirements 25° ±5°C Ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
R81-R84	Propagation delay time (t_{PHL}), SUM, A to $\overline{C}_O$, C_L = 50 pF	XVIII	S	3.0	26.0	3.0	26.0	3.0	28.0	3.0	28.0	ns
R85-R88	Propagation delay time (t_{PHL}), SUM, A to $\overline{C}_O$, C_L = 150 pF	XVIII	Q	4.0	32.5	4.0	32.5	4.0	35.0	4.0	35.0	ns
R89-R92	Propagation delay time (t_{PHL}), DIFF, B to $\overline{C}_O$, C_L = 50 pF	XVIII	S	3.0	30.0	3.0	30.0	3.0	32.0	3.0	32.0	ns
R93-R96	Propagation delay time (t_{PHL}), DIFF, B to $\overline{C}_O$, C_L = 150 pF	XVIII	Q	4.0	37.5	4.0	37.5	4.0	40.0	4.0	40.0	ns
R97-R100	Propagation delay time (t_{PHL}), $\overline{C}_{IN}$ to F, C_L = 50 pF	XVIII	S	3.0	15.0	3.0	15.0	3.0	17.0	3.0	17.0	ns
R101-R104	Propagation delay time (t_{PHL}), $\overline{C}_{IN}$ to F, C_L = 150 pF	XVIII	Q	4.0	18.5	4.0	18.5	4.0	21.0	4.0	21.0	ns
R105	Propagation delay time (t_{PHL}), $\overline{C}_{IN}$ to $\overline{C}_O$, C_L = 50 pF	XVIII	S	3.0	15.0	3.0	15.0	3.0	17.0	3.0	17.0	ns
R106	Propagation delay time (t_{PHL}), $\overline{C}_{IN}$ to $\overline{C}_O$, C_L = 150 pF	XVIII	Q	4.0	21.5	4.0	21.5	4.0	24.0	4.0	24.0	ns
R107-R114	Propagation delay time (t_{PHL}), S to F, C_L = 50 pF	XVIII	S	3.0	40.0	3.0	40.0	3.0	45.0	3.0	45.0	ns
R115-R122	Propagation delay time (t_{PHL}), S to F, C_L = 150 pF	XVIII	Q	4.0	50.0	4.0	50.0	4.0	55.0	4.0	55.0	ns

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TABLE IV. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type (see notes)	Acceptance requirements 25° ±5°C Ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
R123-R130	Propagation delay time (t_{PHL}), S to CG or CP or $\overline{C}_0$, M to F, C_L = 50 pF	XVIII	S	3.0	40.0	3.0	40.0	3.0	45.0	3.0	45.0	ns
R131-R138	Propagation delay time (t_{PHL}), S to CG or CP or $\overline{C}_0$, M to F, C_L = 150 pF	XVIII	Q	4.0	50.0	4.0	50.0	4.0	55.0	4.0	55.0	ns
R139-R144	Propagation delay time (t_{PHL}), $\overline{P}0$-$\overline{P}2$ or $\overline{G}0$-$\overline{G}2$ to C_{N+X} or C_{N+Y} or C_{N+Z}, C_L = 50 pF	XVIII	S	1.0	10.5	1.0	10 5	1.0	11.5	1.0	11.5	ns
R145-R152	Propagation delay time (t_{PHL}), $\overline{P}0$-$\overline{P}3$ to $\overline{P}$, $\overline{P}1$-$\overline{P}3$ to $\overline{G}$, $\overline{G}3$ to $\overline{G}$, C_L = 50 pF	XVIII	S	1.0	12 5	1.0	12 5	1.0	13.0	1.0	13 0	ns
R153-R155	Propagation delay time (t_{PHL}), $\overline{G}1$ to C_{N+Y}, $\overline{G}2$ to C_{N+Z}, C_L = 50 pF	XVIII	S	1.0	10.5	1.0	10 5	1.0	11.5	1.0	11.5	ns
R156-R158	Propagation delay time (t_{PHL}), C_N to C_{N+X} or C_{N+Y} or C_{N+Z}, C_L = 50 pF	XVIII	S	1 0	14.5	1.0	14 5	1.0	15.5	1.0	15 5	ns
R159-R164	Propagation delay time (t_{PHL}), $\overline{P}0$-$\overline{P}2$ or $\overline{G}0$-$\overline{G}2$ to C_{N+X} or C_{N+Y} or C_{N+Z}, C_L = 150 pF	XVIII	Q	3.0	15.5	3.0	15.5	3 0	17.0	3 0	17.0	ns
R165-R172	Propagation delay time (t_{PHL}), $\overline{P}0$-$\overline{P}3$ to $\overline{P}$, $\overline{P}1$-$\overline{P}3$ to $\overline{G}$, $\overline{G}3$ to $\overline{G}$, C_L = 150 pF	XVIII	Q	3 0	17.0	3.0	17.0	3 0	19.0	3.0	19.0	ns
R173-R175	Propagation delay time (t_{PHL}), $\overline{G}1$ to C_{N+Y}, $\overline{G}2$ to C_{N+Z}, C_L = 150 pF	XVIII	Q	3.0	15 5	3.0	15.5	3.0	17.0	3.0	17.0	ns

See notes at end of table.

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TABLE IV. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type (see notes)	Acceptance requirements 25° ±5°C Ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
R176-R178	Propagation delay time (t_{PHL}), C_N to C_{N+X} or C_{N+Y} or C_{N+Z}, C_L = 150 pF	XVIII	Q	3.0	18.0	3.0	18 0	3.0	20.0	3.0	20.0	ns
S1-S13	Output crosstalk, low level	XIX	Q	0	1.0	0	1 0	0	1.0	0	1.0	V
S14-S26	Output crosstalk, low-level duration	XIX	Q	0	500	0	500	0	500	0	500	mV
T1-T13	Output crosstalk, high level	XIX	Q	0	400	0	400	0	400	0	400	mV
U	Power supply transient amplitude	XX	Q	0	500	0	500	0	500	0	500	mV
V	Power supply filter capacitance	XXI	S	3.6	7 1	3.6	7.1	3.6	7.1	3.6	7.1	µF
W1-W23	Input capacitance	XXII	Q	5.0	15.0	5.0	15.0	5.0	15.0	5.0	15.0	pF

NOTES:
1. (X) parameters shall be 100 percent tested during module acceptance tests.
2. (S) parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C
3. (Q) parameters shall be tested during qualification testing only.

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* 4.3 Test equipment. The following test equipment is necessary to conduct acceptance tests for this module. Test equipment shall be as specified in MIL-STD-1665 and that listed below

Equipment	MIL-STD-1665 designator	No. required (max per test figure)
Oscilloscope	Item D05	1
Oscilloscope probe	Item Y01	2
DC ammeter	Item X01	1
DC voltmeter	Item V03	1
Pulse generator	Item J01	1
Pulse generator	Item J05	1
DC power supply	Item P01	4
Megohm bridge	Item R06	1
Capacitance bridge	Item C01	1
Capacitance meter	Item C05	1
Resistors		
249Ω ±1%, 1/4 W		13
590Ω ±1%, 1/4 W		1
51.1Ω ±1%, 1/4 W		1
1 kΩ ±5%, 1/4 W		1
51Ω ±5%, 1/2 W		2
2.49 kΩ ±1%, 1/8 W		13
Value as required for figure 14, note 2, 1/4 W		12
Value as required for figure 14, note 3, 1/4 W		1
Capacitors		
Value as required for figure 14, note 1(c), test type S		13
Value as required for figure 14, note 1(c), test type Q		1
Diode		
1N3600, 1N4150, or equivalent		13

All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table IV

* 4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity (SEMP-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through W shall be performed in any order with the following exceptions. If conditions A, B, C, D, or E are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D, E, where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

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4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through W in table IV shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through V shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 4.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 5. Following the vibration tests, the electrical tests of table IV which are marked with an (X) in the test type column, and test condition V, shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Upon completing the life test, electrical tests D through V shall be performed in the order specified in 4.4.2 using the 0° to 60°C end-of-life requirements of table IV. After the module has been returned to 25°C, electrical tests D through W shall be performed in the order specified in 4.4.2 using the 25° ±5°C end-of-life requirements of table IV.

5. PACKAGING

5.1 Packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table IV, the module is considered to have completed its useful life.

* 6.3 Cross reference of classification. Part number M28787/79-1 supersedes part number M28787/79. The two parts are interchangeable.

6.4 0 volt reference. The reference for test setups will be 0 volt and not necessarily earth ground.

6.5 Decoupling. Decoupling of test voltages and V_{CC} in test setups is permissible except test condition U, power supply transient amplitude test.

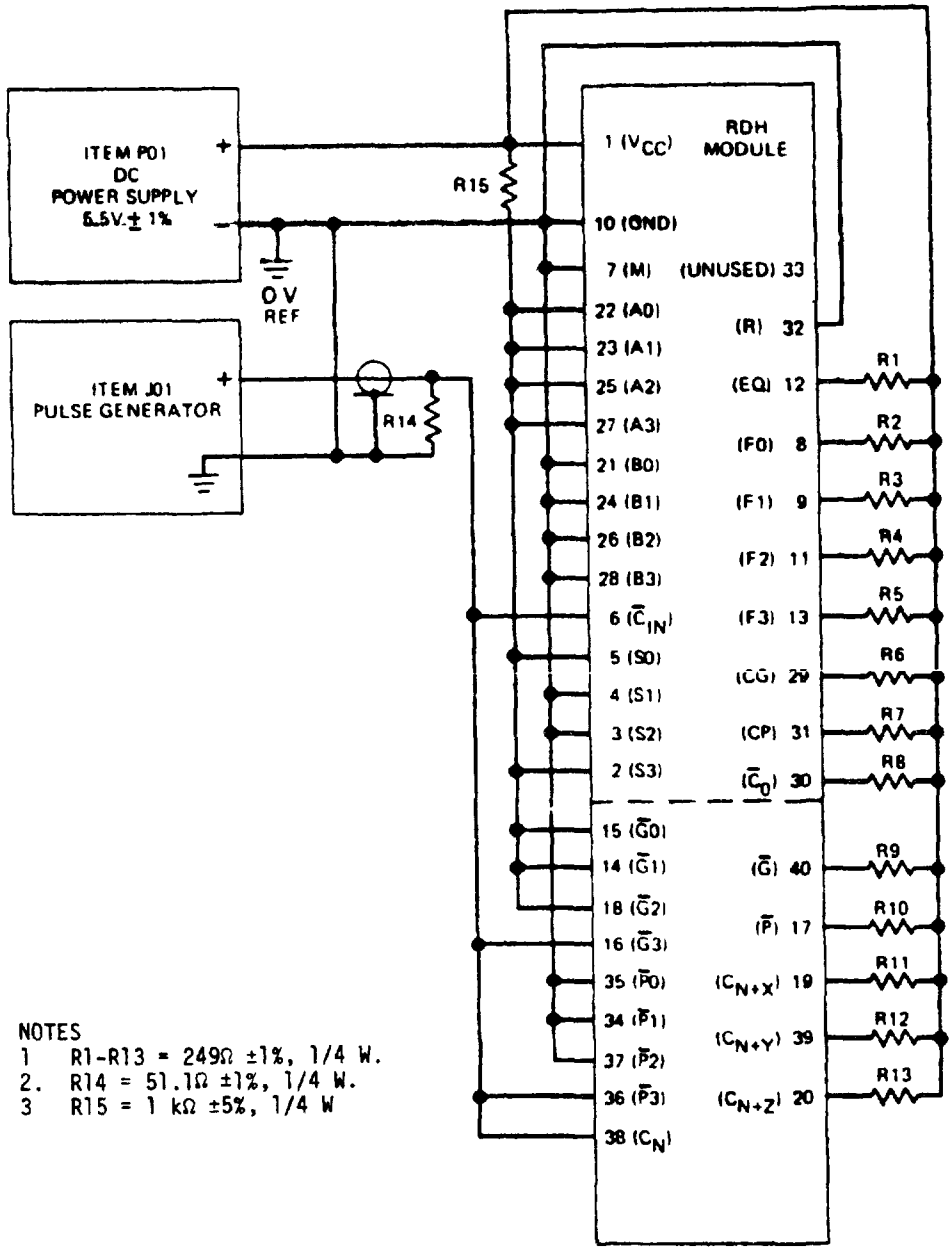
* 6.6 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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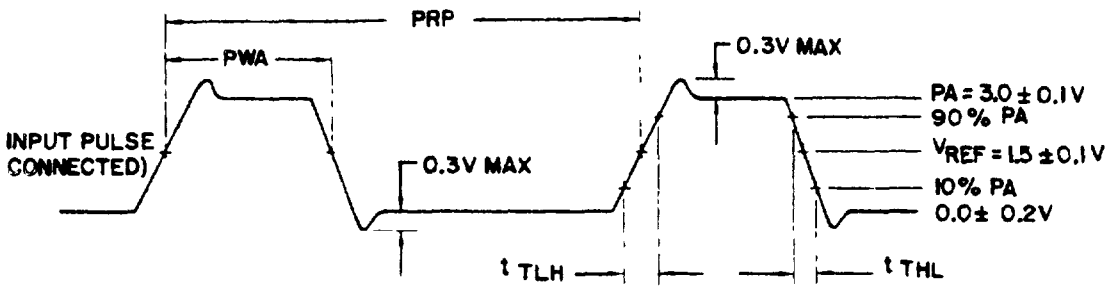
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NOTES
1. R1-R13 = 249Ω ±1%, 1/4 W.
2. R14 = 51.1Ω ±1%, 1/4 W.
3. R15 = 1 kΩ ±5%, 1/4 W

FIGURE 2 Operating temperature and life test circuit

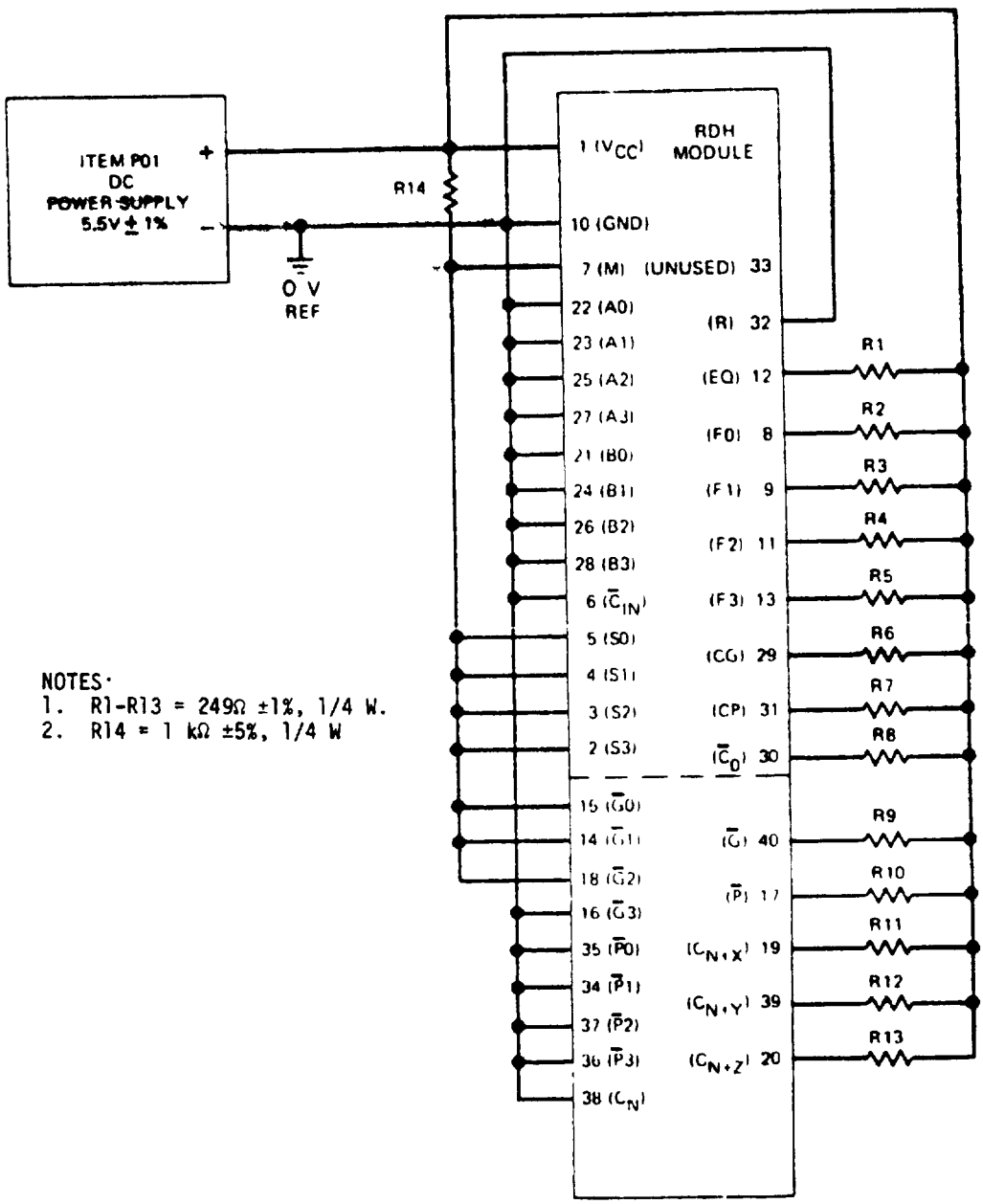
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NOTE: Input pulse parameters
$t_{TLH} = t_{THL} \leq 10$ ns.
PRP = 1,000 ±200 ns.
PWA = 500 ±100 ns

FIGURE 3. Operating temperature and life test circuit waveform

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NOTES:
1. R1-R13 = 249Ω ±1%, 1/4 W.
2. R14 = 1 kΩ ±5%, 1/4 W

FIGURE 4. Component temperature test circuit

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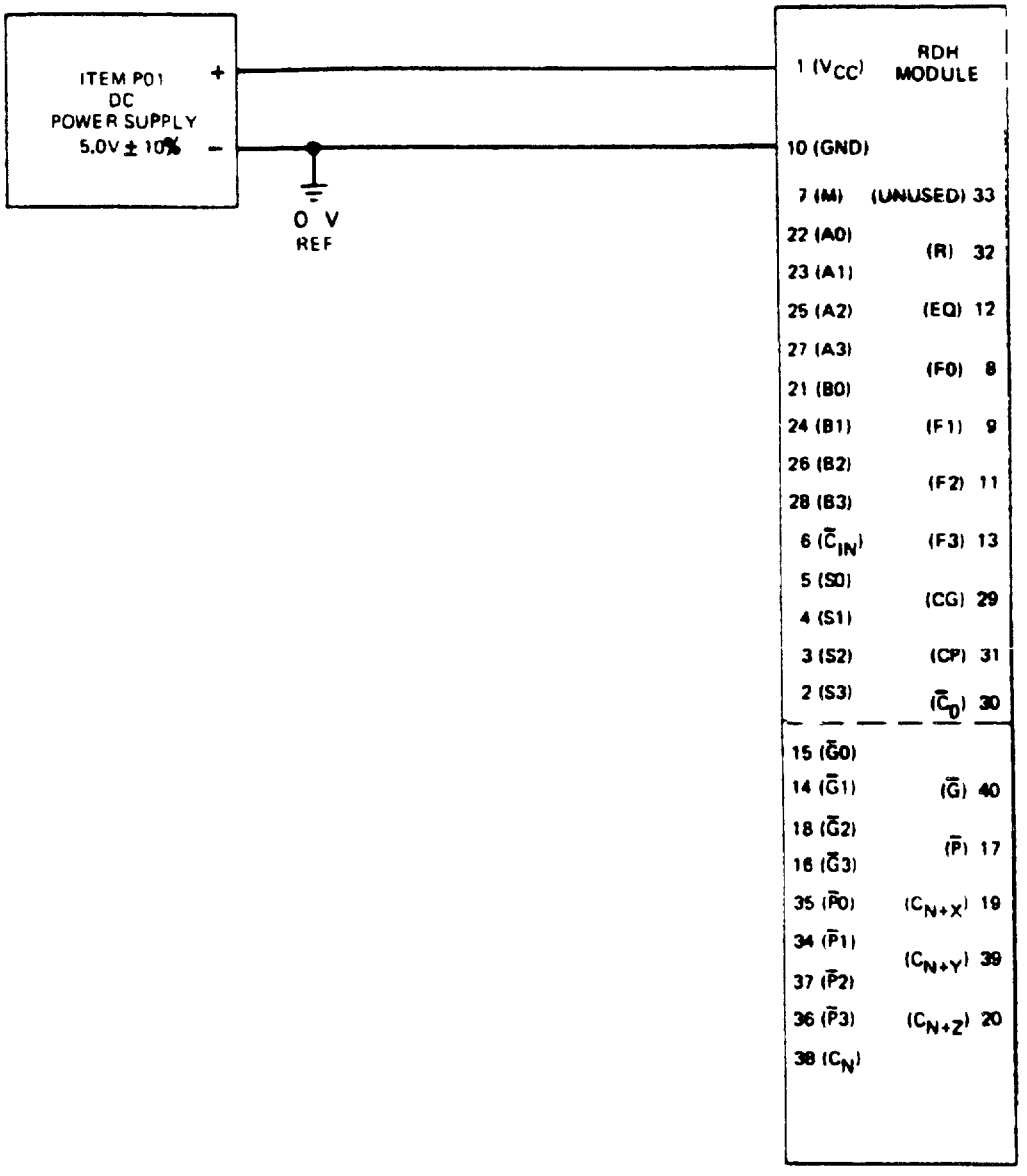


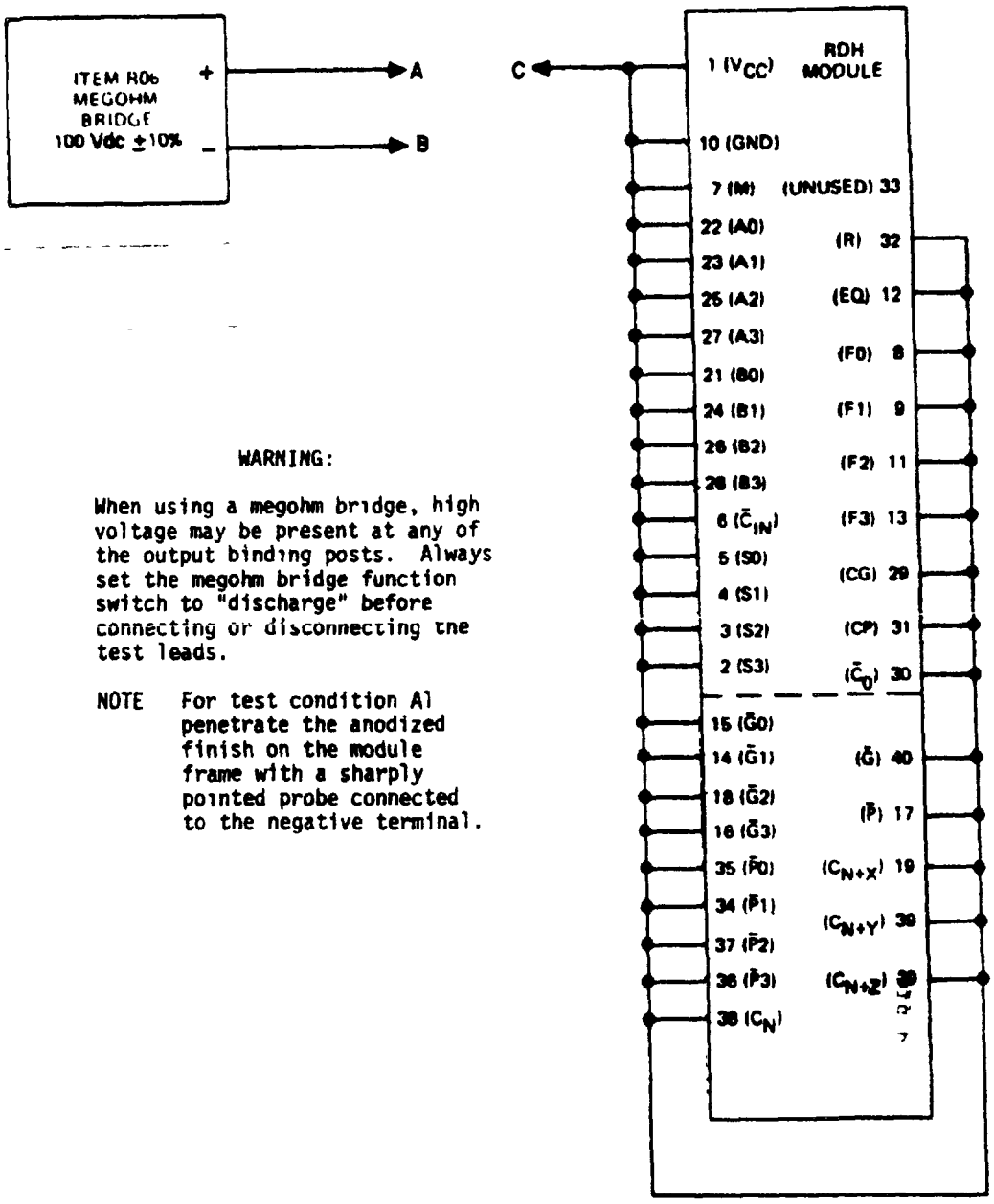
FIGURE 5. Vibration test circuit

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TABLE V. Isolation test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	Isolation	C, 33	Frame	10	∞	MΩ	6
A2	Isolation, unused pin	C	33	10	∞	MΩ	6

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WARNING:

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE For test condition A1 penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

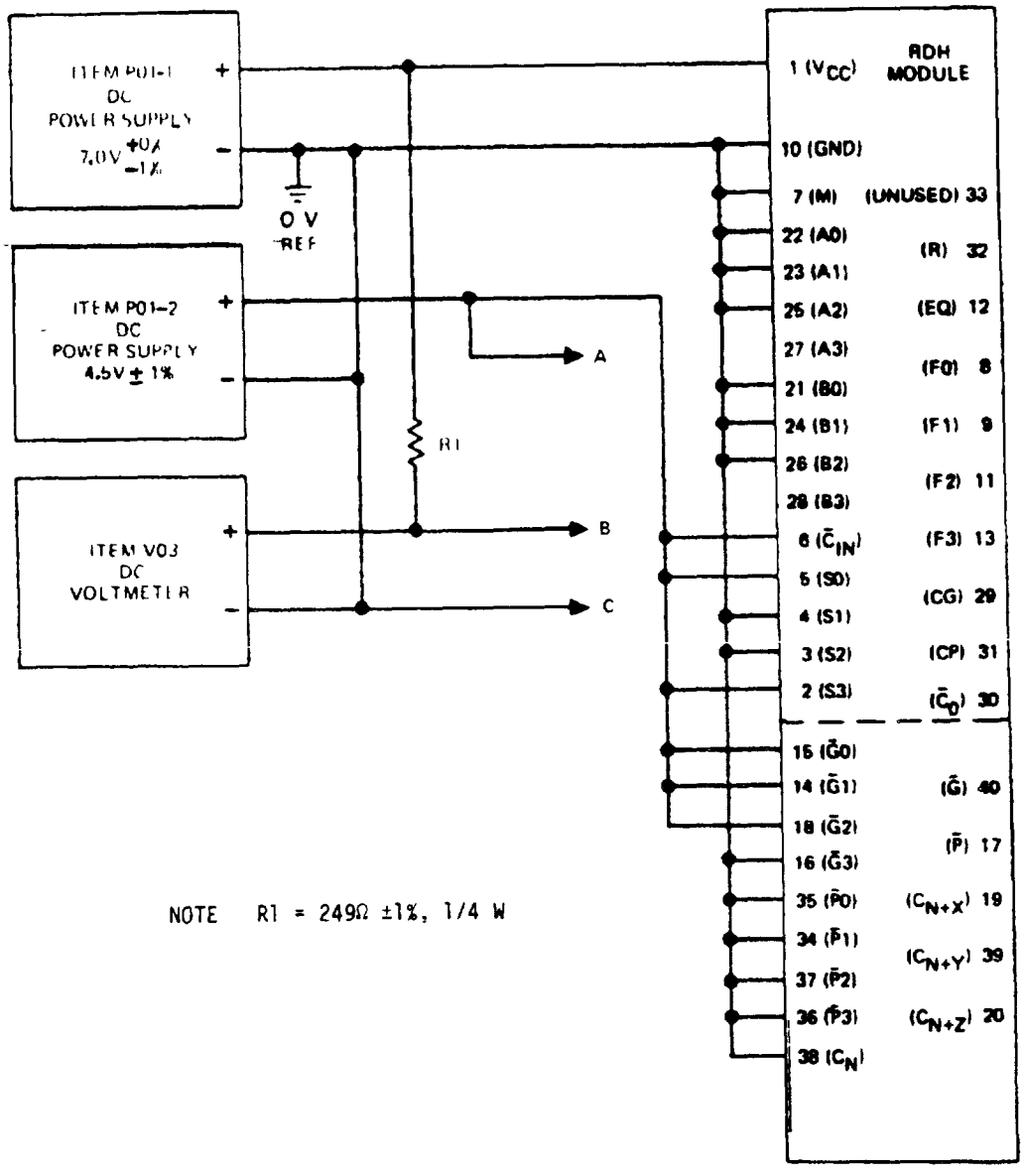
FIGURE 6. Isolation test circuit for table V.

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TABLE VI. V_{OUT} (low-loaded) test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
B1	FO V_{OUT} (low-loaded)	Open	8	27, 28	0.01	1.0	V	7
B2	F1		9					
B3	F2		11					
B4	F3		13					
B5	EQ		12					
B6	CP		31					
B7	CG	Open	29	27, 28				
B8	$\overline{C}_O$	27, 28	30	Open				
B9	C_{N+X}	Open	19					
B10	C_{N+Y}		39					
B11	C_{N+Z}		20					
B12	$\overline{P}$		17					
B13	$\overline{G}$ V_{OUT} (low-loaded)	Open	40	Open	0.01	1.0	V	7

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NOTE R1 = 249Ω ±1%, 1/4 W

FIGURE 7 V_{OUT} (low-loaded) test circuit for table VI

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TABLE VII. Input breakdown current test.

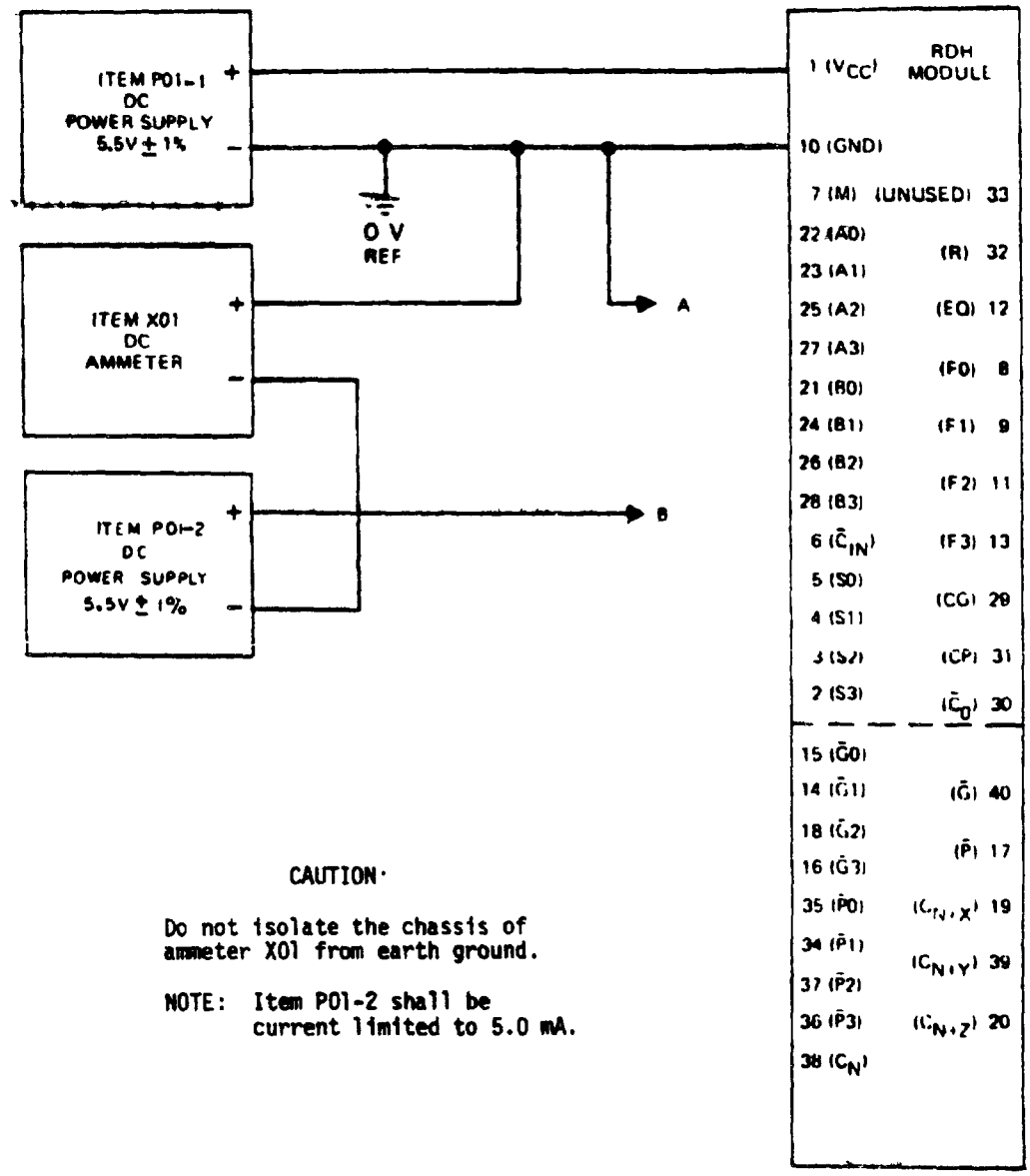
Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	M Input breakdown current	2,3,4,5,6,14,15,16,18,21,22, 23,24,25,26,27,28,34,35,36,37, 38	7	0.0	1.0	mA	8
C2	A3	2,3,4,5,6,7,14,15,16,18,21,22, 23,24,25,26,28,34,35,36,37,38	27				
C3	A2	2,3,4,5,6,7,14,15,16,18,21,22, 23,24,26,27,28,34,35,36,37,38	25				
C4	A1	2,3,4,5,6,7,14,15,16,18,21,22, 24,25,26,27,28,34,35,36,37,38	23				
C5	A0	2,3,4,5,6,7,14,15,16,18,21,23, 24,25,26,27,28,34,35,36,37,38	22				
C6	B3	2,3,4,5,6,7,14,15,16,18,21,22, 23,24,25,26,27,34,35,36,37,38	28				
C7	B2	2,3,4,5,6,7,14,15,16,18,21,22, 23,24,25,27,28,34,35,36,37,38	26				
C8	B1	2,3,4,5,6,7,14,15,16,18,21,22, 23,25,26,27,28,34,35,36,37,38	24				
C9	B0	2,3,4,5,6,7,14,15,16,18,22,23, 24,25,26,27,28,34,35,36,37,38	21				
C10	$\overline{C}_{IN}$	2,4,5,14,15,16,18,21,23,24,25, 26,27,28,34,35,36,37,38	6				
C11	S0	2,3,4,6,7,14,15,18,21,22,23, 24,25,26,27,28,34,35,36,37,38	5				
C12	S1 Input breakdown current	2,3,5,6,7,14,15,16,18,22,23, 25,27,34,35,36,37,38	4	0.0	1.0	mA	8

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TABLE VII. Input breakdown current test - Continued.

Test condition	Parameter		Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
C13	S2	Input breakdown current	2,4,5,6,7,14,15,16,18,22,23, 25,27,34,35,36,37,38	3	0.0	1.0	mA	8
C14	S3		3,4,5,6,7,14,15,16,18,21,22, 23,24,25,26,27,28,34,35,36,37, 38	2				
C15	$\overline{G0}$		2,3,4,5,6,7,14,16,18,21,22,23, 24,25,26,27,28,34,35,36,37	15				
C16	$\overline{G1}$		2,3,4,5,6,7,15,16,18,21,22,23, 24,25,26,27,28,34,35,36,37	14				
C17	$\overline{G2}$		2,3,4,5,6,7,14,15,16,21,22,23, 24,25,26,27,28,34,35,36,37	18				
C18	$\overline{G3}$		2,3,4,5,6,7,14,15,18,21,22,23, 24,25,26,27,28,34,35,36,37	16				
C19	$\overline{P0}$		2,3,4,5,6,7,14,15,16,18,21,22, 23,24,25,26,27,28,34,36,37,38	35				
C20	$\overline{P1}$		2,3,4,5,6,7,14,15,16,18,21,22, 23,24,25,26,27,28,35,36,37,38	34				
C21	$\overline{P2}$		2,3,4,5,6,7,14,15,16,18,21,22, 23,24,25,26,27,28,34,35,36,38	37				
C22	$\overline{P3}$		2,3,4,5,6,7,14,15,16,18,21,22, 23,24,25,26,27,28,34,35,37,38	36				
C23	C_N	Input breakdown current	2,3,4,5,6,7,14,15,16,18,21,22, 23,24,25,26,27,28,34,35,36,37	38	0.0	1.0	mA	8

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

NOTE: Item P01-2 shall be current limited to 5.0 mA.

FIGURE 8. Input breakdown current test circuit for table VII.

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TABLE VIII. Output leakage current (high-level) test.

Test condition	Parameter		Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
D1	EQ	Output leakage current, high level	6,21,22,23,24, 25,26,27,28	2,3,4,5,7	12	0.0	0.25	mA	9
D2	F3				13		1.0		
D3	F2				11				
D4	F1				9				
D5	FO				8				
D6	CG				29				
D7	$\overline{C}_O$		6,21,22,23,24, 25,26,27,28	2,3,4,5,7	30				
D8	CP		2,6,21,22,23, 24,25,26,27,28	3,4,5,7	31				
D9	$\overline{G}$		16,36	14,15,18,34, 35,37,38	40				
D10	$\overline{P}$				17				
D11	C_{N+X}				19				
D12	C_{N+Y}				39				
D13	C_{N+Z}	Output leakage current, high level	16,36	14,15,18,34, 35,37,38	20	0.0	1.0	mA	9

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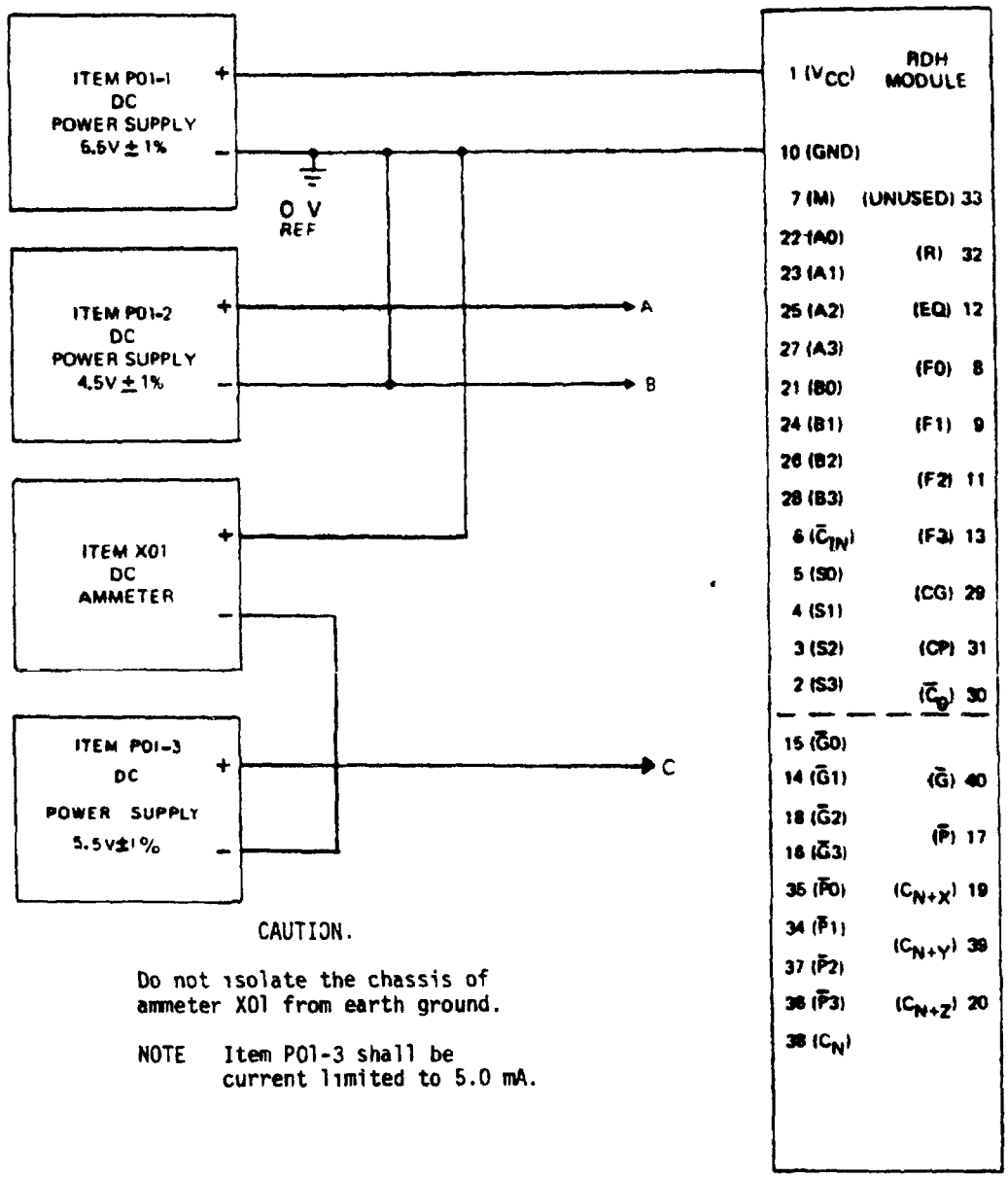


FIGURE 9. Output leakage current (high-level) test circuit for table VIII.

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TABLE IX. Output short circuit current test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
E1	R Output short circuit current	Open	2,3,5,16,18,21,22, 23,24,25,26,27,28, 36,37	32	-7.0	-7.7	mA	10
E2	$\overline{C}_O$	Open	2,3,5,16,18,21,22, 23,24,25,26,27,28, 36,37	30	-40	-100		
E3	FO	Open	2,3,5,16,18,21,22, 23,24,25,26,27,28, 36,37	8				
E4	F1	22	2,3,5,16,18,21,23, 24,25,26,27,28,36, 37	9				
E5	F1	23	2,3,5,16,18,21,22, 24,25,26,27,28,36, 37	9				
E6	F2	25	2,3,5,16,18,21,22, 23,24,26,27,28,36, 37	11				
E7	F2	22, 23	2,3,5,16,18,21,24, 25,26,27,28,36,37	11				
E8	F3	27	2,3,5,16,18,21,22, 23,24,25,26,28,36, 37	13				
E9	F3	22,23, 25	2,3,5,16,18,21,24, 26,27,28,36,37	13				
E10	CP Output short circuit current	3, 22	2,5,16,18,21,23, 24,25,26,27,28,36, 37	31	-40	-100	mA	10

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TABLE IX. Output short circuit current test - Continued.

Test condition	Parameter		Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
E11	CP	Output short circuit current	2,23, 24	3,5,16,18,21,22, 25,26,27,28,36,37	31	-40	-100	mA	10
E12	CG		22,23, 25,27	2,3,5,16,18,21,24, 26,28,36,37	29				
E13	CG		5,21, 24,26, 28	2,3,16,18,22,23, 25,27,36,37	29				
E14	$\overline{G}$		16, 36	2,3,5,18,21,22,23, 24,25,26,27,28,37	40				
E15	$\overline{P}$		36	2,3,5,16,18,21,22, 23,24,25,26,27,28, 37	17				
E16	$\overline{G}$		16,18, 37	2,3,5,21,22,23,24, 25,26,27,28,36	40				
E17	$\overline{P}$		37	2,3,5,16,18,21,22, 23,24,25,26,27,28, 36	17				
E18	C_{N+X}		Open	2,3,5,16,18,21,22, 23,24,25,26,27,28, 36,37	19				
E19	C_{N+Y}		Open	2,3,5,16,18,21,22, 23,24,25,26,27,28, 36,37	39				
E20	C_{N+Z}	Output short circuit current	Open	2,3,5,16,18,21,22, 23,24,25,26,27,28, 36,37	20	-40	-100	mA	10

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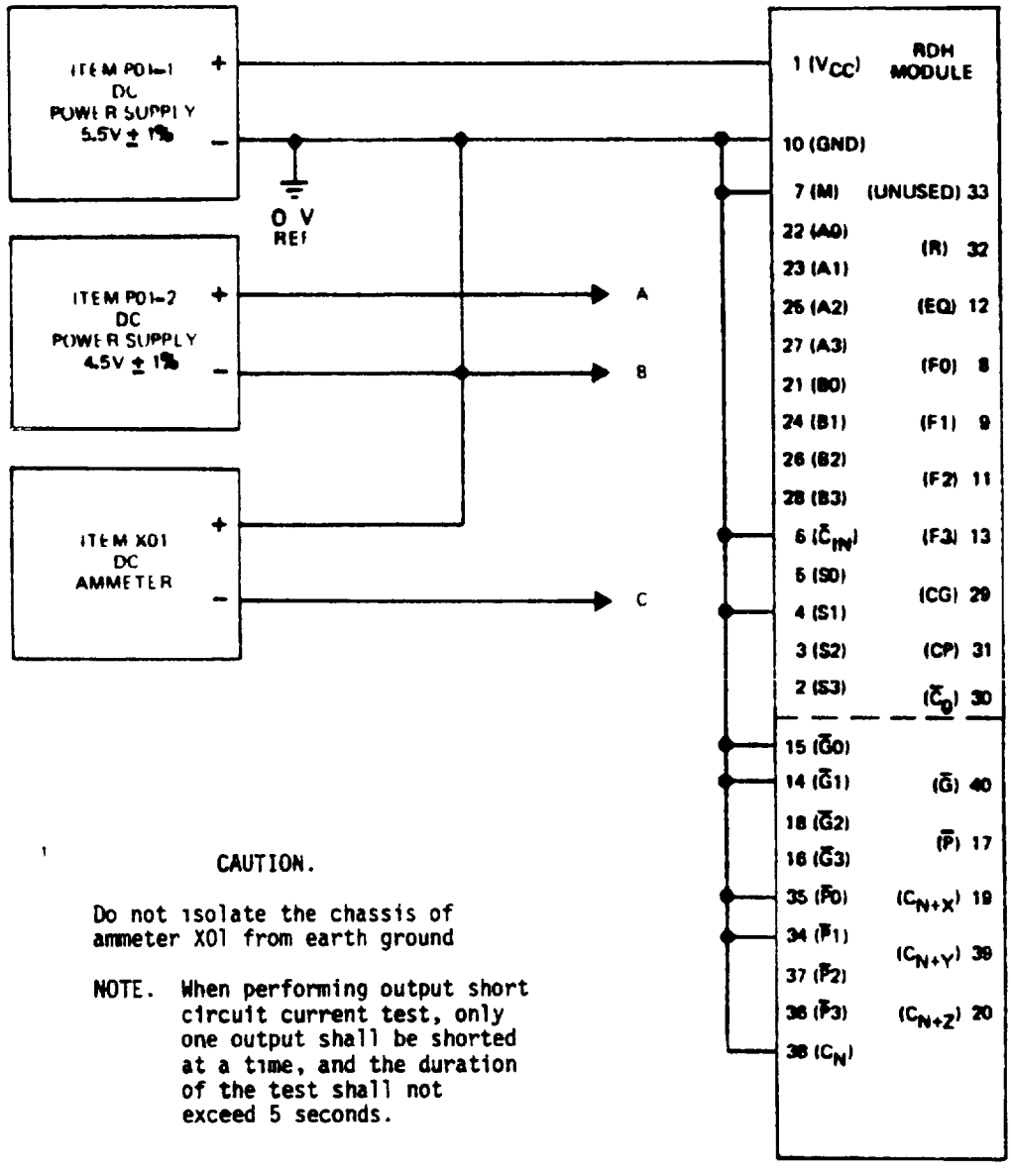


FIGURE 10. Output short circuit current test circuit for table IX.

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TABLE X. Functional test.

Test condition	Parameter	Connect			Output logic states													Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	EQ	F0	F1	F2	F3	CG	CP	$\overline{C}_0$	$\overline{G}$	$\overline{P}$	C_{N+X}	C_{N+Y}	C_{N+Z}	Min	Max		
F1	Functional	2,5,6,18,21,24, 26,27,34,36,38	3,4,7,14,15,16, 22,23,25,28,35, 37	See notes 2 and 3	H	H	H	H	H	H	L	H	L	H	H	H	H	See note 1			11
F2		2,3,15,16,18, 22,23,25,27,34	4,5,6,7,14,21, 24,26,28,35,36, 37,38		H	H	H	H	H	H	H	L	L	H	L	H	H				
F3		2,3,5,6,7,16, 25,26,28,37	4,14,15,18,21, 22,23,24,27,34, 35,36,38		L	H	H	H	L	H	H	L	L	H	H	H	H				
F4		2,4,6,7,14,15, 18,24,28	3,5,16,21,22, 23,25,26,27,34, 35,36,37,38		L	L	H	L	H	L	L	H	L	L	L	L	L				
F5		3,14,16,18,23, 24,25,26,27,28, 34,35,38	2,4,5,6,7,15, 21,22,36,37		H	H	H	H	H	L	L	H	H	H	H	L	L				
F6		2,7,14,16,18, 21,23,24,25,28, 35	3,4,5,6,15,22, 26,27,34,36,37, 38		L	H	H	L	H	L	H	H	L	H	H	H	H				
F7		2,3,5,6,7,14, 15,18,21,23,25, 26,38	4,16,22,24,27, 28,34,35,36,37		L	L	H	H	H	L	H	H	L	L	H	H	H				
F8		2,7,14,15,16, 21,24,25,34,37	3,4,5,6,18,22, 23,26,27,28,35, 36,38		L	H	H	L	H	L	L	H	L	H	L	L	H				
F9		3,16,23,24,25, 27,34,35,38	2,4,5,6,7,14, 15,18,21,22,26, 28,36,37		L	H	H	L	H	H	H	L	L	H	H	H	H				
F10	Functional	2,4,5,6,16,27, 28,34	3,7,14,15,18, 21,22,23,24,25, 26,35,36,37,38	See notes 2 and 3	L	H	H	H	L	H	H	L	L	H	H	H	H	See note 1			11

See notes at end of table.

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TABLE X Functional test - Continued

Test condition	Parameter	Connect			Output logic states													Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	EQ	FO	F1	F2	F3	CG	CP	$\overline{C}_O$	$\overline{G}$	$\overline{P}$	C_{N+X}	C_{N+Y}	C_{N+Z}	Min	Max		
F11	Functional	2,4,14,15,16, 22,23,24,26,27, 28,35,38	3,5,6,7,18,21, 25,34,36,37	See notes 2 and 3	L	L	H	H	L	H	H	L	L	H	L	L	H	See note 1			11
F12		2,4,5,6,18,25, 26,27,28,34,35, 36	3,7,14,15,16, 21,22,23,24,37, 38		L	H	H	L	H	H	H	L	L	H	H	H	H				
F13		2,3,4,18,21,23, 25,37	5,6,7,14,15,16, 22,24,26,27,28, 34,35,36,38		L	H	H	H	L	H	H	L	L	H	H	H	L				
F14		4,5,14,16,18, 21,22,24,25,35, 36,37	2,3,6,7,15,23, 26,27,28,34,38		L	L	H	L	L	H	L	L	H	H	H	H	L				
F15		2,5,6,16,18,21, 22,24,25,35,37	3,4,7,14,15,23, 26,27,28,34,36, 38		L	L	H	L	H	L	H	H	H	H	H	H	L				
F16		3,5,6,14,18,22, 25,26,27,34,35, 36,37,38	2,4,7,15,16,21, 23,24,28		L	L	H	H	L	H	H	L	L	H	H	L	L				
F17		2,5,6,7,14,15, 23,24,26,28,34, 35,36,37	3,4,16,18,21, 22,25,27,38		L	H	H	L	L	H	H	L	L	H	L	L	H				
F18		2,14,15,18,24, 27,28,34,36,37, 38	3,4,5,6,7,16, 21,22,23,25,26, 35		L	H	H	L	L	H	H	L	L	H	H	L	L				
F19	Functional	3,5,6,15,16,21, 22,23,24,27,28, 36,38	2,4,7,14,18,25, 26,34,35,37	See notes 2 and 3	L	H	H	L	H	L	L	H	H	H	H	H	H	See note 1			11

See notes at end of table

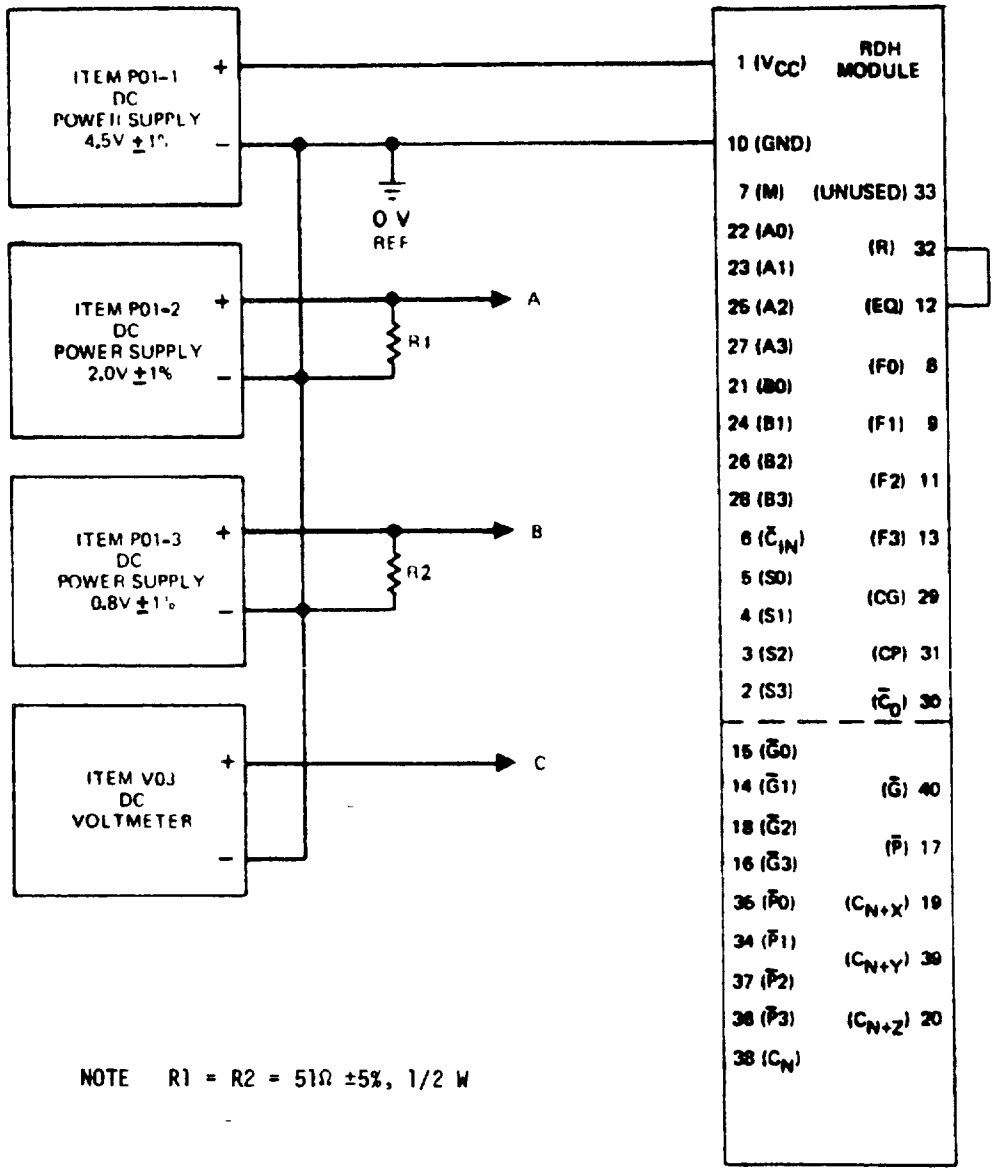
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TABLE X Functional test - Continued.

Test condition	Parameter	Connect			Output logic states													Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	EQ	FO	F1	F2	F3	CG	CP	$\overline{C}_O$	$\overline{G}$	$\overline{P}$	C_{N+X}	C_{N+Y}	C_{N+Z}	Min	Max		
F20	Functional	3,5,6,7,14,15, 21,26,28,34,35	2,4,16,18,22, 23,24,25,27,36, 37,38	See notes 2 and 3	L	L	H	L	L	L	L	H	L	H	L	L	H	See note 1			11
F21		2,3,14,15,16, 18,22,25,26,28, 35,36,38	4,5,6,7,21,23, 24,27,34,37		L	H	H	L	H	L	H	H	H	H	L	L	L				
F22		2,5,6,14,15,18, 23,24,26,27,28, 34,35,36,37,38	3,4,7,16,21,22, 25		L	L	L	L	H	H	H	L	L	H	L	L	L				
F23		2,5,6,15,18,24, 34,35	3,4,7,14,16,21, 22,23,25,26,27, 28,36,37,38		L	L	H	L	L	L	L	H	L	H	L	H	H				
F24		2,4,5,6,7,14, 15,16,18,21,22, 24,26,28,34,36	3,23,25,27,35, 37,38		L	H	L	L	L	H	H	L	H	H	L	L	L				
F25	Functional	2,7,14,15,16, 18,21,22,23,26, 28,35	3,4,5,6,24,25, 27,34,36,37,38	See notes 2 and 3	L	H	L	H	H	L	H	H	H	H	L	L	L	See note 1			11

NOTES:
1. Output logic states: L = low-voltage level = 0.0 to 0.5 V.
   H = high-voltage level = 2.5 to 4.5 V
2. For test conditions F1-F25 verification of the output logic states in note 1 is obtained by connecting voltmeter (V03), point C, to each output pin in succession after input conditions have been accomplished
3. EQ, FO, F1, F2, F3, CG, CP, $\overline{C}_O$, $\overline{G}$, $\overline{P}$, C_{N+X}, C_{N+Y}, C_{N+Z} output logic states are measured at output pins 12, 8, 9, 11, 13, 29, 31, 30, 40, 17, 19, 39, and 20 respectively

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NOTE R1 = R2 = 51Ω ±5%, 1/2 W

FIGURE 11 Functional test circuit for table X

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TABLE XI. Input clamp diode voltage test.

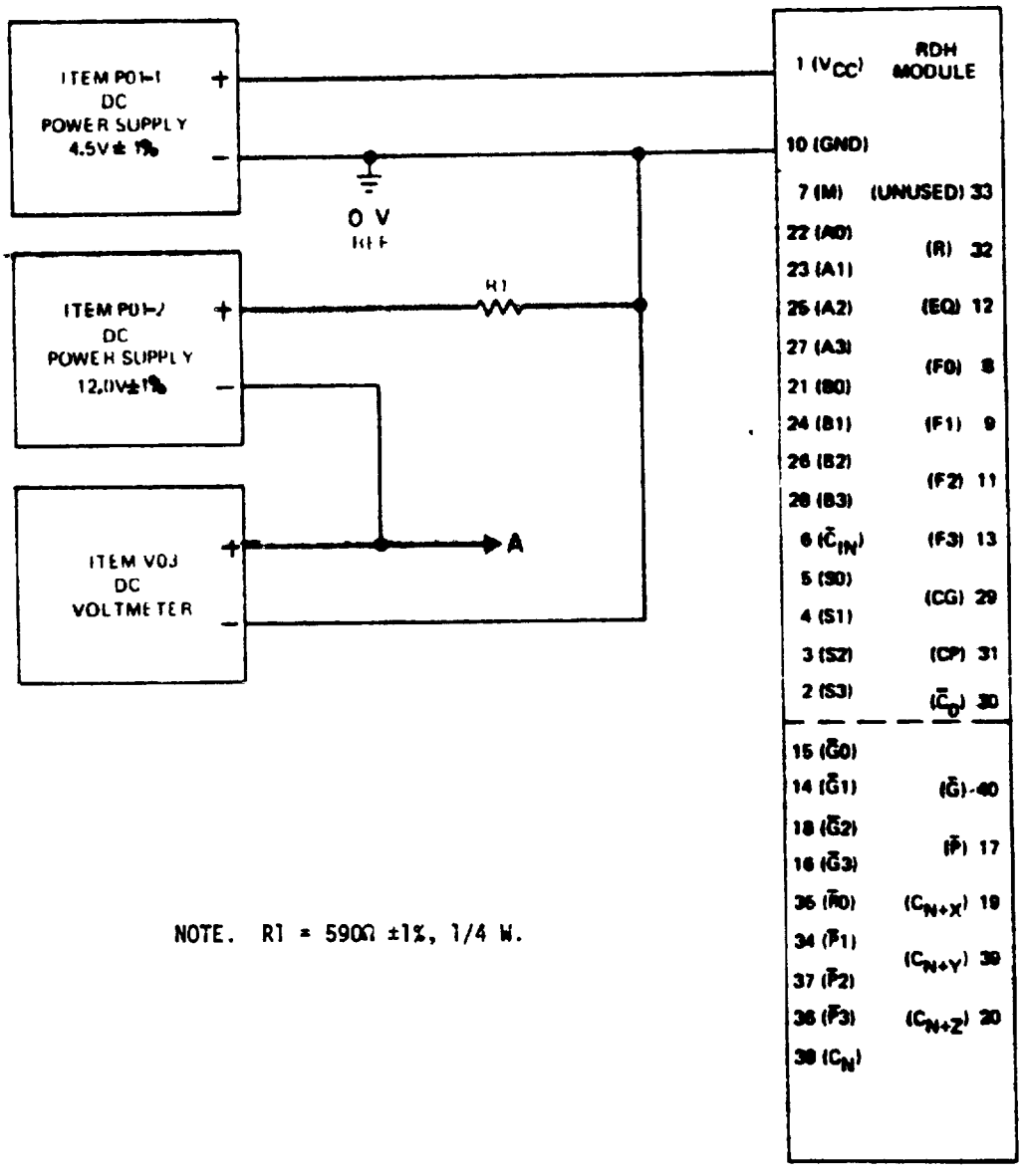
Test condition	Parameter		Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	Min	Max		
G1	M	Input clamp diode voltage	7	-0.4	-1.2	V	12
G2	A3		27				
G3	A2		25				
G4	A1		23				
G5	A0		22				
G6	B3		28				
G7	B2		26				
G8	B1		24				
G9	B0		21				
G10	$\overline{C}_{IN}$		6				
G11	S0		5				
G12	S1		4				
G13	S2		3				
G14	S3		2				
G15	$\overline{G}0$		15				
G16	$\overline{G}1$		14				
G17	$\overline{G}2$		18				
G18	$\overline{G}3$	Input clamp diode voltage	16	-0.4	-1.2	V	12

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TABLE XI. Input clamp diode voltage test - Continued.

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
G19	$\overline{P0}$ Input clamp diode voltage	35	-0.4	-1.2	V	12
G20	$\overline{P1}$	34				
G21	$\overline{P2}$	37				
G22	$\overline{P3}$	36				
G23	CN Input clamp diode voltage	38	-0.4	-1.2	V	12

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NOTE. R1 = 590Ω ±1%, 1/4 W.

FIGURE 12. Input clamp diode voltage test circuit for table XI.

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TABLE XII. Static output voltage (low-level) test.

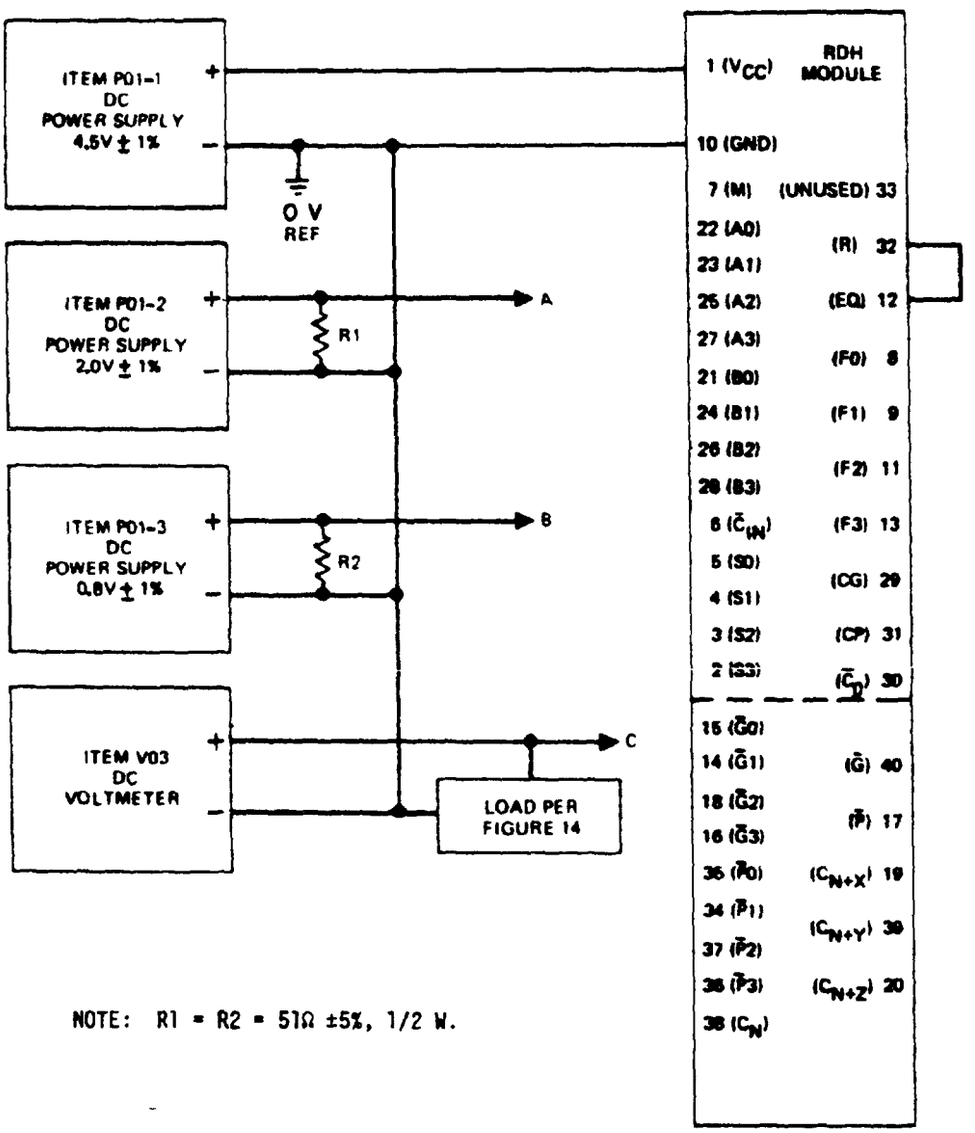
Test condition	Parameter		Connect: A to pin	Connect: B to pin	Connect: C to pin	Initial limits 25° ±5°C Ambient: Min	Initial limits 25° ±5°C Ambient: Max	Unit	Test figure
H1	EQ	Static output voltage, low level	7,22,23,25,27	2,3,4,5,6,21, 24,26,28	12	0.0	0.5	V	13
H2	F3				13				
H3	F2				11				
H4	F1				9				
H5	FO				8				
H6	CP				31				
H7	$\overline{C}_O$		7,22,23,25,27	2,3,4,5,6,21, 24,26,28	30				
H8	CG		7,22,23,25	2,3,4,5,6,21, 24,26,27,28	29				
H9	$\overline{G}$		14,15,18	16,34,35,36, 37,38	40				
H10	$\overline{P}$				17				
H11	C_{N+X}				20				
H12	C_{N+Y}				39				
H13	C_{N+Z}	Static output voltage, low level	14,15,18	16,34,35,36, 37,38	19	0.0	0.5	V	13

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TABLE XIII. Static output voltage (high-level) test.

Test condition	Parameter		Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
J1	EQ	Static output voltage, high level	6,21,22,23,24, 25,26,27,28	2,3,4,5,7	12	2.5	4.5	V	13
J2	F3				13				
J3	F2				11				
J4	F1				9				
J5	F0				8				
J6	CG				29				
J7	$\overline{C}_O$		6,21,22,23,24, 25,26,27,28	2,3,4,5,7	30				
J8	CP		2,6,21,22,23, 24,25,26,27,28	3,4,5,7	31				
J9	$\overline{G}$		16, 36	14,15,18,34, 35,37,38	40				
J10	$\overline{P}$				17				
J11	C_{N+X}				20				
J12	C_{N+Y}				39				
J13	C_{N+Z}	Static output voltage, high level	16, 36	14,15,18,34, 35,37,38	19	2.5	4.5	V	13

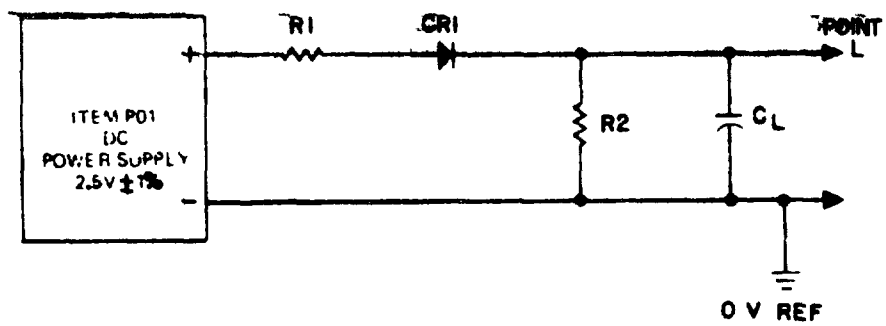
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NOTE: R1 = R2 = 51Ω ±5%, 1/2 W.

FIGURE 13. Static output voltage (low-level and high-level) test circuit for tables XII and XIII.

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NOTES:
1 Load circuit 1 and load circuit 2:

(a) CR1 = 1N3600, 1N4150, or equivalent.
(b) R2 = 2.49 kΩ ±1%, 1/8 W.
(c) Test type S:
C_L = 50 ±5 pF (including probe and parasitic capacitance).
Test type Q:
C_L = 150 ±5 pF (including probe and parasitic capacitance)
(d) Capacitor C_L is not required for dc electrical test conditions H1-H13 and J1-J13.
(e) Lead length from load circuit to module output pin under test is to be less than 1 inch (25.4 mm)

2. Load circuit 1 for all outputs except EQ:

R1 = Select value for a current flow of 20 mA ±5% with point L held at 0.5 V.

3. Load circuit 2 for EQ output.

R1 = Select value for a current flow of 14 mA ±5% with point L held at 0.5 V.

FIGURE 14. RDH logic load circuit.

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TABLE XIV. Input current (high-level) test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
K1	M Input current, high level	7	2,3,4,5,6,14,15,16,18,21, 22,23,24,25,26,27,28,34, 35,36,37,38	0	50	µA	15
K2	A3	27	2,3,4,5,6,7,14,15,16,18, 21,22,23,24,25,26,28,34, 35,36,37,38		150		
K3	A2	25	2,3,4,5,6,7,14,15,16,18, 21,22,23,24,26,27,28,34, 35,36,37,38				
K4	A1	23	2,3,4,5,6,7,14,15,16,18, 21,22,24,25,26,27,28,34, 35,36,37,38				
K5	A0	22	2,3,4,5,6,7,14,15,16,18, 21,23,24,25,26,27,28,34, 35,36,37,38				
K6	B3	28	2,3,4,5,6,7,14,15,16,18, 21,22,23,24,25,26,27,34, 35,36,37,38				
K7	B2	26	2,3,4,5,6,7,14,15,16,18, 21,22,23,24,25,27,28,34, 35,36,37,38				
K8	B1	24	2,3,4,5,6,7,14,15,16,18, 21,22,23,25,26,27,28,34, 35,36,37,38				
K9	B0 Input current, high level	21	2,3,4,5,6,7,14,15,16,18, 22,23,24,25,26,27,28,34, 35,36,37,38	0	150	µA	15

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TABLE XIV. Input current (high-level) test - Continued.

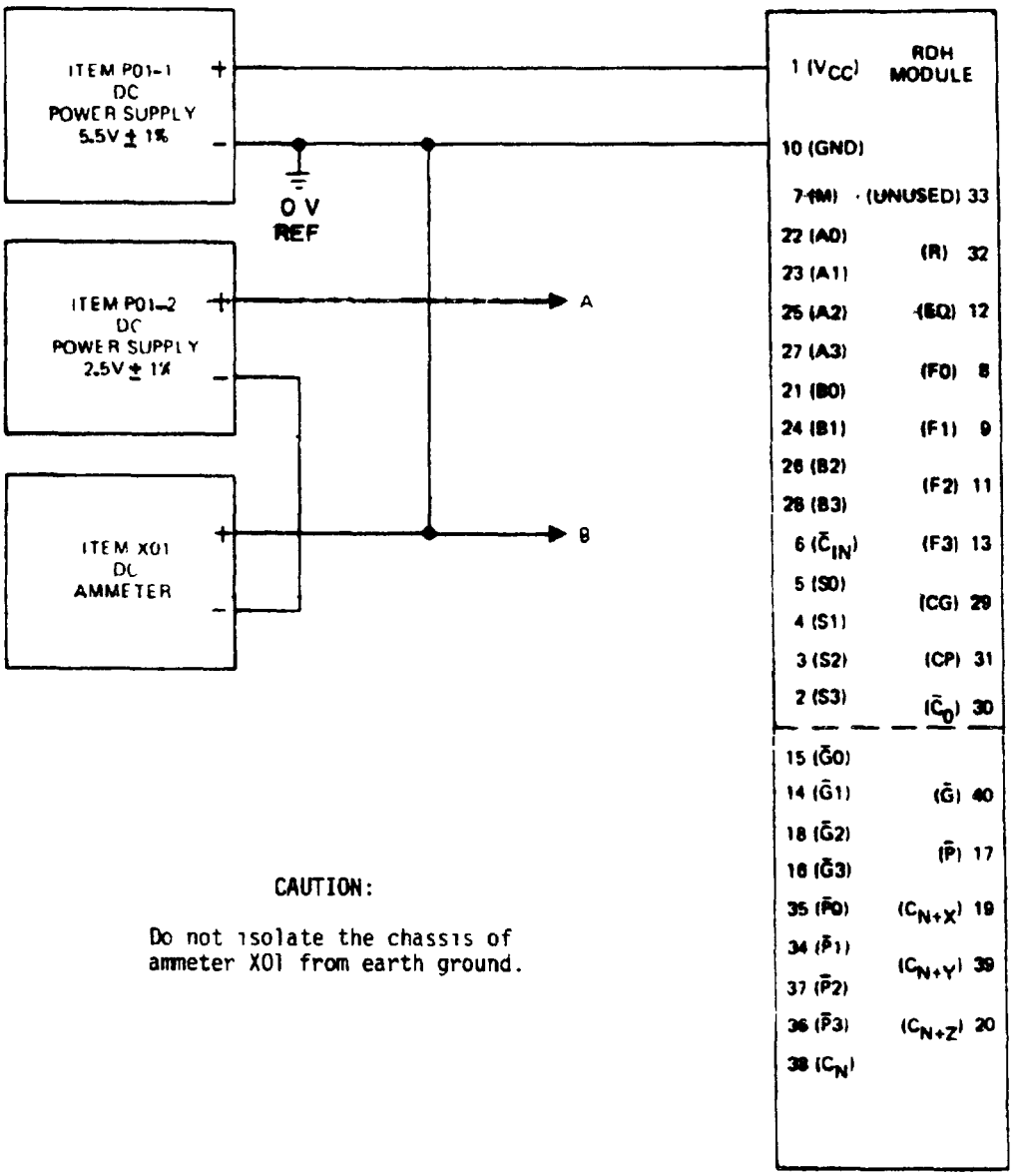
Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
K10	S3 Input current, high level	2	3,4,5,6,7,14,15,16,18,21, 22,23,24,25,26,27,28,34, 35,36,37,38	0	200	μA	15
K11	S2	3	2,4,5,6,7,14,15,16,18,22, 23,25,27,34,35,36,37,38				
K12	S1	4	2,3,5,6,7,14,15,16,18,22, 23,25,27,34,35,36,37,38				
K13	S0	5	2,3,4,6,7,14,15,16,18,21, 22,23,24,25,26,27,28,34, 35,36,37,38		200		
K14	$\overline{C}_{IN}$	6	2,4,5,14,15,16,18,21,23, 24,25,26,27,28,34,35,36, 37,38		250		
K15	C_N	38	2,3,4,5,6,7,14,15,16,18, 21,22,23,24,25,26,27,28, 34,35,36,37		50		
K16	$\overline{P}3$	36	2,3,4,5,6,7,14,15,16,18, 21,22,23,24,25,26,27,28, 34,35,37,38		100		
K17	$\overline{P}2$	37	2,3,4,5,6,7,14,15,16,18, 21,22,23,24,25,26,27,28, 34,35,36,38		150		
K18	$\overline{P}1$ Input current, high level	34	2,3,4,5,6,7,14,15,16,18, 21,22,23,24,25,26,27,28, 35,36,37,38	0	200	μA	15

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TABLE XIV. Input current (high-level) test - Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
K19	$\overline{P0}$ Input current, high level	35	2,3,4,5,6,7,14,15,16,18, 21,22,23,24,25,26,27,28, 34,36,37,38	0	200	µA	15
K20	$\overline{G3}$	16	2,3,4,5,6,7,14,15,18,21, 22,23,24,25,26,27,28,34, 35,36,37		200		
K21	$\overline{G2}$	18	2,3,4,5,6,7,14,15,16,21, 22,23,24,25,26,27,28,34, 35,36,37		350		
K22	$\overline{G1}$	14	2,3,4,5,6,7,15,16,18,21, 22,23,24,25,26,27,28,34, 35,36,37		400		
K23	$\overline{G0}$ Input current, high level	15	2,3,4,5,6,7,14,16,18,21, 22,23,24,25,26,27,28,34, 35,36,37	0	350	µA	15

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

FIGURE 15 Input current (high-level) test circuit for table XIV

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TABLE XV. Input current (low-level) test.

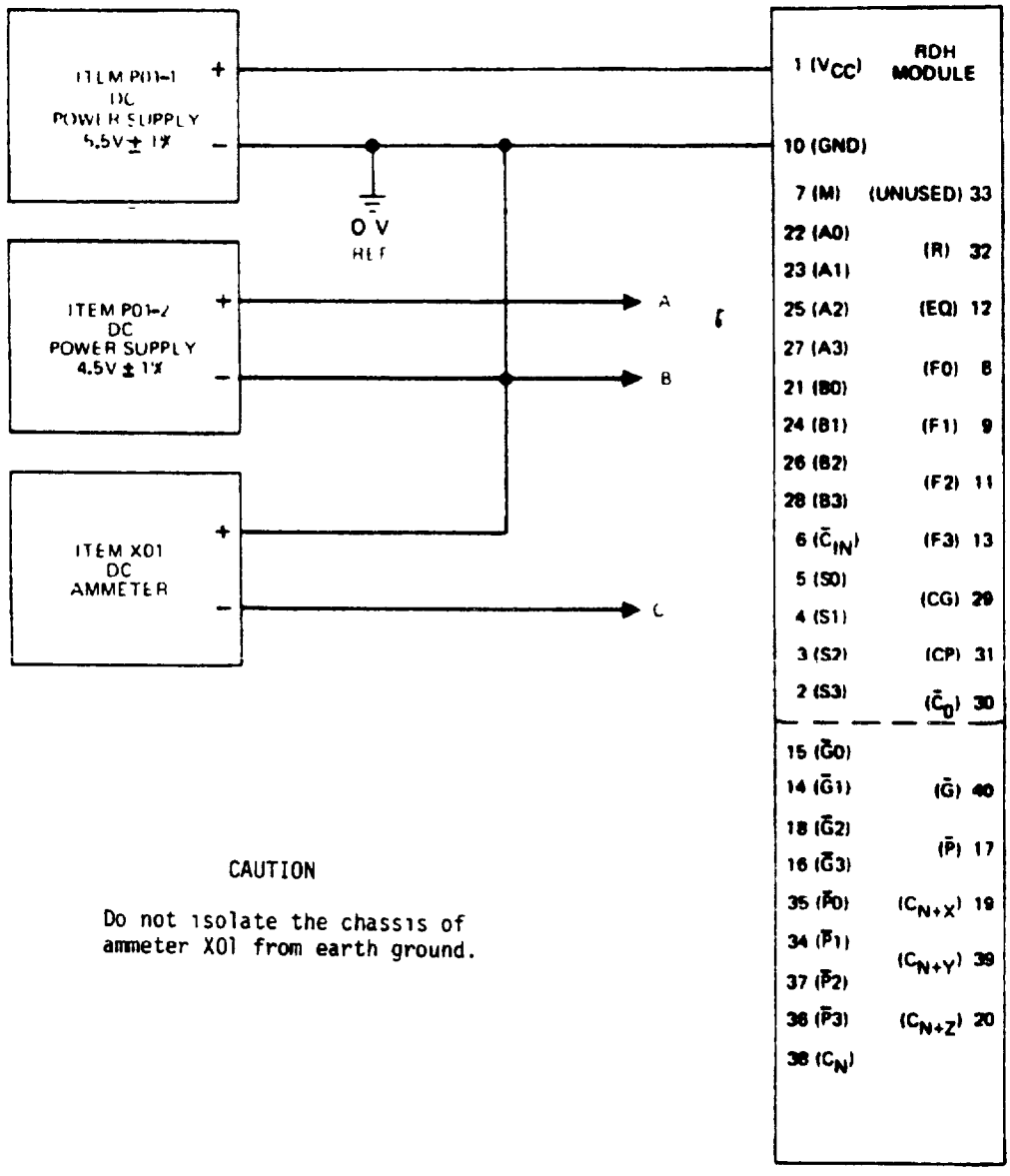
Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
L1	M Input current, low level	Open	Open	7	-0.1	-2.2	mA	16
L2	C_N	Open		38		-2.2		
L3	$\overline{P3}$	16,34,35,37		36		-4.4		
L4	$\overline{P2}$	18,34,35,36		37		-6.6		
L5	A3	Open		27				
L6	A2			25				
L7	A1			23				
L8	A0	Open		22				
L9	B3	2,5,27		28				
L10	B2	2,5,25		26				
L11	B1	2,5,23		24				
L12	B0	2,5,22		21		-6.6		
L13	S3	21,22,23,24, 25,26,27,28	Open	2		-8.8		
L14	S2	22,23,25,27	21,24, 26,28	3				
L15	S1	Open	21,24, 26,28	4				
L16	S0 Input current, low level	21,24,26,28	Open	5	-0.1	-8.8	mA	16

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:57Z
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TABLE XV. Input current (low-level) test - Continued.

Test condition	Parameter		Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
L17	$\overline{P}1$	Input current, low level	14,35,36,37	Open	34	-0.1	-8.8	mA	16
L18	$\overline{P}0$		15,34,36,37	Open	35		-8.8		
L19	$\overline{G}3$		14,15,18,34, 36,37	Open	16		-8.8		
L20	$\overline{C}_{IN}$		Open	7	6		-11.0		
L21	$\overline{G}2$		14,15,16,34, 35,37	38	18		-15.4		
L22	$\overline{G}0$		14,16,18,35	38	15		-15.4		
L23	$\overline{G}1$	Input current, low level	15,16,18,34, 35	38	14	-0.1	-17.6	mA	16

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CAUTION

Do not isolate the chassis of ammeter X01 from earth ground.

FIGURE 16. Input current (low-level) test circuit for table XV.

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TABLE XVI. Power supply current (low-level and high-level) test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
M1	Power supply current, low-level	14,15,18	16,22,23, 25,27,36	100	319	mA	17
M2	Power supply current, high-level	16,22,23, 25,27,36	14,15,18	90	270	mA	17

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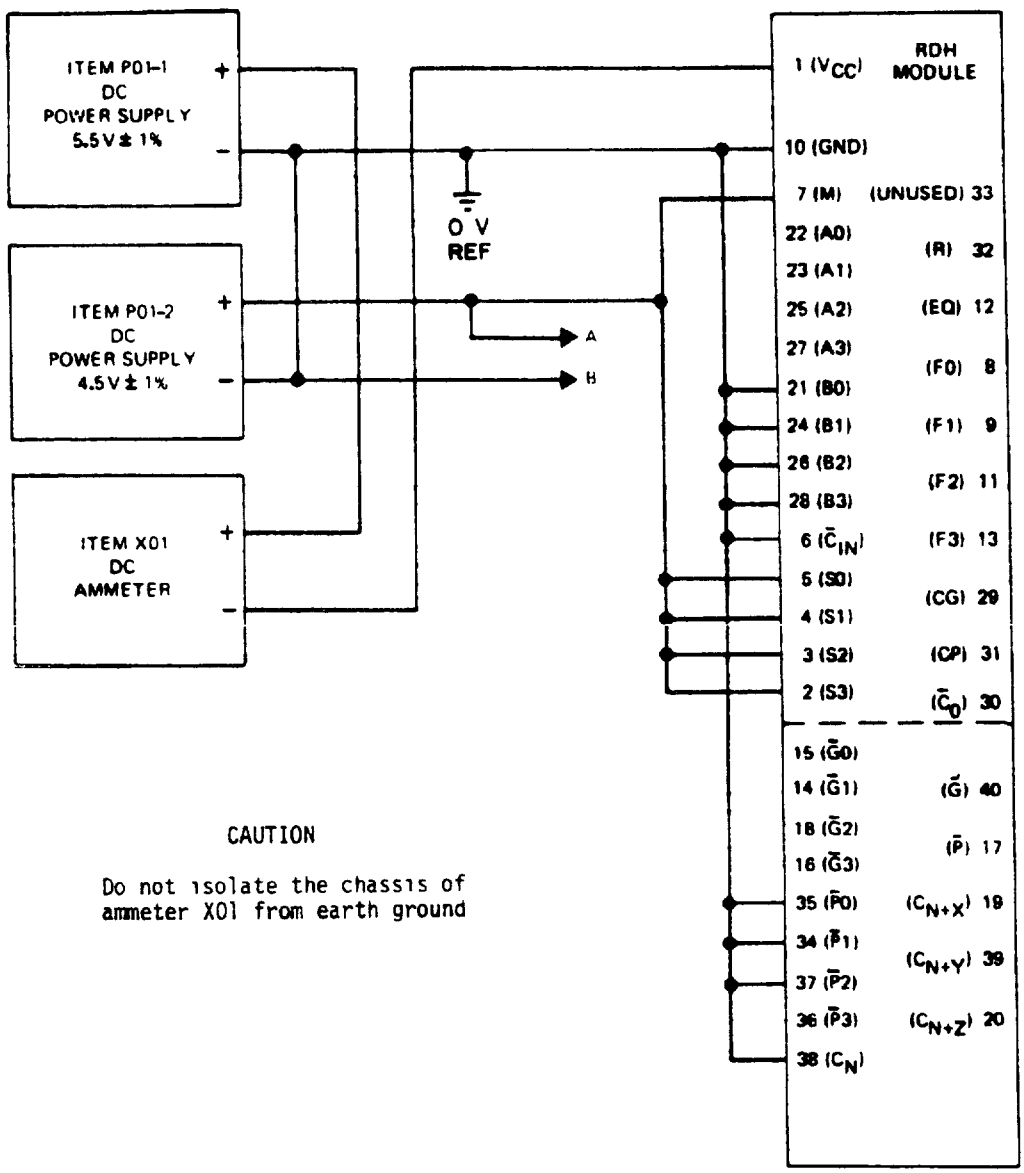


FIGURE 17. Power supply current (low-level and high-level) test circuit for table XVI.

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TABLE XVII. Transition time test.

Test condition	Parameter		Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
N1	EQ	Transition time (t_{TLH}), C_L = 50 pF	2,4,6,7, 24,26,28	3,5,22, 23,25,27	21	12	2.0	28.0	ns	18,19
N2	F3		6,21,24, 26,28	2,3,4,7, 22,23,25, 27	5	13		15.0		
N3	F2					11				
N4	F1					9				
N5	FO		6,21,24, 26,28	2,3,4,7, 22,23,25, 27	5	8				
N6	CP		6,22,23, 24,25,26, 27,28	2,4,5,7, 21	3	31				
N7	CG		6,21,23, 24,25,26, 27,28	2,3,4,5,7	22	29				
N8	$\overline{C}_O$		21,22,23, 24,25,26, 27,28	2,3,4,5,7	6	30				
N9	C_{N+X}		14,15,16, 18,36	34,35,37	38	19				
N10	C_{N+Y}		14,15,16, 18,36	34,35,37	38	39				
N11	C_{N+Z}	Transition time (t_{TLH}), C_L = 50 pF	14,15,16, 18,36	34,35,37	38	20	2.0	15.0	ns	18,19

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TABLE XVII. Transition time test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
N12	$\overline{G}$	Transition time (t_{TLH}), C_L = 50 pF	16,34,35, 37	14,15,18	36	40	2.0	15.0	ns	18,19
N13	$\overline{P}$	Transition time (t_{TLH}), C_L = 50 pF	14,15,16, 18	34,35,37	36	17		15.0		
N14	EQ	Transition time (t_{TLH}), C_L = 150 pF	2,4,6,7, 24,26,28	3,5,22, 23,25,27	21	12		80.0		
N15	F3		6,21,24, 26,28	2,3,4,7, 22,23,25, 27	5	13		27.0		
N16	F2					11				
N17	F1					9				
N18	FO		6,21,24, 26,28	2,3,4,7, 22,23,25, 27	5	8				
N19	CP		6,22,23, 24,25,26, 27,28	2,4,5,7, 21	3	31				
N20	CG		6,21,23, 24,25,26, 27,28	2,3,4,5,7	22	29				
N21	$\overline{C}_O$		21,22,23, 24,25,26, 27,28	2,3,4,5,7	6	30				
N22	C_{N+X}	Transition time (t_{TLH}), C_L = 150 pF	14,15,16, 18,36	34,35,37	38	19	2.0	27.0	ns	18,19

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TABLE XVII. Transition time test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
N23	C_{N+Y}	Transition time (t_{TLH}), C_L = 150 pF	14,15,16, 18,36	34,35,37	38	39	2.0	27.0	ns	18,19
N24	C_{N+Z}		14,15,16, 18,36	34,35,37	38	20				
N25	$\overline{G}$		16,34,35, 37	14,15,18	36	40				
N26	$\overline{P}$	Transition time (t_{TLH}), C_L = 150 pF	14,15,16, 18	34,35,37	36	17		27.0		
P1	EQ	Transition time (t_{THL}), C_L = 50 pF	2,4,6,7, 24,26,28	3,5,22, 23,25,27	21	12		10.0		
* P2	F3		6,21,24, 26,28	2,3,4,7, 22,23,25, 27	5	13		8.0		
P3	F2					11				
P4	F1					9				
P5	F0		6,21,24, 26,28	2,3,4,7, 22,23,25, 27	5	8				
P6	CP		6,22,23, 24,25,26, 27,28	2,4,5,7, 21	3	31				
* P7	CG	Transition time (t_{THL}), C_L = 50 pF	6,21,23, 24,25,26, 27,28	2,3,4,5,7	22	29	2.0	8.0	ns	18,19

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TABLE XVII. Transition time test - Continued.

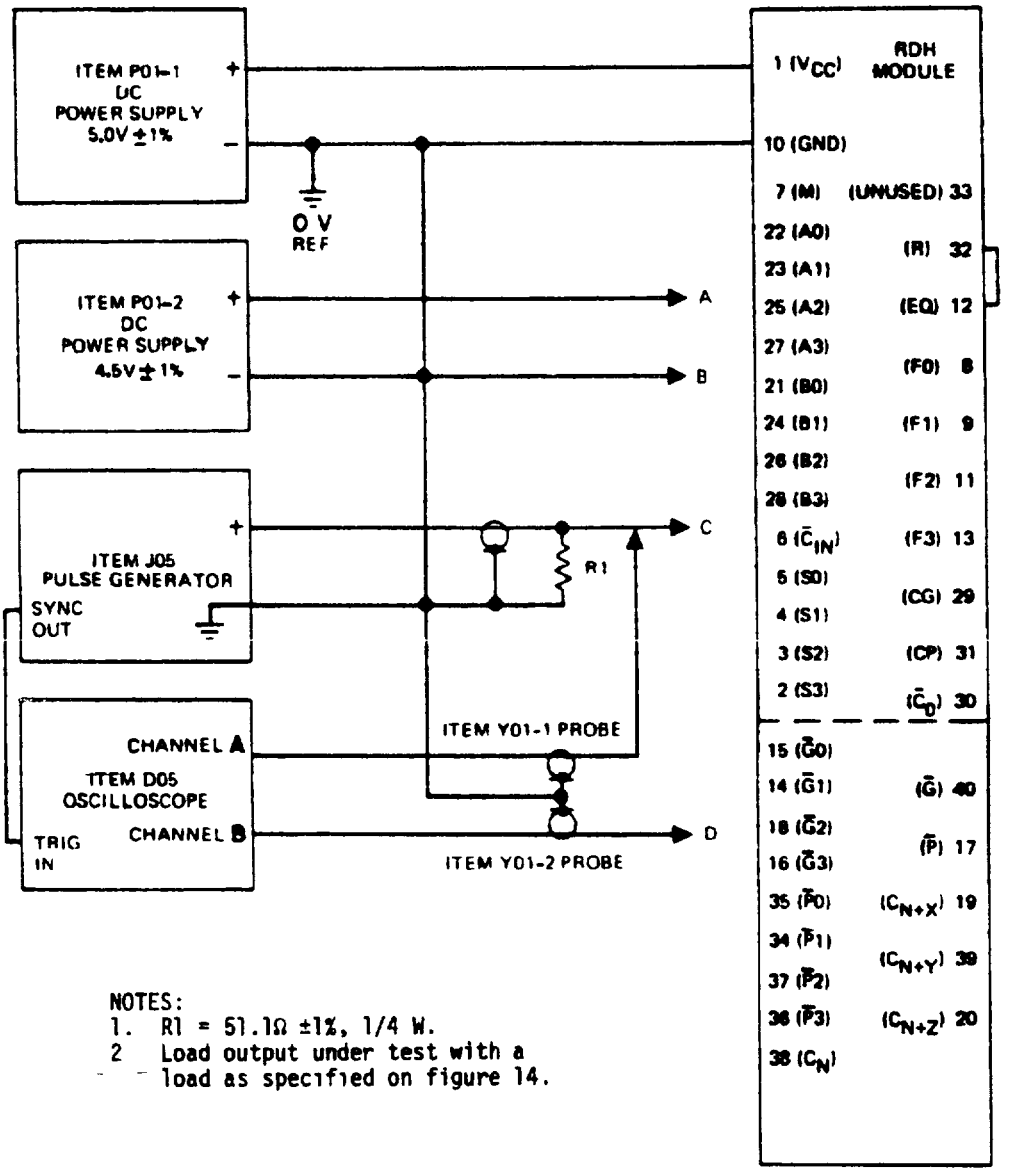
	Test condition	Parameter		Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
				A to pin	B to pin	C to pin	D to pin	Min	Max		
*	P8	$\overline{C}_O$	Transition time (t_{THL}), C_L = 50 pF	21,22,23, 24,25,26, 27,28	2,3,4,5,7	6	30	2.0	8.0	ns	18,19
	P9	C_{N+X}		14,15,16, 18,36	34,35,37	38	19				
	P10	C_{N+Y}		14,15,16, 18,36	34,35,37	38	39				
	P11	C_{N+Z}		14,15,16, 18,36	34,35,37	38	20				
	P12	$\overline{G}$		16,34,35, 37	14,15,18	36	40				
*	P13	$\overline{P}$	Transition time (t_{THL}), C_L = 50 pF	14,15,16, 18	34,35,37	36	17		8.0		
	P14	EQ	Transition time (t_{THL}), C_L = 150 pF	2,4,6,7, 24,26,28	3,5,22, 23,25,27	21	12		20.0		
*	P15	F3		6,21,24, 26,28	2,3,4,7, 22,23,25, 27	5	13		13.0		
	P16	F2					11				
	P17	F1					9				
*	P18	F0	Transition time (t_{THL}), C_L = 150 pF	6,21,24, 26,28	2,3,4,7, 22,23,25, 27	5	8	2.0	13.0	ns	18,19

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TABLE XVII. Transition time test - Continued.

	Test condition	Parameter		Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
				A to pin	B to pin	C to pin	D to pin	Min	Max		
*	P19	CP	Transition time (t_{THL}), C_L = 150 pF	6,22,23, 24,25,26, 27,28	2,4,5,7, 21	3	31	2.0	13.0	ns	18,19
	P20	CG		6,21,23, 24,25,26, 27,28	2,3,4,5,7	22	29				
	P21	$\overline{C}_O$		21,22,23, 24,25,26, 27,28	2,3,4,5,7	6	30				
	P22	C_{N+X}		14,15,16, 18,36	34,35,37	38	19				
	P23	C_{N+Y}		14,15,16, 18,36	34,35,37	38	39				
	P24	C_{N+Z}		14,15,16, 18,36	34,35,37	38	20				
	P25	$\overline{G}$		16,34,35, 37	14,15,18	36	40				
*	P26	$\overline{P}$	Transition time (t_{THL}), C_L = 150 pF	14,15,16, 18	34,35,37	36	17	2.0	13.0	ns	18,19

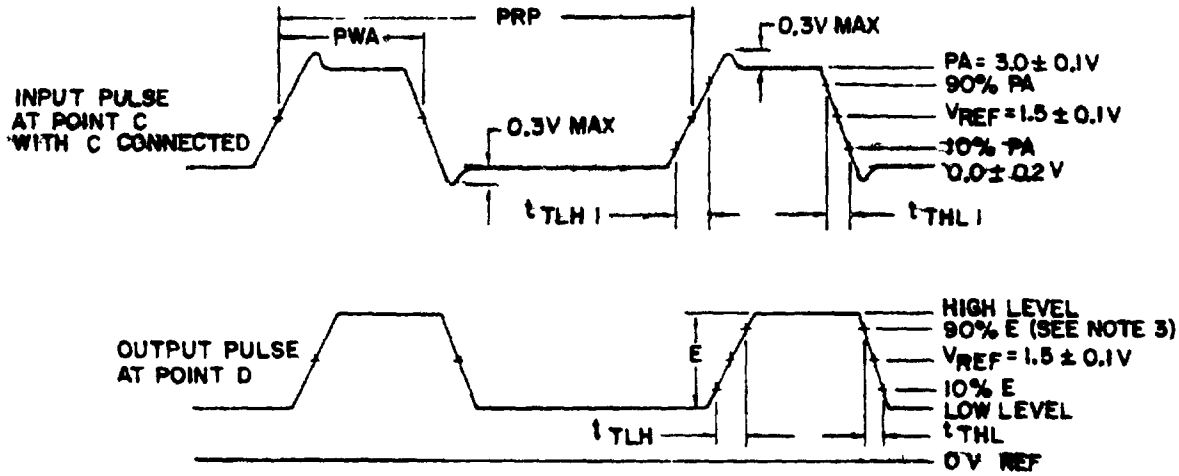
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NOTES:
1. R1 = 51.1Ω ±1%, 1/4 W.
2 Load output under test with a load as specified on figure 14.

FIGURE 18. Transition time test circuit for table XVII

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NOTES:
1. Input pulse parameters:
   $t_{TLH1} = t_{THL1} = 4 \pm 1$ ns.
   PWA = 500 ±50 ns.
   PRP = 1,000 ±100 ns.
2. Before making measurements specified in table XVII, verify that the output pulse high level is $\geq$ 2.5 V and the low level is $\leq$ 0.5 V.
3. Transition time t_{TLH} for EQ output is measured between 10% E and 50% E points on output waveform.
4. The lead length of electrical conductors between resistor R1 and input pin, probes to input and output pins, and load circuit to output pin shall be less than 1 inch (25.4 mm).
5. All voltages are referenced to ground.

FIGURE 19. Transition time test circuit waveforms for table XVII.

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TABLE XVIII. Propagation delay time test.

Test condition	Parameter		Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
Q1	A0 to F0	Propagation delay time (t_{PLH}), SUM mode, C_L = 50 pF	2,5,6	3,4,7,21,23, 24,25,26,27, 28	22	8	3.0	21.0	ns	20,21
Q2	A1 to F1			3,4,7,21,22, 24,25,26,27, 28	23	9				
Q3	A2 to F2			3,4,7,21,22, 23,24,26,27, 28	25	11				
Q4	A3 to F3		2,5,6	3,4,7,21,22, 23,24,25,26, 28	27	13				20,21
Q5	B0 to F0		2,5,24,26, 28	3,4,6,7,22, 23,25,27	21	8		21.0		20,22
Q6	B0 to F1					9		31.0		
Q7	B0 to F2					11		31.0		
Q8	B0 to F3	Propagation delay time (t_{PLH}), SUM mode, C_L = 50 pF	2,5,24,26, 28	3,4,6,7,22, 23,25,27	21	13	3.0	31.0		20,22
Q9	A0 to F0	Propagation delay time (t_{PLH}), SUM mode, C_L = 150 pF	2,5,6	3,4,7,21,23, 24,25,26,27, 28	22	8	4.0	23.5		20,21
Q10	A1 to F1	Propagation delay time (t_{PLH}), SUM mode, C_L = 150 pF	2,5,6	3,4,7,21,22, 24,25,26,27, 28	23	9	4.0	23.5	ns	20,21

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TABLE XVIII. Propagation delay time test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
Q11	A2 to F2	Propagation delay time (t_{PLH}), SUM mode, C_L = 150 pF	2,5,6	3,4,7,21,22, 23,24,26,27, 28	25	11	4.0	23.5	ns	20,21
Q12	A3 to F3		2,5,6	3,4,7,21,22, 23,24,25,26, 28	27	13		23.5		20,21
Q13	B0 to F0		2,5,24,26, 28	3,4,6,7,22, 23,25,27	21	8		23.5		20,22
Q14	B0 to F1					9		34.0		
Q15	B0 to F2					11		34.0		
Q16	B0 to F3	Propagation delay time (t_{PLH}), SUM mode, C_L = 150 pF	2,5,24,26, 28	3,4,6,7,22, 23,25,27	21	13	4.0	34.0		20,22
Q17	A0 to F0	Propagation delay time (t_{PLH}), DIFF mode, C_L = 50 pF	3,4,23,25, 27	2,5,6,7,21, 24,26,28	22	8	3.0	24.0		20,21
Q18	A1 to F1		3,4,22,25, 27		23	9				
Q19	A2 to F2		3,4,22,23, 27		25	11				
Q20	A3 to F3	Propagation delay time (t_{PLH}), DIFF mode, C_L = 50 pF	3,4,22,23, 25	2,5,6,7,21, 24,26,28	27	13	3.0	24.0	ns	20,21

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TABLE XVIII. Propagation delay time test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
Q21	B0 to F0	Propagation delay time (t_{PLH}), DIFF mode, C_L = 50 pF	3,4,23,24, 25,26,27, 28	2,5,6,7,22	21	8	3.0	24.0	ns	20,21
Q22	B0 to F1					9		34.0		
Q23	B0 to F2					11		34.0		
Q24	B0 to F3	Propagation delay time (t_{PLH}), DIFF mode, C_L = 50 pF	3,4,23,24, 25,26,27, 28	2,5,6,7,22	21	13	3.0	34.0		
Q25	A0 to F0	Propagation delay time (t_{PLH}), DIFF mode, C_L = 150 pF	3,4.23,25, 27	2,5,6,7,21, 24,26,28	22	8	4.0	27.5		
Q26	A1 to F1		3,4,22,25, 27		23	9				
Q27	A2 to F2		3,4,22,23, 27		25	11				
Q28	A3 to F3		3,4,22,23, 25	2,5,6,7,21, 24,26,28	27	13				
Q29	B0 to F0	Propagation delay time (t_{PLH}), DIFF mode, C_L = 150 pF	3,4,23,24, 25,26,27, 28	2,5,6,7,22	21	8	4.0	27.5	ns	20,21

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TABLE XVIII. Propagation delay time test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
Q30	B0 to F1	Propagation delay time (t_{PLH}), DIFF mode, C_L = 150 pF	3,4,23,24, 25,26,27, 28	2,5,6,7,22	21	9	4.0	37.0	ns	20,21
Q31	B0 to F2	Propagation delay time (t_{PLH}), DIFF mode, C_L = 150 pF	3,4,23,24, 25,26,27, 28	2,5,6,7,22		11	4.0	37.0		20,21
Q32	B0 to F3	Propagation delay time (t_{PLH}), DIFF mode, C_L = 150 pF	3,4,23,24, 25,26,27, 28	2,5,6,7,22		13	4.0	37.0		20,21
Q33	B0 to F0	Propagation delay time (t_{PLH}), LOGIC mode, C_L = 50 pF	3,4,7,22	2,5,6,23,24, 25,26,27,28	21	8	3.0	24.0		20,22
Q34	B1 to F1		3,4,7,23	2,5,6,21,22, 25,26,27,28	24	9				
Q35	B2 to F2		3,4,7,25	2,5,6,21,22, 23,24,27,28	26	11				
Q36	B3 to F3	Propagation delay time (t_{PLH}), LOGIC mode, C_L = 50 pF	3,4,7,27	2,5,6,21,22, 23,24,25,26	28	13	3.0	24.0		
Q37	B0 to F0	Propagation delay time (t_{PLH}), LOGIC mode, C_L = 150 pF	3,4,7,22	2,5,6,23,24, 25,26,27,28	21	8	4.0	27.5	ns	20,22

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TABLE XVIII. Propagation delay time test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
Q38	B1 to F1	Propagation delay time (t_{PLH}), LOGIC mode, C_L = 150 pF	3,4,7,23	2,5,6,21,22, 25,26,27,28	24	9	4.0	27.5	ns	20,22
Q39	B2 to F2	Propagation delay time (t_{PLH}), LOGIC mode, C_L = 150 pF	3,4,7,25	2,5,6,21,22, 23,24,27,28	26	11	4.0	27.5		
Q40	B3 to F3	Propagation delay time (t_{PLH}), LOGIC mode, C_L = 150 pF	3,4,7,27	2,5,6,21,22, 23,24,25,26	28	13	4.0	27.5		
Q41	B0 to E0	Propagation delay time (t_{PLH}), DIFF mode, C_L = 50 pF	3,4,22,23, 25,27	2,5,6,7,24, 26,28	21	12	3.0	29.0		
Q42	B1 to E0			2,5,6,7,21, 26,28	24					
Q43	B2 to E0			2,5,6,7,21, 24,28	26					
Q44	B3 to E0	Propagation delay time (t_{PLH}), DIFF mode, C_L = 50 pF		2,5,6,7,21, 24,26	28		3.0	29.0		
Q45	B0 to E0	Propagation delay time (t_{PLH}), DIFF mode, C_L = 150 pF	3,4,22,23, 25,27	2,5,6,7,24, 26,28	21	12	4.0	35.5	ns	20,22

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TABLE XVIII. Propagation delay time test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
Q46	B1 to EQ	Propagation delay time (t_{PLH}), DIFF mode, C_L = 150 pF	3,4,22,23, 25,27	2,5,6,7,21, 26,28	24	12	4.0	35.5	ns	20,22
Q47	B2 to EQ	Propagation delay time (t_{PLH}), DIFF mode, C_L = 150 pF	3,4,22,23, 25,27	2,5,6,7,21, 24,28	26	12	4.0	35.5		20,22
Q48	B3 to EQ	Propagation delay time (t_{PLH}), DIFF mode, C_L = 150 pF	3,4,22,23, 25,27	2,5,6,7,21, 24,26	28	12	4.0	35.5		20,22
Q49	A0 to CG	Propagation delay time (t_{PLH}), SUM mode, C_L = 50 pF	2,5,24,26, 28	3,4,6,7,21, 23,25,27	22	29	3.0	15.0		20,21
Q50	A1 to CG		2,5,21,26, 28	3,4,6,7,22, 24,25,27	23					
Q51	A2 to CG		2,5,21,24, 28	3,4,6,7,22, 23,26,27	25					
Q52	A3 to CG	Propagation delay time (t_{PLH}), SUM mode, C_L = 50 pF	2,5,21,24, 26	3,4,6,7,22, 23,25,28	27		3.0	15.0		
Q53	A0 to CG	Propagation delay time (t_{PLH}), SUM mode, C_L = 150 pF	2,5,24,26, 28	3,4,6,7,21, 23,25,27	22	29	4.0	17.5	ns	20,21

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TABLE XVIII. Propagation delay time test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
Q54	A1 to CG	Propagation delay time (t_{PLH}), SUM mode, C_L = 150 pF	2,5,21,26, 28	3,4,6,7,22, 24,25,27	23	29	4.0	17.5	ns	20,21
Q55	A2 to CG	Propagation delay time (t_{PLH}), SUM mode, C_L = 150 pF	2,5,21,24, 28	3,4,6,7,22, 23,26,27	25		4.0	17.5		20,21
Q56	A3 to CG	Propagation delay time (t_{PLH}), SUM mode, C_L = 150 pF	2,5,21,24, 26	3,4,6,7,22, 23,25,28	27		4.0	17.5		20,21
Q57	B0 to CG	Propagation delay time (t_{PLH}), DIFF mode, C_L = 50 pF	3,4	2,5,6,7,22, 23,24,25,26, 27,28	21		3.0	19.0		20,22
Q58	B1 to CG			2,5,6,7,21, 22,23,25,26, 27,28	24					
Q59	B2 to CG			2,5,6,7,21, 22,23,24,25, 27,28	26					
Q60	B3 to CG	Propagation delay time (t_{PLH}), DIFF mode, C_L = 50 pF		2,5,6,7,21, 22,23,24,25, 26,27	28		3.0	19.0		
Q61	B0 to CG	Propagation delay time (t_{PLH}), DIFF mode, C_L = 150 pF	3,4	2,5,6,7,22, 23,24,25,26, 27,28	21	29	4.0	21.5	ns	20,22

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TABLE XVIII. Propagation delay time test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
Q62	B1 to CG	Propagation delay time (t_{PLH}), DIFF mode, C_L = 150 pF	3,4	2,5,6,7,21, 22,23,25,26, 27,28	24	29	4.0	21.5	ns	20,22
Q63	B2 to CG	Propagation delay time (t_{PLH}), DIFF mode, C_L = 150 pF	3,4	2,5,6,7,21, 22,23,24,25, 27,28	26	29	4.0	21.5		20,22
Q64	B3 to CG	Propagation delay time (t_{PLH}), DIFF mode, C_L = 150 pF	3,4	2,5,6,7,21, 22,23,24,25, 26,27	28	29	4.0	21.5		20,22
Q65	A0 to CP	Propagation delay time (t_{PLH}), SUM mode, C_L = 50 pF	2,5,21	3,4,6,7,23, 24,25,26,27, 28	22	31	3.0	15.0		20,21
Q66	A1 to CP		2,5,24	3,4,6,7,21, 22,25,26,27, 28	23					
Q67	A2 to CP		2,5,26	3,4,6,7,21, 22,23,24,27, 28	25					
Q68	A3 to CP	Propagation delay time (t_{PLH}), SUM mode, C_L = 50 pF	2,5,28	3,4,6,7,21, 22.23,24,25, 26	27		3.0	15.0		
Q69	A0 to CP	Propagation delay time (t_{PLH}), SUM mode, C_L = 150 pF	2,5,21	3,4,6,7,23, 24,25,26,27, 28	22	31	4.0	17 5	ns	20,21

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TABLE XVIII. Propagation delay time test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
Q70	A1 to CP	Propagation delay time (t_{PLH}), SUM mode, C_L = 150 pF	2,5,24	3,4,6,7,21, 22,25,26,27, 28	23	31	4.0	17.5	ns	20,21
Q71	A2 to CP	Propagation delay time (t_{PLH}), SUM mode, C_L = 150 pF	2,5,26	3,4,6,7,21, 22,23,24,27, 28	25		4.0	17.5		20,21
Q72	A3 to CP	Propagation delay time (t_{PLH}), SUM mode, C_L = 150 pF	2,5,28	3,4,6,7,21, 22,23,24,25, 26	27		4.0	17.5		20,21
Q73	B0 to CP	Propagation delay time (t_{PLH}), DIFF mode, C_L = 50 pF	3,4,22	2,5,6,7,23, 24,25,26,27, 28	21		3.0	19.0		20,22
Q74	B1 to CP		3,4,23	2,5,6,7,21, 22,25,26,27, 28	24					
Q75	B2 to CP		3,4,25	2,5,6,7,21, 22,23,24,27, 28	26					
Q76	B3 to CP	Propagation delay time (t_{PLH}), DIFF mode, C_L = 50 pF	3,4,27	2,5,6,7,21, 22,23,24,25, 26	28		3.0	19.0		
Q77	B0 to CP	Propagation delay time (t_{PLH}), DIFF mode, C_L = 150 pF	3,4,22	2,5,6,7,23, 24,25,26,27, 28	21	31	4.0	21.5	ns	20,22

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TABLE XVIII. Propagation delay time test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
Q78	BI to CP	Propagation delay time (t_{PLH}), DIFF mode, C_L = 150 pF	3,4,23	2,5,6,7,21, 22,25,26,27, 28	24	31	4.0	21.5	ns	20,22
Q79	B2 to CP	Propagation delay time (t_{PLH}), DIFF mode, C_L = 150 pF	3,4,25	2,5,6,7,21, 22,23,24,27, 28	26	31	4.0	21.5		
Q80	B3 to CP	Propagation delay time (t_{PLH}), DIFF mode, C_L = 150 pF	3,4,27	2,5,6,7,21, 22,23,24,25, 26	28	31	4.0	21.5		
Q81	A0 to $\overline{C}_0$	Propagation delay time (t_{PLH}), SUM mode, C_L = 50 pF	2,5,24,26, 28	3,4,6,7,21, 23,25,27	22	30	3.0	26.0		
Q82	A1 to $\overline{C}_0$		2,5,21,26, 28	3,4,6,7,22, 24,25,27	23					
Q83	A2 to $\overline{C}_0$		2,5,21,24, 28	3,4,6,7,22, 23,26,27	25					
Q84	A3 to $\overline{C}_0$	Propagation delay time (t_{PLH}), SUM mode, C_L = 50 pF	2,5,21,24, 26	3,4,6,7,22, 23,25,28	27		3.0	26.0		
Q85	A0 to $\overline{C}_0$	Propagation delay time (t_{PLH}), SUM mode, C_L = 150 pF	2,5,24,26, 28	3,4,6,7,21, 23,25,27	22	30	4.0	32.5	ns	20,22

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TABLE XVIII. Propagation delay time test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
Q86	A1 to $\overline{C}_0$	Propagation delay time (t_{PLH}), SUM mode, C_L = 150 pF	2,5,21,26, 28	3,4,6,7,22, 24,25,27	23	30	4.0	32.5	ns	20,22
Q87	A2 to $\overline{C}_0$	Propagation delay time (t_{PLH}), SUM mode, C_L = 150 pF	2,5,21,24, 28	3,4,6,7,22, 23,26,27	25		4.0	32.5		20,22
Q88	A3 to $\overline{C}_0$	Propagation delay time (t_{PLH}), SUM mode, C_L = 150 pF	2,5,21,24, 26	3,4,6,7,22, 23,25,28	27		4.0	32.5		20,22
Q89	B0 to $\overline{C}_0$	Propagation delay time (t_{PLH}), DIFF mode, C_L = 50 pF	3,4	2,5,6,7,22, 23,24,25,26, 27,28	21		3.0	30.0		20,21
Q90	B1 to $\overline{C}_0$			2,5,6,7,21, 22,23,25,26, 27,28	24					
Q91	B2 to $\overline{C}_0$			2,5,6,7,21, 22,23,24,25, 27,28	26					
Q92	B3 to $\overline{C}_0$	Propagation delay time (t_{PLH}), DIFF mode, C_L = 50 pF		2,5,6,7,21, 22,23,24,25, 26,27	28		3.0	30.0		
Q93	B0 to $\overline{C}_0$	Propagation delay time (t_{PLH}), DIFF mode, C_L = 150 pF	3,4	2,5,6,7,22, 23,24,25,26, 27,28	21	30	4.0	37.5	ns	20,21

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TABLE XVIII. Propagation delay time test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
Q94	B1 to $\overline{C}_0$	Propagation delay time (t_{PLH}), DIFF mode, C_L = 150 pF	3,4	2,5,6,7,21, 22,23,25,26, 27,28	24	30	4.0	37.5	ns	20,21
Q95	B2 to $\overline{C}_0$	Propagation delay time (t_{PLH}), DIFF mode, C_L = 150 pF	3,4	2,5,6,7,21, 22,23,24,25, 27,28	26	30	4.0	37.5		
Q96	B3 to $\overline{C}_0$	Propagation delay time (t_{PLH}), DIFF mode, C_L = 150 pF	3,4	2,5,6,7,21, 22,23,24,25, 26,27	28	30	4.0	37.5		
Q97	$\overline{C}_{IN}$ to F0	Propagation delay time (t_{PLH}), C_L = 50 pF	2,5,22,23, 25,27	3,4,7,21,24, 26,28	6	8	3.0	15.0		
Q98	$\overline{C}_{IN}$ to F1					9				
Q99	$\overline{C}_{IN}$ to F2					11				
Q100	$\overline{C}_{IN}$ to F3	Propagation delay time (t_{PLH}), C_L = 50 pF				13	3.0	15.0		
Q101	$\overline{C}_{IN}$ to F0	Propagation delay time (t_{PLH}), C_L = 150 pF				8	4.0	18.5		
Q102	$\overline{C}_{IN}$ to F1	Propagation delay time (t_{PLH}), C_L = 150 pF	2,5,22,23, 25,27	3,4,7,21,24, 26,28	6	9	4.0	18.5	ns	20,21

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TABLE XVIII. Propagation delay time test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
Q103	$\overline{C}_{IN}$ to F2	Propagation delay time (t_{PLH}), C_L = 150 pF	2,5,22,23, 25,27	3,4,7,21,24, 26,28	6	11	4.0	18.5	ns	20,21
Q104	$\overline{C}_{IN}$ to F3	Propagation delay time (t_{PLH}), C_L = 150 pF				13	4.0	18.5		
Q105	$\overline{C}_{IN}$ to $\overline{C}_O$	Propagation delay time (t_{PLH}), C_L = 50 pF				30	3.0	15.0		
Q106	$\overline{C}_{IN}$ to $\overline{C}_O$	Propagation delay time (t_{PLH}), C_L = 150 pF	2,5,22,23, 25,27	3,4,7,21,24, 26,28	6	30	4.0	21.5		
Q107	S0 to F0	Propagation delay time (t_{PLH}), C_L = 50 pF	6,21,24, 26,28	2,3,4,7,22, 23,25,27	5	8	3.0	40.0		
Q108	S0 to F1		21	2,3,4,6,7, 22,23,24,25, 26,27,28		9				
Q109	S0 to F2		22,24	2,3,4,6,7, 21,23,25,26, 27,28		11				
Q110	S0 to F3		22,24,25	2,3,4,6,7, 21,23,26,27, 28		13				
Q111	S0 to F3	Propagation delay time (t_{PLH}), C_L = 50 pF	22,23,26	2,3,4,6,7, 21,24,25,27, 28	5	13	3.0	40.0	ns	20,21

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TABLE XVIII. Propagation delay time test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
Q112	S1 to F1	Propagation delay time (t_{PLH}), C_L = 50 pF	6	2,3,5,7,21, 22,23,24,25, 26,27,28	4	9	3.0	40.0	ns	20,21
Q113	S2 to F1	Propagation delay time (t_{PLH}), C_L = 50 pF	6,22	2,4,5,7,21, 23,24,25,26, 27,28	3	9	3.0	40.0		
Q114	S3 to F3	Propagation delay time (t_{PLH}), C_L = 50 pF	6,7,21,22, 23,24,25, 26,27,28	3,4,5	2	13	3.0	40.0		
Q115	S0 to F0	Propagation delay time (t_{PLH}), C_L = 150 pF	6,21,24, 26,28	2,3,4,7,22, 23,25,27	5	8	4.0	50.0		
Q116	S0 to F1		21	2,3,4,6,7, 22,23,24,25, 26,27,28		9				
Q117	S0 to F2		22,24	2,3,4,6,7, 21,23,25,26, 27,28		11				
Q118	S0 to F3		22,24,25	2,3,4,6,7, 21,23,26,27, 28		13				
Q119	S0 to F3		22,23,26	2,3,4,6,7, 21,24,25,27, 28	5	13				
Q120	S1 to F1	Propagation delay time (t_{PLH}), C_L = 150 pF	6	2,3,5,7,21, 22,23,24,25, 26,27,28	4	9	4.0	50.0	ns	20,21

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TABLE XVIII. Propagation delay time test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
Q121	S2 to F1	Propagation delay time (t_{PLH}), C_L = 150 pF	6,22	2,4,5,7,21, 23,24,25,26, 27,28	3	9	4.0	50.0	ns	20,21
Q122	S3 to F3	Propagation delay time (t_{PLH}), C_L = 150 pF	6,7,21,22, 23,24,25, 26,27,28	3,4,5	2	13	4.0	50.0		
Q123	S0 to CG	Propagation delay time (t_{PLH}), C_L = 50 pF	22,23,25, 28	2,3,4,6,7, 21,24,26,27	5	29	3.0	40.0		
Q124	S2 to CG		21,22,23, 24,27	2,4,5,6,7, 25,26,28	3	29				
Q125	S2 to CP		22,23,24, 25,26,27, 28	2,4,5,6,7, 21		31				
Q126	S2 to CP		21,22,23, 25,26,27, 28	2,4,5,6,7, 24		31				20,21
Q127	S2 to $\overline{C}_0$		21,22,25, 27,28	2,4,5,6,7, 23,24,26		30				20,22
Q128	S2 to $\overline{C}_0$		21,23,25, 26,27,28	2,4,5,6,7, 22,24	3	30				
Q129	M to F1		6,22,23, 25,27	2,3,4,5,21, 24,26,28	7	9				
Q130	M to F3	Propagation delay time (t_{PLH}), C_L = 50 pF	6,22,23, 25,27	2,3,4,5,21, 24,26,28	7	13	3.0	40.0	ns	20,22

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TABLE XVIII. Propagation delay time test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
Q131	S0 to CG	Propagation delay time (t_{PLH}), C_L = 150 pF	22,23,25, 28	2,3,4,6,7, 21,24,26,27	5	29	4.0	50.0	ns	20,21
Q132	S2 to CG		21,22,23, 24,27	2,4,5,6,7, 25,26,28	3	29				
Q133	S2 to CP		22,23,24, 25,26,27, 28	2,4,5,6,7, 21		31				
Q134	S2 to CP		21,22,23, 25,26,27, 28	2,4,5,6,7, 24		31				20,21
Q135	S2 to $\overline{C}_0$		21,22,25, 27,28	2,4,5,6,7, 23,24,26		30				20,22
Q136	S2 to $\overline{C}_0$		21,23,25, 26,27,28	2,4,5,6,7, 22,24	3	30				
Q137	M to F1		6,22,23, 25,27	2,3,4,5,21, 24,26,28	7	9				
Q138	M to F3	Propagation delay time (t_{PLH}), C_L = 150 pF	6,22,23, 25,27	2,3,4,5,21, 24,26,28	7	13	4.0	50.0		
Q139	$\overline{P}0$ to C_{N+X}	Propagation delay time (t_{PLH}), C_L = 50 pF	15,38	Open	35	19	1.0	8.5		
Q140	$\overline{P}1$ to C_{N+Y}	Propagation delay time (t_{PLH}), C_L = 50 pF	14,38	15	34	39	1.0	8.5	ns	20,22

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TABLE XVIII. Propagation delay time test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
Q141	$\overline{P}2$ to C_{N+Z}	Propagation delay time (t_{PLH}), C_L = 50 pF	18,38	14,15	37	20	1.0	8.5	ns	20,22
Q142	$\overline{G}0$ to C_{N+X}		Open	35,38	15	19				
Q143	$\overline{G}1$ to C_{N+Y}		15	34,35,38	14	39				
Q144	$\overline{G}2$ to C_{N+Z}		14,15	34,35,37,38	18	20		8.5		20,22
Q145	$\overline{P}0$ to $\overline{P}$		Open	34,36,37	35	17		10.5		20,21
Q146	$\overline{P}1$ to $\overline{P}$			35,36,37	34					
Q147	$\overline{P}2$ to $\overline{P}$			34,35,36	37					
Q148	$\overline{P}3$ to $\overline{P}$		Open	34,35,37	36	17				
Q149	$\overline{P}1$ to $\overline{G}$		14,16,18	15,36,37	34	40				
Q150	$\overline{P}2$ to $\overline{G}$		14,16,18	15,34,36	37					
Q151	$\overline{P}3$ to $\overline{G}$		14,16,18	15,34,37	36					
Q152	$\overline{G}3$ to $\overline{G}$		14,15,18	34,36,37	16	40		10.5		20,21
Q153	$\overline{G}1$ to C_{N+Y}		15,18,35	34,36,37	14	39		8.5		20,22
Q154	$\overline{G}2$ to C_{N+Z}		14,15,35	34,36,37	18	20		8.5		20,22
Q155	$\overline{G}2$ to C_{N+Z}		14,15,34	35,36,37	18	20		8.5		20,22
Q156	C_N to C_{N+X}	Propagation delay time (t_{PLH}), C_L = 50 pF	14,15,16, 18	34,35,36,37	38	19	1.0	11.0	ns	20,21

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TABLE XVIII. Propagation delay time test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
Q157	C_N to C_{N+Y}	Propagation delay time (t_{PLH}), C_L = 50 pF	14,15,16, 18	34,35,36,37	38	39	1.0	11.0	ns	20,21
Q158	C_N to C_{N+Z}	Propagation delay time (t_{PLH}), C_L = 50 pF	14,15,16, 18	34,35,36,37	38	20	1.0	11.0		20,21
Q159	$\overline{P}0$ to C_{N+X}	Propagation delay time (t_{PLH}), C_L = 150 pF	15,38	Open	35	19	3.0	12.5		20,22
Q160	$\overline{P}1$ to C_{N+Y}		14,38	15	34	39				
Q161	$\overline{P}2$ to C_{N+Z}		18,38	14,15	37	20				
Q162	$\overline{G}0$ to C_{N+X}		Open	35,38	15	19				
Q163	$\overline{G}1$ to C_{N+Y}		15	34,35,38	14	39				
Q164	$\overline{G}2$ to C_{N+Z}		14,15	34,35,37,38	18	20		12.5		20,22
Q165	$\overline{P}0$ to $\overline{P}$		Open	34,36,37	35	17		13.0		20,21
Q166	$\overline{P}1$ to $\overline{P}$			35,36,37	34					
Q167	$\overline{P}2$ to $\overline{P}$			34,35,36	37					
Q168	$\overline{P}3$ to $\overline{P}$		Open	34,35,37	36	17				
Q169	$\overline{P}1$ to $\overline{G}$	Propagation delay time (t_{PLH}), C_L = 150 pF	14,16,18	15,36,37	34	40	3.0	13.0	ns	20,21

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TABLE XVIII. Propagation delay time test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
Q170	$\overline{P}2$ to $\overline{G}$	Propagation delay time (t_{PLH}), C_L = 150 pF	14,16,18	15,34,36	37	40	3.0	13.0	ns	20,21
Q171	$\overline{P}3$ to $\overline{G}$		14,16,18	15,34,37	36	40		13.0		20,21
Q172	$\overline{G}3$ to $\overline{G}$		14,15,18	34,36,37	16	40		13.0		20,21
Q173	$\overline{G}1$ to C_{N+Y}		15,18,35	34,36,37	14	39		12.5		20,22
Q174	$\overline{G}2$ to C_{N+Z}		14,15,35	34,36,37	18	20		12.5		20,22
Q175	$\overline{G}2$ to C_{N+Z}		14,15,34	35,36,37	18	20		12.5		20,22
Q176	C_N to C_{N+X}		14,15,16, 18	34,35,36,37	38	19		16.0		20,21
Q177	C_N to C_{N+Y}		14,15,16, 18	34,35,36,37	38	39		16.0		
Q178	C_N to C_{N+Z}	Propagation delay time (t_{PLH}), C_L = 150 pF	14,15,16, 18	34,35,36,37	38	20		16.0		
R1	A0 to F0	Propagation delay time (t_{PHL}), SUM mode, C_L = 50 pF	2,5,6	3,4,7,21,23, 24,25,26,27, 28	22	8		21.0		
R2	A1 to F1	Propagation delay time (t_{PHL}), SUM mode, C_L = 50 pF	2,5,6	3,4,7,21,22, 24,25,26,27, 28	23	9	3.0	21.0	ns	20,21

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TABLE XVIII. Propagation delay time test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
R3	A2 to F2	Propagation delay time (t_{PHL}), SUM mode, C_L = 50 pF	2,5,6	3,4,7,21,22, 23,24,26,27, 28	25	11	3.0	21.0	ns	20,21
R4	A3 to F3		2,5,6	3,4,7,21,22, 23,24,25,26, 28	27	13		21.0		20,21
R5	B0 to F0		2,5,24,26, 28	3,4,6,7,22, 23,25,27	21	8		21.0		20,22
R6	B0 to F1					9		31.0		
R7	B0 to F2					11		31.0		
R8	B0 to F3	Propagation delay time (t_{PHL}), SUM mode, C_L = 50 pF	2,5,24,26, 28	3,4,6,7,22, 23,25,27	21	13	3.0	31.0		20,22
R9	A0 to F0	Propagation delay time (t_{PHL}), SUM mode, C_L = 150 pF	2,5,6	3,4,7,21,23, 24,25,26,27, 28	22	8	4.0	23.5		20,21
R10	A1 to F1			3,4,7,21,22, 24,25,26,27, 28	23	9				
R11	A2 to F2			3,4,7,21,22, 23,24,26,27, 28	25	11				
R12	A3 to F3	Propagation delay time (t_{PHL}), SUM mode, C_L = 150 pF	2,5,6	3,4,7,21,22, 23,24,25,26, 28	27	13	4.0	23.5	ns	20,21

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TABLE XVIII. Propagation delay time test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
R13	B0 to F0	Propagation delay time (t_{PHL}), SUM mode, C_L = 150 pF	2,5,24,26, 28	3,4,6,7,22, 23,25,27	21	8	4.0	23.5	ns	20,22
R14	B0 to F1					9		34.0		
R15	B0 to F2					11		34.0		
R16	B0 to F3	Propagation delay time (t_{PHL}), SUM mode, C_L = 150 pF	2,5,24,26, 28	3,4,6,7,22, 23,25,27	21	13	4.0	34.0		20,22
R17	A0 to F0	Propagation delay time (t_{PHL}), DIFF mode, C_L = 50 pF	3,4,23,25, 27	2,5,6,7,21, 24,26,28	22	8	3.0	26.0		20,21
R18	A1 to F1		3,4,22,25, 27		23	9				
R19	A2 to F2		3,4,22,23, 27		25	11				
R20	A3 to F3		3,4,22,23, 25	2,5,6,7,21, 24,26,28	27	13				
R21	B0 to F0		3,4,23,24, 25,26,27, 28	2,5,6,7,22	21	8		26.0		
R22	B0 to F1	Propagation delay time (t_{PHL}), DIFF mode, C_L = 50 pF	3,4,23,24, 25,26,27, 28	2,5,6,7,22	21	9	3.0	36.0	ns	20,21

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TABLE XVIII. Propagation delay time test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
R23	B0 to F2	Propagation delay time (t_{PHL}), DIFF mode, C_L = 50 pF	3,4,23,24, 25,26,27, 28	2,5,6,7,22	21	11	3.0	36.0	ns	20,21
R24	B0 to F3	Propagation delay time (t_{PHL}), DIFF mode, C_L = 50 pF	3,4,23,24, 25,26,27, 28	2,5,6,7,22	21	13	3.0	36.0		
R25	A0 to F0	Propagation delay time (t_{PHL}), DIFF mode, C_L = 150 pF	3,4,23,25, 27	2,5,6,7,21, 24,26,28	22	8	4.0	29.5		
R26	A1 to F1		3,4,22,25, 27		23	9				
R27	A2 to F2		3,4,22,23, 27		25	11				
R28	A3 to F3		3,4,22,23, 25	2,5,6,7,21, 24,26,28	27	13				
R29	B0 to F0		3,4,23,24, 25,26,27, 28	2,5,6,7,22	21	8		29.5		
R30	B0 to F1					9		39.0		
R31	B0 to F2					11		39.0		
R32	B0 to F3	Propagation delay time (t_{PHL}), DIFF mode, C_L = 150 pF	3,4,23,24, 25,26,27, 28	2,5,6,7,22	21	13	4.0	39.0	ns	20,21

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TABLE XVIII. Propagation delay time test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
R33	B0 to F0	Propagation delay time (t_{PHL}), LOGIC mode, C_L = 50 pF	3,4,7,22	2,5,6.23,24, 25,26.27,28	21	8	3.0	26.0	ns	20,22
R34	B1 to F1		3,4,7,23	2,5,6.21,22, 25,26.27,28	24	9				
R35	B2 to F2		3,4,7,25	2,5,6.21,22, 23,24.27,28	26	11				
R36	B3 to F3	Propagation delay time (t_{PHL}), LOGIC mode, C_L = 50 pF	3,4,7,27	2,5,6,21,22, 23,24,25,26	28	13	3.0	26.0		
R37	B0 to F0	Propagation delay time (t_{PHL}), LOGIC mode, C_L = 150 pF	3,4,7,22	2,5,6.23,24, 25,26.27,28	21	8	4.0	29.5		
R38	B1 to F1		3,4,7,23	2,5,6,21,22, 25,26.27,28	24	9				
R39	B2 to F2		3,4,7,25	2,5,6.21,22, 23,24.27,28	26	11				
R40	B3 to F3	Propagation delay time (t_{PHL}), LOGIC mode, C_L = 150 pF	3,4,7,27	2,5,6.21,22, 23,24.25,26	28	13	4.0	29.5		
R41	B0 to EQ	Propagation delay time (t_{PHL}), DIFF mode, C_L = 50 pF	3,4,22,23, 25,27	2,5,6.7,24, 26,28	21	12	3.0	40.0	ns	20,22

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TABLE XVIII. Propagation delay time test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
R42	B1 to EQ	Propagation delay time (t_{PHL}), DIFF mode, C_L = 50 pF	3,4,22,23, 25,27	2,5,6,7,21, 26,28	24	12	3.0	40.0	ns	20,22
R43	B2 to EQ	Propagation delay time (t_{PHL}), DIFF mode, C_L = 50 pF		2,5,6,7,21, 24,28	26		3.0	40.0		
R44	B3 to EQ	Propagation delay time (t_{PHL}), DIFF mode, C_L = 50 pF		2,5,6,7,21, 24,26	28		3.0	40.0		
R45	B0 to EQ	Propagation delay time (t_{PHL}), DIFF mode, C_L = 150 pF		2,5,6,7,24, 26,28	21		4.0	44.5		
R46	B1 to EQ			2,5,6,7,21, 26,28	24					
R47	B2 to EQ			2,5,6,7,21, 24,28	26					
R48	B3 to EQ	Propagation delay time (t_{PHL}), DIFF mode, C_L = 150 pF	3,4,22,23, 25,27	2,5,6,7,21, 24,26	28	12	4.0	44.5		20,22
R49	A0 to CG	Propagation delay time (t_{PHL}), SUM mode, C_L = 50 pF	2,5,24,26, 28	3,4,6,7,21, 23,25,27	22	29	3.0	15.0	ns	20,21

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TABLE XVIII. Propagation delay time test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
R50	A1 to CG	Propagation delay time (t_{PHL}), SUM mode, C_L = 50 pF	2,5,21,26, 28	3,4,6,7,22, 24,25,27	23	29	3.0	15.0	ns	20,21
R51	A2 to CG	Propagation delay time (t_{PHL}), SUM mode, C_L = 50 pF	2,5,21,24, 28	3,4,6,7,22, 23,26,27	25		3.0	15.0		
R52	A3 to CG	Propagation delay time (t_{PHL}), SUM mode, C_L = 50 pF	2,5,21,24, 26	3,4,6,7,22, 23,25,28	27		3.0	15.0		
R53	A0 to CG	Propagation delay time (t_{PHL}), SUM mode, C_L = 150 pF	2,5,24,26, 28	3,4,6,7,21, 23,25,27	22		4.0	17.5		
R54	A1 to CG		2,5,21,26, 28	3,4,6,7,22, 24,25,27	23					
R55	A2 to CG		2,5,21,24, 28	3,4,6,7,22, 23,26,27	25					
R56	A3 to CG	Propagation delay time (t_{PHL}), SUM mode, C_L = 150 pF	2,5,21,24, 26	3,4,6,7,22, 23,25,28	27		4.0	17.5		20,21
R57	B0 to CG	Propagation delay time (t_{PHL}), DIFF mode, C_L = 50 pF	3,4	2,5,6,7,22, 23,24,25,26, 27,28	21	29	3.0	19.0	ns	20,22

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TABLE XVIII. Propagation delay time test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
R58	B1 to CG	Propagation delay time (t_{PHL}), DIFF mode, C_L = 50 pF	3,4	2,5,6,7,21, 22,23,25,26, 27,28	24	29	3.0	19.0	ns	20,22
R59	B2 to CG	Propagation delay time (t_{PHL}), DIFF mode, C_L = 50 pF		2,5,6,7,21, 22,23,24,25, 27,28	26		3.0	19.0		
R60	B3 to CG	Propagation delay time (t_{PHL}), DIFF mode, C_L = 50 pF		2,5,6,7,21, 22,23,24,25, 26,27	28		3.0	19.0		
R61	B0 to CG	Propagation delay time (t_{PHL}), DIFF mode, C_L = 150 pF		2,5,6,7,22, 23,24,25,26, 27,28	21		4.0	21.5		
R62	B1 to CG			2,5,6,7,21, 22,23,25,26, 27,28	24					
R63	B2 to CG			2,5,6,7,21, 22,23,24,25, 27,28	26					
R64	B3 to CG	Propagation delay time (t_{PHL}), DIFF mode, C_L = 150 pF	3,4	2,5,6,7,21, 22,23,24,25, 26,27	28	29	4.0	21.5		20,22
R65	A0 to CP	Propagation delay time (t_{PHL}), SUM mode, C_L = 50 pF	2,5,21	3,4,6,7,23, 24,25,26,27, 28	22	31	3.0	15.0	ns	20,21

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TABLE XVIII. Propagation delay time test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
R66	A1 to CP	Propagation delay time (t_{PHL}), SUM mode, C_L = 50 pF	2,5,24	3,4,6,7,21, 22,25,26,27, 28	23	31	3.0	15.0	ns	20,21
R67	A2 to CP	Propagation delay time (t_{PHL}), SUM mode, C_L = 50 pF	2,5,26	3,4,6,7,21, 22,23,24,27, 28	25		3.0	15.0		
R68	A3 to CP	Propagation delay time (t_{PHL}), SUM mode, C_L = 50 pF	2,5,28	3,4,6,7,21, 22,23,24,25, 26	27		3.0	15.0		
R69	A0 to CP	Propagation delay time (t_{PHL}), SUM mode, C_L = 150 pF	2,5,21	3,4,6,7,23, 24,25,26,27, 28	22		4.0	17.5		
R70	A1 to CP		2,5,24	3,4,6,7,21, 22,25,26,27, 28	23					
R71	A2 to CP		2,5,26	3,4,6,7,21, 22,23,24,27, 28	25					
R72	A3 to CP	Propagation delay time (t_{PHL}), SUM mode, C_L = 150 pF	2,5,28	3,4,6,7,21, 22,23,24,25, 26	27		4.0	17.5		20,21
R73	B0 to CP	Propagation delay time (t_{PHL}), DIFF mode, C_L = 50 pF	3,4,22	2,5,6,7,23, 24,25,26,27, 28	21	31	3.0	19.0	ns	20,22

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TABLE XVIII. Propagation delay time test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
R74	B1 to CP	Propagation delay time (t_{PHL}), DIFF mode, C_L = 50 pF	3,4,23	2,5,6,7,21, 22,25,26,27, 28	24	31	3.0	19.0	ns	20,22
R75	B2 to CP	Propagation delay time (t_{PHL}), DIFF mode, C_L = 50 pF	3,4,25	2,5,6,7,21, 22,23,24,27, 28	26		3.0	19.0		
R76	B3 to CP	Propagation delay time (t_{PHL}), DIFF mode, C_L = 50 pF	3,4,27	2,5,6,7,21, 22,23,24,25, 26	28		3.0	19.0		
R77	B0 to CP	Propagation delay time (t_{PHL}), DIFF mode, C_L = 150 pF	3,4,22	2,5,6,7,23, 24,25,26,27, 28	21		4.0	21.5		
R78	B1 to CP		3,4,23	2,5,6,7,21, 22,25,26,27, 28	24					
R79	B2 to CP		3,4,25	2,5,6,7,21, 22,23,24,27, 28	26					
R80	B3 to CP	Propagation delay time (t_{PHL}), DIFF mode, C_L = 150 pF	3,4,27	2,5,6,7,21, 22,23,24,25, 26	28	31	4.0	21.5		
R81	A0 to $\overline{C}_0$	Propagation delay time (t_{PHL}), SUM mode, C_L = 50 pF	2,5,24,26, 28	3,4,6,7,21, 23,25,27	22	30	3.0	26.0	ns	20,22

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TABLE XVIII. Propagation delay time test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
R82	A1 to $\overline{C}_0$	Propagation delay time (t_{PHL}), SUM mode, C_L = 50 pF	2,5,21,26, 28	3,4,6,7,22, 24,25,27	23	30	3.0	26.0	ns	20,22
R83	A2 to $\overline{C}_0$	Propagation delay time (t_{PHL}), SUM mode, C_L = 50 pF	2,5,21,24, 28	3,4,6,7,22, 23,26,27	25		3.0	26.0		
R84	A3 to $\overline{C}_0$	Propagation delay time (t_{PHL}), SUM mode, C_L = 50 pF	2,5,21,24, 26	3,4,6,7,22, 23,25,28	27		3.0	26.0		
R85	A0 to $\overline{C}_0$	Propagation delay time (t_{PHL}), SUM mode, C_L = 150 pF	2,5,24,26, 28	3,4,6,7,21, 23,25,27	22		4.0	32.5		
R86	A1 to $\overline{C}_0$		2,5,21,26, 28	3,4,6,7,22, 24,25,27	23					
R87	A2 to $\overline{C}_0$		2,5,21,24, 28	3,4,6,7,22, 23,26,27	25					
R88	A3 to $\overline{C}_0$	Propagation delay time (t_{PHL}), SUM mode, C_L = 150 pF	2,5,21,24, 26	3,4,6,7,22, 23,25,28	27		4.0	32.5		20,22
R89	B0 to $\overline{C}_0$	Propagation delay time (t_{PHL}), DIFF mode, C_L = 50 pF	3,4	2,5,6,7,22, 23,24,25,26, 27,28	21	30	3.0	30.0	ns	20,21

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TABLE XVIII. Propagation delay time test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
R90	B1 to $\overline{C}_0$	Propagation delay time (t_{PHL}), DIFF mode, C_L = 50 pF	3,4	2,5,6,7,21, 22,23,25,26, 27,28	24	30	3.0	30.0	ns	20,21
R91	B2 to $\overline{C}_0$	Propagation delay time (t_{PHL}), DIFF mode, C_L = 50 pF		2,5,6,7,21, 22,23,24,25, 27,28	26		3.0	30.0		
R92	B3 to $\overline{C}_0$	Propagation delay time (t_{PHL}), DIFF mode, C_L = 50 pF		2,5,6,7,21, 22,23,24,25, 26,27	28		3.0	30.0		
R93	B0 to $\overline{C}_0$	Propagation delay time (t_{PHL}), DIFF mode, C_L = 150 pF		2,5,6,7,22, 23,24,25,26, 27,28	21		4.0	37.5		
R94	B1 to $\overline{C}_0$			2,5,6,7,21, 22,23,25,26, 27,28	24					
R95	B2 to $\overline{C}_0$			2,5,6,7,21, 22,23,24,25, 27,28	26					
R96	B3 to $\overline{C}_0$	Propagation delay time (t_{PHL}), DIFF mode, C_L = 150 pF	3,4	2,5,6,7,21, 22,23,24,25, 26,27	28	30	4.0	37.5		
R97	$\overline{C}_{IN}$ to FO	Propagation delay time (t_{PHL}), C_L = 50 pF	2,5,22,23, 25,27	3,4,7,21,24, 26,28	6	8	3.0	15.0	ns	20,21

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TABLE XVIII. Propagation delay time test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
R98	$\overline{C}_{IN}$ to F1	Propagation delay time (t_{PHL}), C_L = 50 pF	2,5,22,23, 25,27	3,4,7,21,24, 26,28	6	9	3.0	15.0	ns	20,21
R99	$\overline{C}_{IN}$ to F2	Propagation delay time (t_{PHL}), C_L = 50 pF				11	3.0	15.0		
R100	$\overline{C}_{IN}$ to F3	Propagation delay time (t_{PHL}), C_L = 50 pF				13	3.0	15.0		
R101	$\overline{C}_{IN}$ to F0	Propagation delay time (t_{PHL}), C_L = 150 pF				8	4.0	18.5		
R102	$\overline{C}_{IN}$ to F1					9				
R103	$\overline{C}_{IN}$ to F2					11				
R104	$\overline{C}_{IN}$ to F3	Propagation delay time (t_{PHL}), C_L = 150 pF				13	4.0	18.5		
R105	$\overline{C}_{IN}$ to $\overline{C}_O$	Propagation delay time (t_{PHL}), C_L = 50 pF				30	3.0	15.0		
R106	$\overline{C}_{IN}$ to $\overline{C}_O$	Propagation delay time (t_{PHL}), C_L = 150 pF	2,5,22,23, 25,27	3,4,7,21,24, 26,28	6	30	4.0	21.5		
R107	S0 to F0	Propagation delay time (t_{PHL}), C_L = 50 pF	6,21,24, 26,28	2,3,4,7,22, 23,25,27	5	8	3.0	40.0	ns	20,21

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TABLE XVIII. Propagation delay time test - Continued.

Test condition	Parameter		Connect				Initial limits 25° +5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
R108	S0 to F1	Propagation delay time (t_{PHL}), C_L = 50 pF	21	2,3,4,6,7, 22,23,24,25, 26,27,28	5	9	3.0	40.0	ns	20,21
R109	S0 to F2		22,24	2,3,4,6,7, 21,23,25,26, 27,28		11				
R110	S0 to F3		22,24,25	2,3,4,6,7, 21,23,26,27, 28		13				
R111	S0 to F3		22,23,26	2,3,4,6,7, 21,24,25,27, 28	5	13				
R112	S1 to F1		6	2,3,5,7,21, 22,23,24,25, 26,27,28	4	9				
R113	S2 to F1		6,22	2,4,5,7,21, 23,24,25,26, 27,28	3	9				
R114	S3 to F3	Propagation delay time (t_{PHL}), C_L = 50 pF	6,7,21,22, 23,24,25, 26,27,28	3,4,5	2	13	3.0	40.0		
R115	S0 to F0	Propagation delay time (t_{PHL}), C_L = 150 pF	6,21,24, 26,28	2,3,4,7,22, 23,25,27	5	8	4.0	50.0		
R116	S0 to F1	Propagation delay time (t_{PHL}), C_L = 150 pF	21	2,3,4,6,7, 22,23,24,25, 26,27,28	5	9	4.0	50.0	ns	20,21

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TABLE XVIII. Propagation delay time test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
R117	S0 to F2	Propagation delay time (t_{PHL}), C_L = 150 pF	22,24	2,3,4,6,7, 21,23,25,26, 27,28	5	11	4.0	50.0	ns	20,21
R118	S0 to F3		22,24,25	2,3,4,6,7, 21,23,26,27, 28	5	13				
R119	S0 to F3		22,23,26	2,3,4,6,7, 21,24,25,27, 28	5	13				
R120	S1 to F1		6	2,3,5,7,21, 22,23,24,25, 26,27,28	4	9				
R121	S2 to F1		6,22	2,4,5,7,21, 23,24,25,26, 27,28	3	9				
R122	S3 to F3	Propagation delay time (t_{PHL}), C_L = 150 pF	6,7,21,22, 23,24,25, 26,27,28	3,4,5	2	13	4.0	50.0		
R123	S0 to CG	Propagation delay time (t_{PHL}), C_L = 50 pF	22,23,25, 28	2,3,4,6,7, 21,24,26,27	5	29	3.0	40.0		
R124	S2 to CG	Propagation delay time (t_{PHL}), C_L = 50 pF	21,22,23, 24,27	2,4,5,6,7, 25,26,28	3	29	3.0	40.0		
R125	S2 to CP	Propagation delay time (t_{PHL}), C_L = 50 pF	22,23,24, 25,26,27, 28	2,4,5,6,7, 21	3	31	3.0	40.0	ns	20,21

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TABLE XVIII. Propagation delay time test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
R126	S2 to CP	Propagation delay time (t_{PHL}), C_L = 50 pF	21,22,23, 25,26,27, 28	2,4,5,6,7,24	3	31	3.0	40.0	ns	20,21
R127	S2 to $\overline{C}_0$		21,22,25, 27,28	2,4,5,6,7, 23,24,26	3	30				20,22
R128	S2 to $\overline{C}_0$		21,23,25, 26,27,28	2,4,5,6,7, 22,24	3	30				
R129	M to F1		6,22,23, 25,27	2,3,4,5,21, 24,26,28	7	9				
R130	M to F3	Propagation delay time (t_{PHL}), C_L = 50 pF	6,22,23, 25,27	2,3,4,5,21, 24,26,28	7	13	3.0	40.0		20,22
R131	S0 to CG	Propagation delay time (t_{PHL}), C_L = 150 pF	22,23,25, 28	2,3,4,6,7, 21,24,26,27	5	29	4.0	50.0		20,21
R132	S2 to CG		21,22,23, 24,27	2,4,5,6,7, 25,26,28	3	29				
R133	S2 to CP		22,23,24, 25,26,27, 28	2,4,5,6,7,21		31				
R134	S2 to CP		21,22,23, 25,26,27, 28	2,4,5,6,7,24		31				20,21
R135	S2 to $\overline{C}_0$		21,22,25, 27,28	2,4,5,6,7, 23,24,26		30				20,22
R136	S2 to $\overline{C}_0$	Propagation delay time (t_{PHL}), C_L = 150 pF	21,23,25, 26,27,28	2,4,5,6,7, 22,24	3	30	4.0	50.0	ns	20,22

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TABLE XVIII. Propagation delay time test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
R137	M to F1	Propagation delay time (t_{PHL}), C_L = 150 pF	6,22,23, 25,27	2,3,4,5,21, 24,26,28	7	9	4.0	50.0	ns	20,22
R138	M to F3	Propagation delay time (t_{PHL}), C_L = 150 pF	6,22,23, 25,27	2,3,4,5,21, 24,26,28	7	13	4.0	50.0		
R139	$\overline{P}0$ to C_{N+X}	Propagation delay time (t_{PHL}), C_L = 50 pF	15,38	Open	35	19	1.0	10.5		
R140	$\overline{P}1$ to C_{N+Y}		14,38	15	34	39				
R141	$\overline{P}2$ to C_{N+Z}		18,38	14,15	37	20				
R142	$\overline{G}0$ to C_{N+X}		Open	35,38	15	19				
R143	$\overline{G}1$ to C_{N+Y}		15	34,35,38	14	39				
R144	$\overline{G}2$ to C_{N+Z}		14,15	34,35,37,38	18	20		10 5		20,22
R145	$\overline{P}0$ to $\overline{P}$		Open	34,36,37	35	17		12.5		20,21
R146	$\overline{P}1$ to $\overline{P}$			35,36,37	34					
R147	$\overline{P}2$ to $\overline{P}$			34,35,36	37					
R148	$\overline{P}3$ to $\overline{P}$		Open	34,35,37	36	17				
R149	$\overline{P}1$ to $\overline{G}$	Propagation delay time (t_{PHL}), C_L = 50 pF	14,16,18	15,36,37	34	40	1.0	12.5	ns	20,21

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TABLE XVIII Propagation delay time test - Continued.

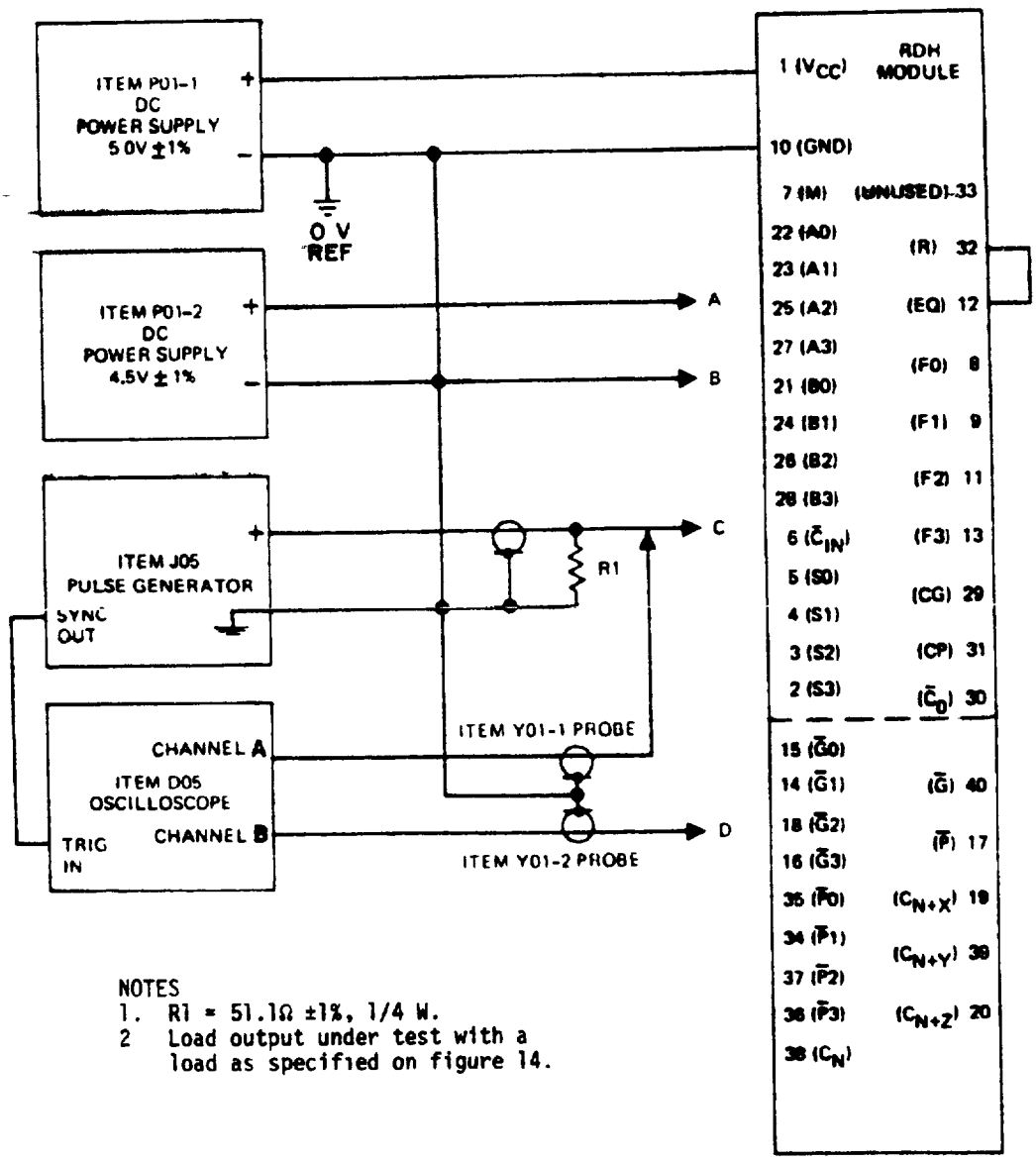
Test condition	Parameter		Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
R150	$\overline{P}2$ to $\overline{G}$	Propagation delay time (t_{PHL}), C_L = 50 pF	14,16,18	15,34,36	37	40	1.0	12.5	ns	20,21
R151	$\overline{P}3$ to $\overline{G}$		14,16,18	15,34,37	36	40		12.5		20,21
R152	$\overline{G}3$ to $\overline{G}$		14,15,18	34,36,37	16	40		12.5		20,21
R153	$\overline{G}1$ to C_{N+Y}		15,18,35	34,36,37	14	39		10.5		20,22
R154	$\overline{G}2$ to C_{N+Z}		14,15,35	34,36,37	18	20		10.5		20,22
R155	$\overline{G}2$ to C_{N+Z}		14,15,34	35,36,37	18	20		10.5		20,22
R156	C_N to C_{N+X}		14,15,16, 18	34,35,36,37	38	19		14.5		20,21
R157	C_N to C_{N+Y}		14,15,16, 18	34,35,36,37	38	39		14.5		20,21
R158	C_N to C_{N+Z}	Propagation delay time (t_{PHL}), C_L = 50 pF	14,15,16, 18	34,35,36,37	38	20	1.0	14.5		20,21
R159	$\overline{P}0$ to C_{N+X}	Propagation delay time (t_{PHL}), C_L = 150 pF	15,38	Open	35	19	3.0	15.5		20,22
R160	$\overline{P}1$ to C_{N+Y}		14,38	15	34	39				
R161	$\overline{P}2$ to C_{N+Z}		18,38	14,15	37	20				
R162	$\overline{G}0$ to C_{N+X}	Propagation delay time (t_{PHL}), C_L = 150 pF	Open	35,38	15	19	3.0	15.5	ns	20,22

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TABLE XVIII. Propagation delay time test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
R163	$\overline{G}1$ to C_{N+Y}	Propagation delay time (t_{PHL}), C_L = 150 pF	15	34,35,38	14	39	3.0	15.5	ns	20,22
R164	$\overline{G}2$ to C_{N+Z}		14,15	34,35,37,38	18	20		15.5		20,22
R165	$\overline{P}0$ to $\overline{P}$		Open	34,36,37	35	17		17.0		20,21
R166	$\overline{P}1$ to $\overline{P}$			35,36,37	34					
R167	$\overline{P}2$ to $\overline{P}$			34,35,36	37					
R168	$\overline{P}3$ to $\overline{P}$		Open	34,35,37	36	17				
R169	$\overline{P}1$ to $\overline{G}$		14,16,18	15,36,37	34	40				
R170	$\overline{P}2$ to $\overline{G}$		14,16,18	15,34,36	37					
R171	$\overline{P}3$ to $\overline{G}$		14,16,18	15,34,37	36					
R172	$\overline{G}3$ to $\overline{G}$		14,15,18	34,36,37	16	40		17.0		20,21
R173	$\overline{G}1$ to C_{N+Y}		15,18,35	34,36,37	14	39		15.5		20,22
R174	$\overline{G}2$ to C_{N+Z}		14,15,35	34,36,37	18	20		15.5		20,22
R175	$\overline{G}2$ to C_{N+Z}		14,15,34	35,36,37	18	20		15.5		20,22
R176	C_N to C_{N+X}		14,15,16, 18	34,35,36,37	38	19		18.0		20,21
R177	C_N to C_{N+Y}		14,15,16, 18	34,35,36,37	38	39		18.0		20,21
R178	C_N to C_{N+Z}	Propagation delay time (t_{PHL}), C_L = 150 pF	14,15,16, 18	34,35,36,37	38	20	3.0	18.0	ns	20,21

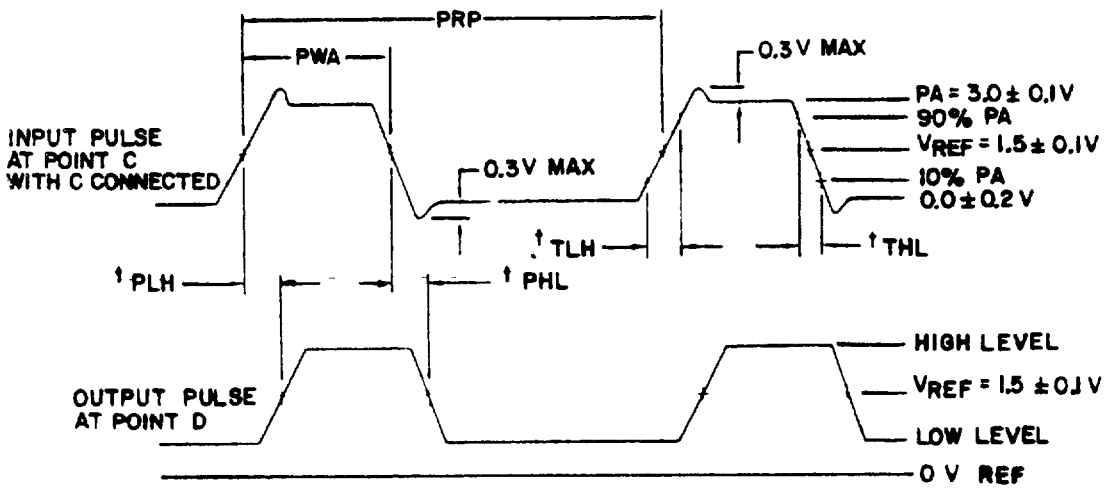
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NOTES
1. R1 = 51.1Ω ±1%, 1/4 W.
2 Load output under test with a load as specified on figure 14.

FIGURE 20 Propagation delay time test circuit for table XVIII

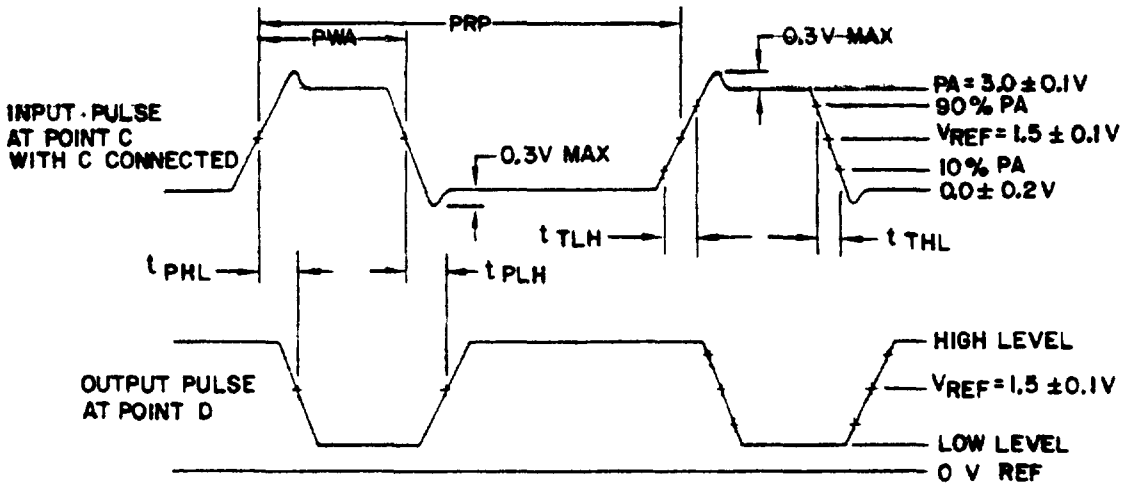
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NOTES:
1. Input pulse parameters.
   $t_{TLH} = t_{THL} = 4 \pm 1$ ns
   PWA = 100 ±10 ns.
   PRP = 1,000 ±100 ns.
2. Before making measurements specified in table XVIII, verify that the output pulse high level is $\geq$ 2.5 V and the low level is $\leq$ 0.5 V.
3. The lead length of electrical conductors between resistor R1 and input pin, between probes and input and output pins, and between load circuit and output pin, shall be less than 1 inch (25.4 mm)
4. All voltages are referenced to ground.

FIGURE 21. Propagation delay time test circuit waveforms for table XVIII.

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NOTES
1. Input pulse parameters:
   $t_{TLH} = t_{THL} = 4 \pm 1$ ns.
   PWA = 100 ±10 ns.
   PRP = 1,000 ±100 ns
2. Before making measurements specified in table XVIII, verify that the output pulse high level is $\geq$ 2 5 V and the low level is $\leq$ 0.5 V
3. The lead length of electrical conductors between resistor R1 and input pin, between probes and input and output pins; and between load circuit and output pin, shall be less than 1 inch (25.4 mm).
4. All voltages are referenced to ground

FIGURE 22 Propagation delay time test circuit waveforms for table XVIII.

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TABLE XIX. Output crosstalk (low-level, high-level and duration) test.

Test condition	Parameter		Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
S1	FO	Output crosstalk, low level	6,7,14,15, 18,21,22,23, 25,27	2,4,5,24,26, 28,34,35,37	3,16, 36,38	8	0	1.0	V	23,24
S2	F1		6,7,14,15, 18,22,23,24, 25,27	2,4,5,21,26, 28,34,35,37		9				
S3	F2		6,7,14,15, 18,22,23,25, 26,27	2,4,5,21,24, 28,34,35,37		11				
S4	F3		6,7,14,15, 18,22,23,25, 27,28	2,4,5,21,24, 26,34,35,37		13				
S5	EQ		6,7,14,15, 18,22,23,25, 27,28	2,4,5,21,24, 26,34,35,37		12				
S6	$\overline{C}_O$		7,14,15,18, 22,23,25,27	2,4,5,6,21, 24,26,28,34, 35,37	3,16, 36,38	30				
S7	CP		2,5,14,15, 18,22,23,25, 27	3,4,7,21,24, 26,28,34,35, 37	6,16, 36,38	31				
S8	CG		6,7,14,15, 18,22,23,25	2,4,5,21,24, 26,27,28,34, 35,37	3,16, 36,38	29				
S9	$\overline{P}$	Output crosstalk, low level	2,5,14,15, 18,22,23,25, 27	3,4,7,21,24, 26,28,34,35, 36,37	6,16, 38	17	0	1.0	V	23,24

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TABLE XIX. Output crosstalk (low-level, high-level and duration) test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
S10	$\overline{G}$	Output crosstalk, low level	2,5,14,15, 18,22,23,25, 27	3,4,7,16,21, 24,26,28,34, 35,37	6,36, 38	40	0	1.0	V	23,24
S11	C_{N+X}		2,5,15,16, 18,22,23,25, 27	3,4,7,21,24, 26,28,34,35, 37,38	6,14, 36	19				
S12	C_{N+Y}		2,5,14,15, 16,22,23,25, 27	3,4,7,21,24, 26,28,34,36, 37,38	6,18, 35	39				
S13	C_{N+Z}	Output crosstalk, low level	2,5,14,15, 18,22,23,25, 27	3,4,7,21,24, 26,28,34,36, 37,38	6,16, 35	20		1.0	V	23,24
S14	F0	Output crosstalk, low-level duration	6,7,14,15, 18,21,22,23, 25,27	2,4,5,24,26, 28,34,35,37	3,16, 36,38	8		500	mV	23,25
S15	F1		6,7,14,15, 18,22,23,24, 25,27	2,4,5,21,26, 28,34,35,37	3,16, 36,38	9				
S16	F2		6,7,14,15, 18,22,23,25, 26,27	2,4,5,21,24, 28,34,35,37	3,16, 36,38	11				
S17	F3		6,7,14,15, 18,22,23,25, 27,28	2,4,5,21,24, 26,34,35,37	3,16, 36,38	13				
S18	EQ	Output crosstalk, low-level duration	6,7,14,15, 18,22,23,25, 27,28	2,4,5,21,24, 26,34,35,37	3,16, 36,38	12	0	500	mV	23,25

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TABLE XIX. Output crosstalk (low-level, high-level and duration) test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
S19	$\overline{C}_O$	Output crosstalk, low-level duration	7,14,15,18, 22,23,25,27	2,4,5,6,21, 24,26,28,34, 35,37	3,16, 36,38	30	0	500	mV	23,25
S20	CP		2,5,14,15, 18,22,23,25, 27	3,4,7,21,24, 26,28,34,35, 37	6,16, 36,38	31				
S21	CG		6,7,14,15, 18,22,23,25	2,4,5,21,24, 26,27,28,34, 35,37	3,16, 36,38	29				
S22	$\overline{P}$		2,5,14,15, 18,22,23,25, 27	3,4,7,21,24, 26,28,34,35, 36,37	6,16, 38	17				
S23	$\overline{G}$		2,5,14,15, 18,22,23,25, 27	3,4,7,16,21, 24,26,28,34, 35,37	6,36, 38	40				
S24	C_{N+X}		2,5,15,16, 18,22,23,25, 27	3,4,7,21,24, 26,28,34,35, 37,38	6,14, 36	19				
S25	C_{N+Y}		2,5,14,15, 16,22,23,25, 27	3,4,7,21,24, 26,28,34,36, 37,38	6,18, 35	39				
S26	C_{N+Z}	Output crosstalk, low-level duration	2,5,14,15, 18,22,23,25, 27	3,4,7,21,24, 26,28,34,36, 37,38	6,16, 35	20		500		23,25
T1	FO	Output crosstalk, high level	6,7,14,15, 18,23,25,27	2,4,5,21,22, 24,26,28,34, 35,37	3,16, 36,38	8	0	400	mV	23,24

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TABLE XIX. Output crosstalk (low-level, high-level and duration) test - Continued.

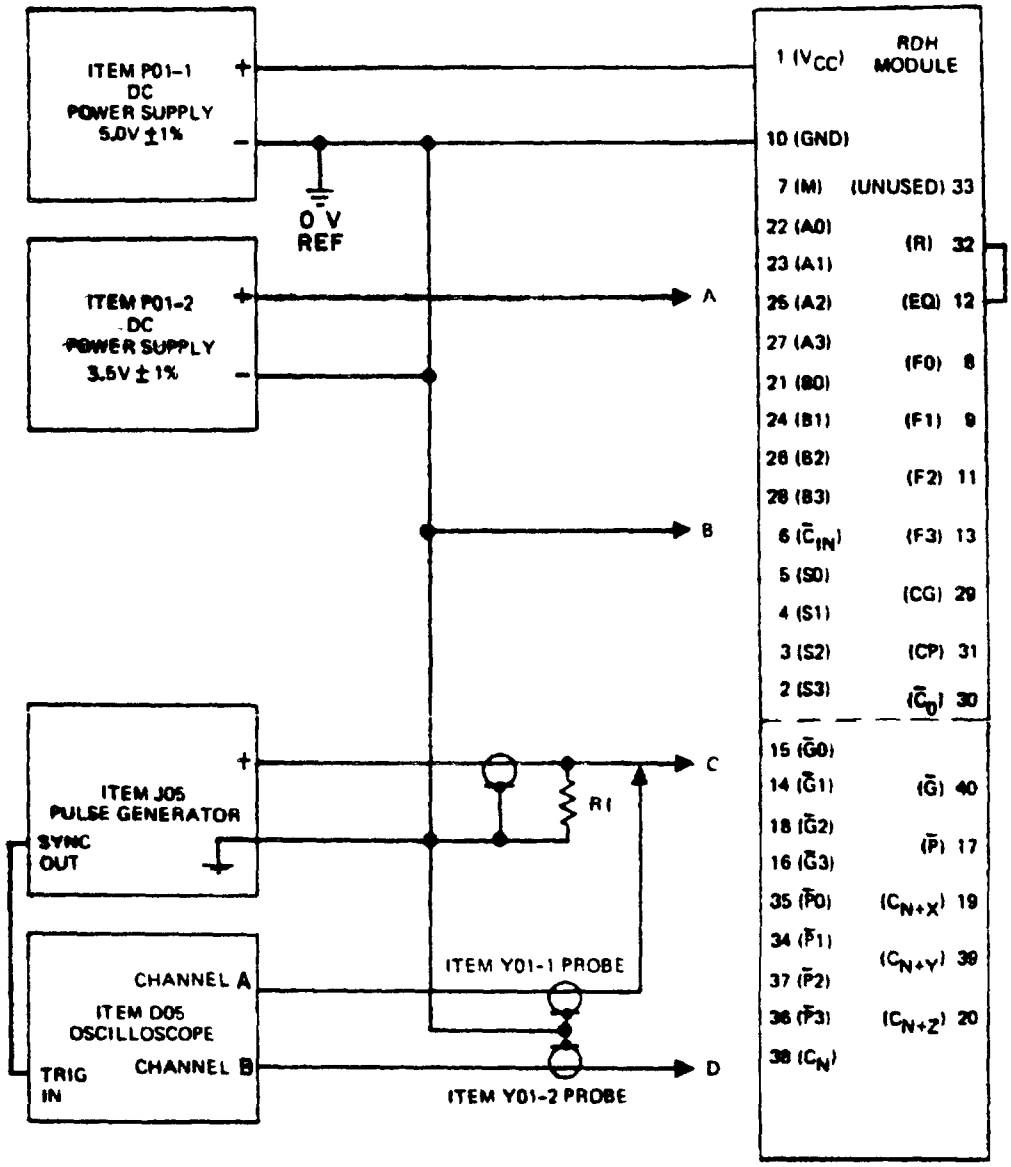
Test condition	Parameter		Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
T2	F1	Output crosstalk, high level	6,7,14,15, 18,22,25,27	2,4,5,21,23, 24,26,28,34, 35,37	3,16, 36,38	9	0	400	mV	23,24
T3	F2		6,7,14,15, 18,22,23,27	2,4,5,21,24, 25,26,28,34, 35,37	3,16, 36,38	11				
T4	F3		6,7,14,15, 18,22,23,25	2,4,5,21,24, 26,27,28,34, 35,37	3,16, 36,38	13				
* T5	EQ		3,4,6,14,15, 18,21,22,23, 24,25,26,27, 28	2,5,7,34,35, 37	16,36, 38	12				
T6	$\overline{C}_O$		6,14,15,18, 21,24,26,28	2,3,4,7,22, 23,25,27,34, 35,37	5,16, 36,38	30				
T7	CG		6,7,14,15, 18,22,23,25, 27	2,4,5,21,24, 26,28,34,35, 37	3,16, 36,38	29				
T8	CP		2,6,7,14,15, 18,21,22,23, 25,27	4,5,24,26, 28,34,35,37	3,16, 36,38	31				
T9	$\overline{P}$		2,5,14,15, 18,22,23,25, 27,36	3,4,7,21,24, 26,28,34,35, 37	6,16, 38	17				
T10	$\overline{G}$	Output crosstalk, high level	2,5,14,15, 16,18,22,23, 25,27	3,4,7,21,24, 26,28,34,35, 37	6,36, 38	40	0	400	mV	23,24

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TABLE XIX. Output crosstalk (low-level, high-level and duration) test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
T11	C_{N+X}	Output crosstalk, high level	6,14,16,18, 22,23,25,27	2,3,4,5,15, 21,24,26,28, 35,37,38	7,34, 36	19	0	400	mV	23,24
T12	C_{N+Y}	Output crosstalk, high level	6,16,18,22, 23,25,27	2,3,4,5,14, 21,24,26,28, 34,35,36,38	7,15, 37	39	0	400	mV	23,24
T13	C_{N+Z}	Output crosstalk, high level	6,14,16,22, 23,25,27	2,3,4,5,18, 21,24,26,28, 34,35,37,38	7,15, 36	20	0	400	mV	23,24

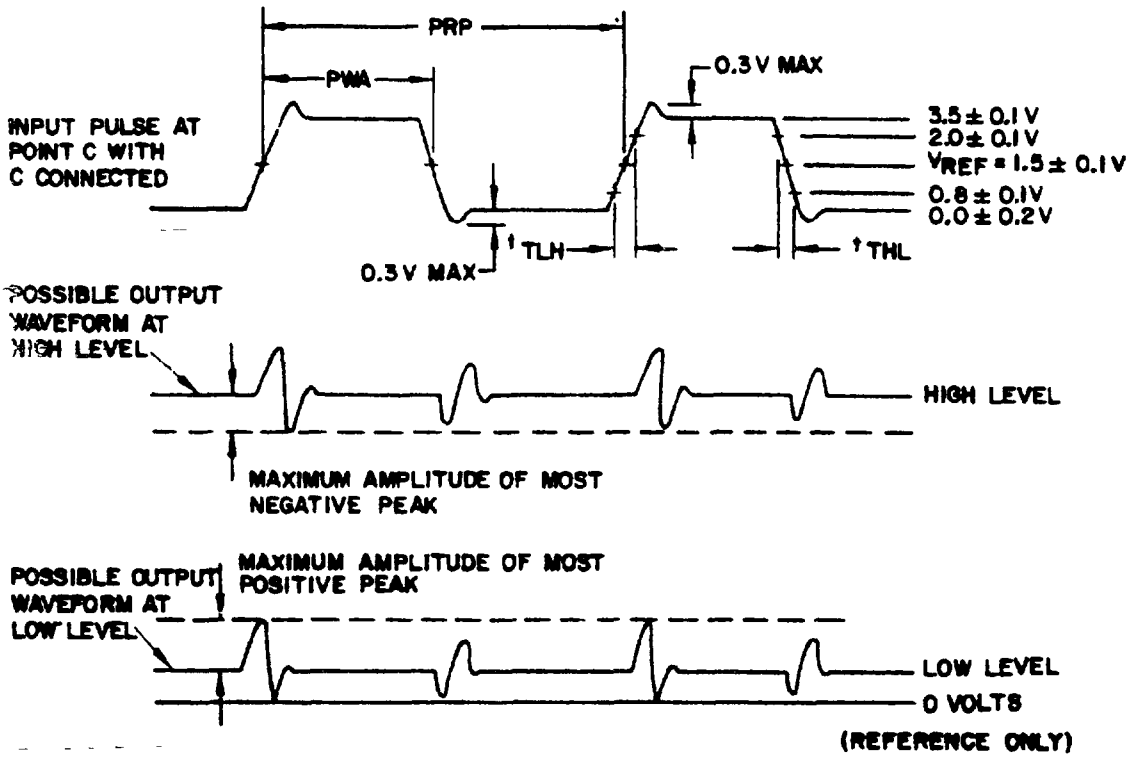
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:57Z
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NOTES:
1. R1 = 51.1Ω ±1%, 1/4 W
2. Load all outputs as specified on figure 14 except capacitor C_L = 50 ±5 pF for table XIX.

FIGURE 23. Output crosstalk (low-level, high-level and duration) test circuit for table XIX.

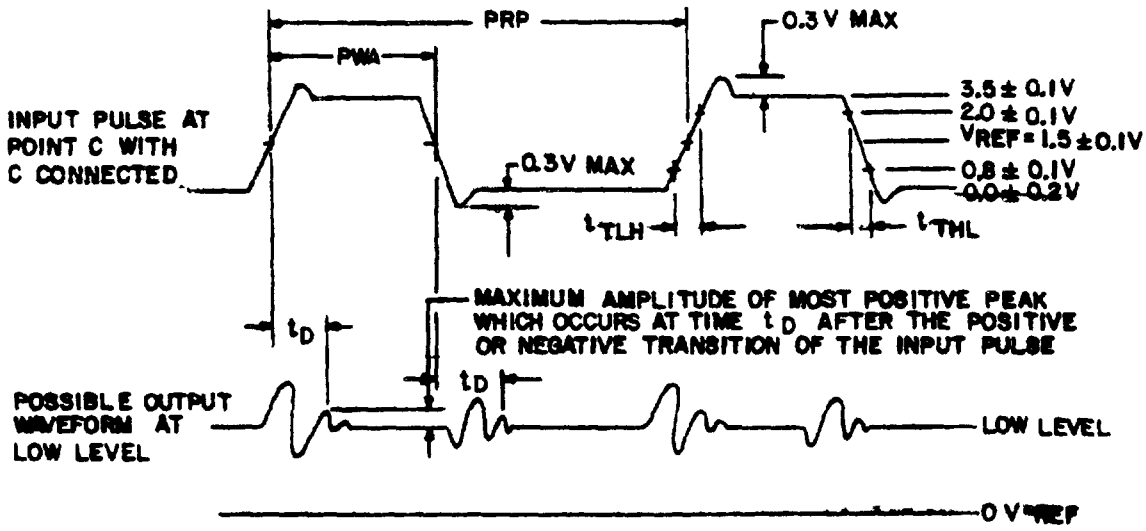
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NOTES:
1. Input pulse parameters
   $t_{THL} = t_{TLH} = 3.5 \pm 1.5$ ns (verified at each input pin).
   PWA = 250 ±50 ns.
   PRP = 500 ±50 ns.
2. The lead length of electrical conductors between resistor R1 and input pins; between probes Y01-1 and Y01-2 and output pins; between load circuits and output pins; and between ground leads of probes Y01-1 and Y01-2 and test fixture and module ground, shall be less than 1 inch (25.4 mm).

FIGURE 24. Output crosstalk (low-level and high-level) test circuit waveforms for table XIX.

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NOTES:
1. Input pulse parameters:
   $t_{TLH} = t_{THL} = 3.5 \pm 1.5$ ns (verified at each input pin).
   PW = 250 ±25 ns.
   PRP = 500 ±50 ns.
   t_D = 45 ns.
2. The lead length of electrical conductors between resistor R1 and input pins; between probes Y01-1 and Y01-2 and input and output pins; between load circuits and output pins; and between ground leads of probes Y01-1 and Y01-2 and test fixture and module ground pins, shall be less than 1 inch (25.4 mm).

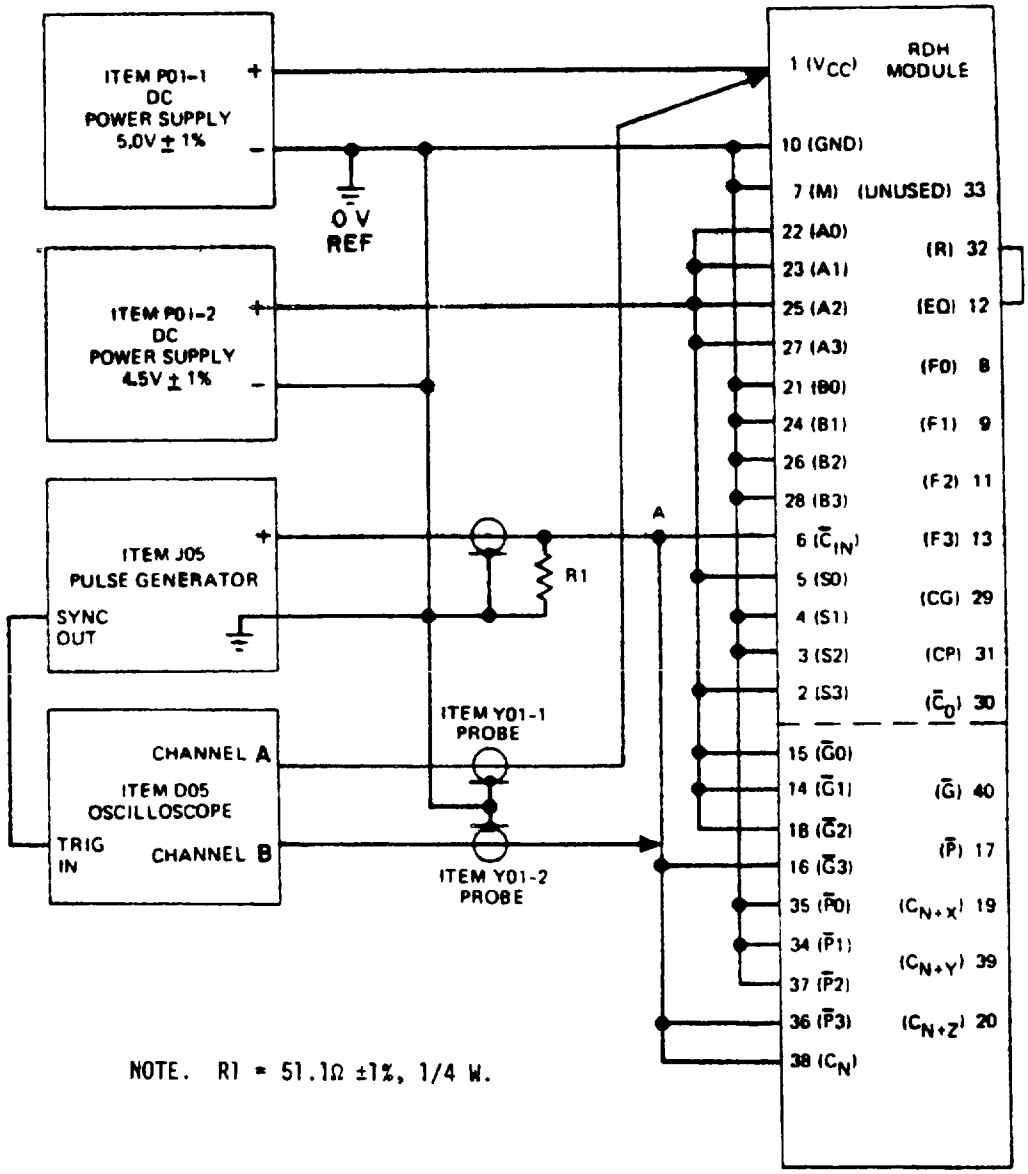
FIGURE 25. Output crosstalk duration test circuit waveforms for table XIX.

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TABLE XX. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25° ±5°C Ambient		Unit	Test figure
		Min	Max		
U	Power supply transient amplitude	0	500	mV	26,27

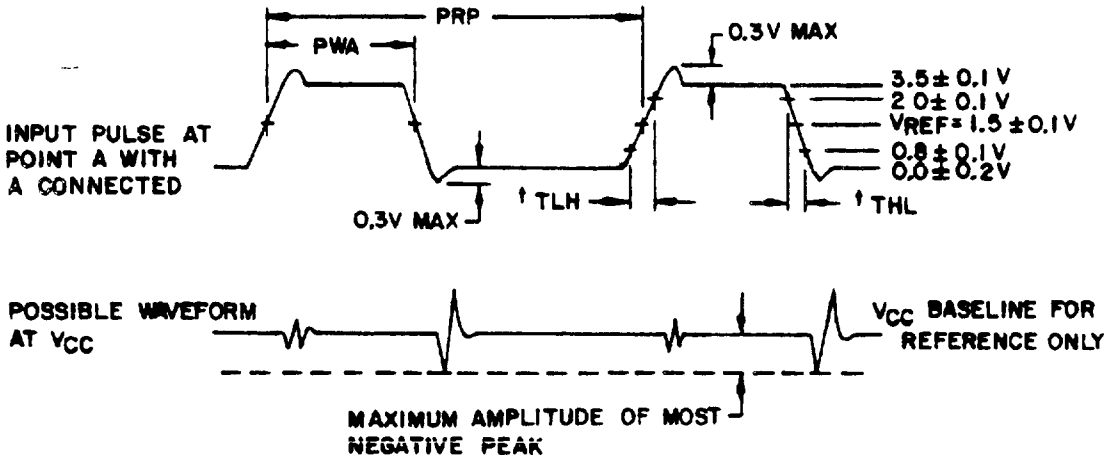
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NOTE. R1 = 51.1Ω ±1%, 1/4 W.

FIGURE 26. Power supply transient amplitude test circuit for table XX

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NOTE. Input pulse parameters.
$t_{TLH} = t_{THL} = 3.5 \pm 1.5$ ns (verified at each input pin)
PWA = 500 ±50 ns.
PRP = 1,000 ±200 ns

FIGURE 27. Power supply transient amplitude test circuit waveforms for table XX

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TABLE XXI. Power supply filter capacitance test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
V	Power supply filter capacitance	Open	1	3.6	7.1	µF	28

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TABLE XXII. Input capacitance test.

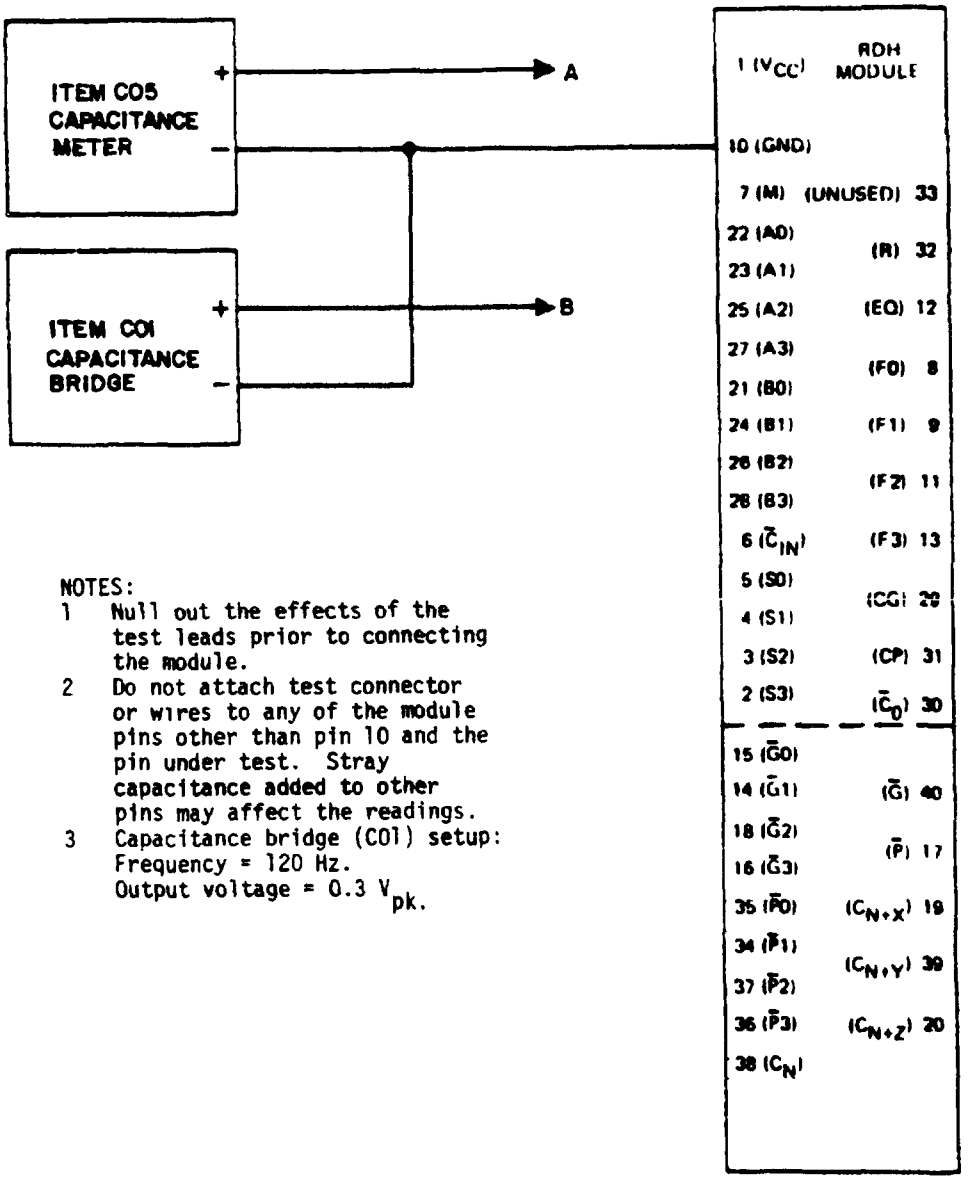
Test condition	Parameter		Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
W1	M	Input capacitance	7	Open	5.0	15.0	pF	28
W2	A3		27					
W3	A2		25					
W4	A1		23					
W5	A0		22					
W6	B3		28					
W7	B2		26					
W8	B1		24					
W9	B0		21					
W10	$\overline{C}_{IN}$		6					
W11	S0		5					
W12	S1		4					
W13	S2		3					
W14	S3		2					
W15	$\overline{G}0$		15					
W16	$\overline{G}1$		14					
W17	$\overline{G}2$		18					
W18	$\overline{G}3$	Input capacitance	16	Open	5.0	15.0	pF	28

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TABLE XXII. Input capacitance test - Continued.

Test condition	Parameter		Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
W19	$\overline{P0}$	Input capacitance	35	Open	5.0	15.0	pF	28
W20	$\overline{P1}$		34					
W21	$\overline{P2}$		37					
W22	$\overline{P3}$		36					
W23	C_N	Input capacitance	38	Open	5.0	15.0	pF	28

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NOTES:
1. Null out the effects of the test leads prior to connecting the module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.
3. Capacitance bridge (C01) setup: Frequency = 120 Hz. Output voltage = 0.3 V_{pk}.

FIGURE 28. Power supply filter and input capacitance test circuit for tables XXI and XXII.

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